# Capstone Project Final Report

## LongK System

| | | |
|---|---|---|
| **Group Members** | Bùi Hoàng Nam | SE04266 |
| | Nguyễn Đình Long | SE04039 |
| | Văn Ngọc Ánh | SE03674 |
| | Nguyễn Huy Quang | SE03792 |
| | Tiêu Quang Thắng | SE04101 |
| **Supervisor** | Lecturer: Trần Quý Ban | |

**Hanoi, August 27, 2018**

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In addition, there are also many other people who have directly or indirectly helped us: our families and friends, book authors, Japanese language researchers, etc.. From the bottom of our heart, thank you all.

# Acronyms

| Acronym | Definition | Note |
|---|---|---|
| **CSRF** | Cross-Site Request Forgery | |
| **DBMS** | Database Management System | |
| **IDE** | Integrated Development Environment | |
| **JS** | JavaScript | |
| **RAM** | Random Access Memory | |
| **SQL** | Structured Query Language | |
| **UI** | User Interface | |

| | | |
|---|---|---|
| **UML** | Unified Modeling Language | |
| **URL** | Uniform Resource Locator | |
| **CSS** | Cascading Style Sheets | |

# Document Conventions

## Date Formatting

All dates in the document are written in yyyy/mm/dd format.

For example, 2017/01/02 means January 2nd, 2017.

## Tables and Figures

Tables and figures are numbered in the format (chapter number)-(# of table or figure).

For example, "Table 2-3" means the third table of the second chapter.

# Chapter 1 : Introduction

## 1.1 Purpose

This chapter provides overview of the project. Including Project information, Project members and Proposal for idea.

## 1.2 Project information

Project name: ShopCar

Project code: LK

Project type: Web

Project group name: LongK Team

Timeline: 14/05/2018 – 20/8/2018

## 1.3 Project members

### 1.3.1 Supervisor

| Full name | E-mail | Title |
|---|---|---|
| Trần Quý Ban | Bantq@fe.edu.vn | Lecturer |

*Table 1-1: Supervisor information*

### 1.3.2 Team Members

| Full name | Student ID | E-mail | Roll |
|---|---|---|---|
| Bùi Hoàng Nam | SE04266 | NamBHSE04266@fpt.edu.vn | Leader |
| Tiêu Quang Thắng | SE04101 | ThangTQSE04101@fpt.edu.vn | Member |
| Văn Ngọc Ánh | SE03674 | AnhNVSE03674@fpt.edu.vn | Member |
| Nguyễn Đình Long | SE04039 | LongNDSE04039@fpt.edu.vn | Member |
| Nguyễn Huy Quang | SE03792 | QuangNHSE03792@fpt.edu.vn | Member |

*Table 1-2: Team member information*

### 1.3.3 Roles and Responsibilities

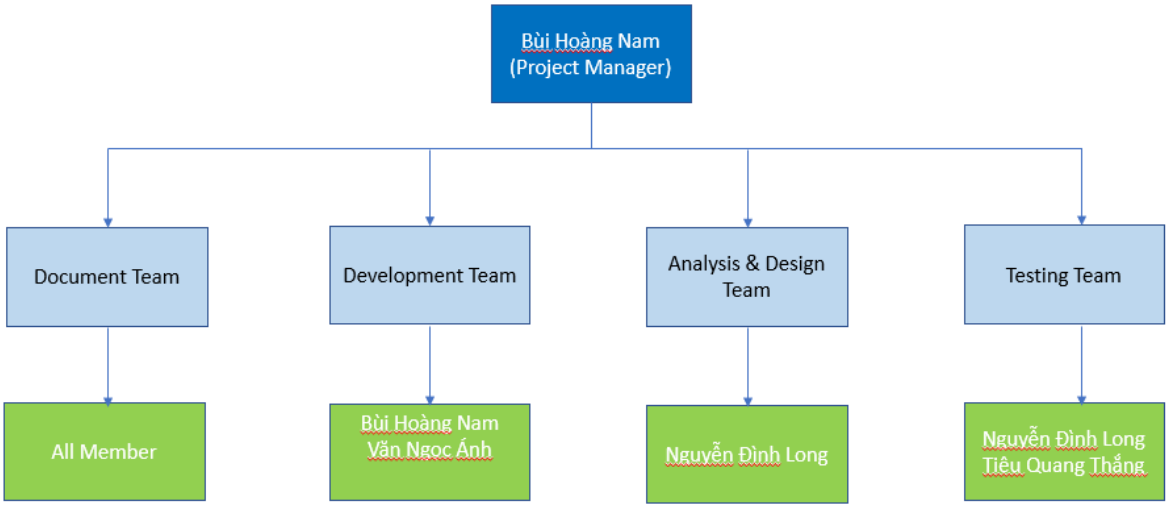



*Figure 1-1: Roles and Responsibilities*

# 1.4 Background

Today, the need for travel is very high, the purchase of cars to serve the family's moving. Every year, the number of selling new cars is up to 300000 units, and hundreds of thousands of used cars are sold. But not everyone has the money to buy a new car, so buying an old car is a smart choice.

# 1.5 The Current System

- Create a website to buy and sell a car and accessories.
- Manage user with admin account and user account.
- Support guest to buy a product.
- Easy to find out product of LongK System.
- Poste for free.

# 1.6 The Proposed System

### 1.6.1 Idea

Create a website named Shop Car, that allows the seller to manage their products. That allows buyers to find the item which they are looking for, it can be car or car's accessories. And buyers can compare the two cars if they are wondering.

### 1.6.2 System Functions

This list below summarizes the main functions of the:

| | |
|---|---|
| **Admin** | • Manager car<br>• Manager accessories<br>• Manager user<br>• Manager post |
| **User** | • View Car<br>• View Accessories<br>• Buy Accessories<br>• View information account<br>• View Detail Car<br>• View Detail Accessories<br>• Checkout<br>• Login<br>• Logout<br>• Facebook Comment<br>• Post topic sell car<br>• Add to Cart<br>• View Transaction information |
| **Guest** | • Register<br>• View Car<br>• View Accessories<br>• Buy Accessories |

| | |
|---|---|
| | • Post topic sell car<br>• Facebook Comment<br>• View Detail Car<br>• View Detail Accessories |

*Table 1-3 Main functions*

### 1.6.3 Technical Approach

### 1.6.4 Out-of-Scope Functions

Due to time constraints, the following features are not included in the initial release of the project but not permanently excluded, and will be developed in future versions of the project:

- Don’t have template for news
- Link service to online check out
- Mini chat bot to communication with customer
- Statistical quantities

# Chapter 2 : Project Plan

## 2.1 Project Organization

### 2.1.1 Purpose

This chapter provides an overview of the project plan, including software process model, project organization and project management plan.

### 2.1.2 Software Process Model

WaterFall model

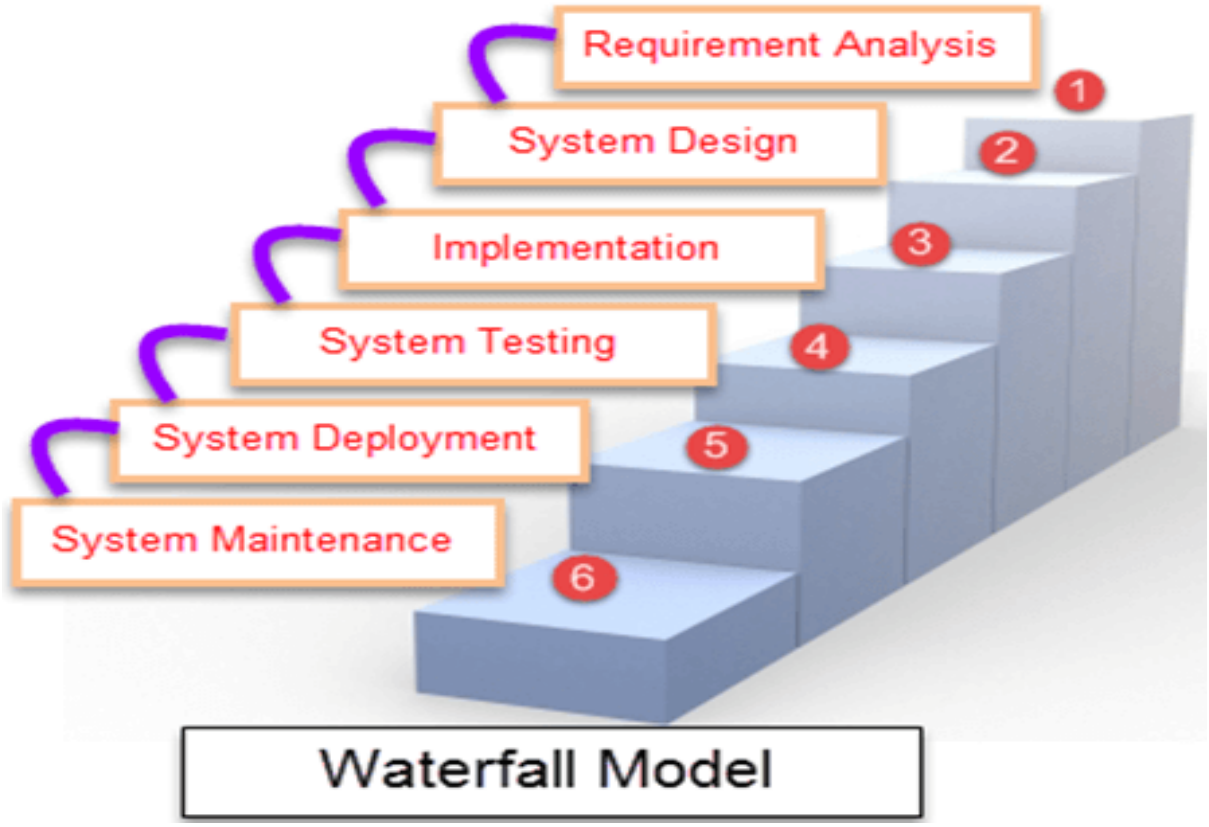



*Figure 2-1: Iterative and Incremental Software Process Model (image: IBM)*

### 2.1.3 Roles and Responsibilities

### 2.1.4 Organizational Structure

| Role | Responsibility |
|---|---|
| Project Manager | Responsible for managing the project as a whole and leading the team; planning, defining scope, developing schedules, coordinating communication, writing documentation, analyzing and managing risks; generally responsible for keeping the project team focused on the right goal. |
| Technical Leader | Responsible for choosing which technologies will be used, resolving difficult technical problems, training other members ensuring all members are capable of coding, assisting |

| | |
|---|---|
| | the Project Manager in assigning developmentbased tasks, keeping track of development work done by other developers. |
| Quality Assurance Manager | Responsible for ensuring that the project meets quality standards and requirements. |
| Test Leader | Responsible for planning, monitoring and controlling testing activities, estimating the testing to be done, ensuring test environment and configuration, writing summaries and test reports. |
| Content Manager | Responsible for writing the website's content, as well as inserting dictionary data and lesson data. |
| Developer | Responsible for coding the product. |
| Tester | Responsible for conducting tests. |

*Table 2-1: Organization Structure*

### 2.1.5 Project Team Members

| Team Member | Roles |
|---|---|
| Bùi Hoàng Nam | Project Manager , Test Leader, Content Manager, Quality Assurance Manager |
| Tiêu Quang Thắng | Tester, Content Manager |
| Nguyễn Đình Long | Tester, Content Manager |
| Văn Ngọc Ánh | Developer, Technique Leader |
| Nguyễn Huy Quang | Developer |

*Table 2-2: Project Team Members*

### 2.1.6 Tools and Techniques

| | |
|---|---|
| Programming languages | JavaScript, Java, ReactJS |
| Framework | Spring Tool Suite |
| Software architecture | MVC |
| Version control | BitBucket |

| | |
|---|---|
| IDEs/Editors | Spring |
| UML tools | Astah Professional 7.1.0, draw.io |
| DBMS | MySQL |
| Project management tool | Microsoft Project |
| Development process | Test-driven development |
| Testing tools | Postman |
| Communication tools | Skype, Facebook |
| Files management tools | Google Driver |

*Table 2- 3: Tools and Techniques*

### 2.1.7 Techniques

### 2.1.8 Front-end slide

- Programming languages: JavaScript , HTML, CSS, Java
- Framework: React
- User interface Library: ReactJS.
- Architecture: MVC

### 2.1.9 Back-end side:

- Programming languages: Java
- Framework: Srping Tool Suite
- Architecture: MVC

### 2.1.10 Database side:

- MySQL

# 2.2 Project Management Plan

## 2.2.1 Tasks

| | Task Mode | Task Name | Duration | Start | Finish | Predecessors | Resource Names |
|---|---|---|---|---|---|---|---|
| 1 | | LongK Project | 81.06 days | Mon 5/14/18 | Tue 8/14/18 | | NamBH,LongND |
| 2 | | Project Initial | 13.13 days | Mon 5/14/18 | Sat 5/26/18 | | |
| 3 | | First meeting | 1 day | Mon 5/14/18 | Mon 5/14/18 | | AnhVN,LongND, |
| 4 | | Determine project scope | 2 days | Mon 5/14/18 | Tue 5/15/18 | 3SS | LongND,NamBH |
| 5 | | Team weekly meeting 1 | 1 day | Mon 5/14/18 | Mon 5/14/18 | 3SS | AnhVN, LongND,NamBH, |
| 6 | | List requirements about first topic | 2 days | Mon 5/14/18 | Wed 5/16/18 | 5 | LongND,NamBH |
| 7 | | Second meeting | 1 day | Wed 5/16/18 | Thu 5/17/18 | 6 | AnhVN,LongND, |
| 8 | | Change topic of project | 2 days | Wed 5/16/18 | Thu 5/17/18 | 7SS | AnhVN, LongND,NamBH, |
| 9 | | List requirements | 2 days | Thu 5/17/18 | Mon 5/21/18 | 8 | AnhVN,LongND, |
| 10 | | Create report 1 | 1 day | Mon 5/21/18 | Mon 5/21/18 | | AnhVN,LongND, |
| 11 | | Finish project initial | 1 day | Sat 5/26/18 | Sat 5/26/18 | | AnhVN,LongND, |
| 12 | | Project Planning | 16.19 days | Thu 5/31/18 | Mon 6/18/18 | | |
| 13 | | Create report 2 (Project Plan) | 4 days | Thu 5/31/18 | Tue 6/5/18 | | |
| 14 | | Supervisor weekly meeting 2 | 1 day | Tue 6/5/18 | Tue 6/5/18 | 13 | |
| 15 | | Team weekly meeting 2 | 1 day | Tue 6/5/18 | Tue 6/5/18 | 14SS | |
| 16 | | List up user requirement | 1 day | Tue 6/5/18 | Wed 6/6/18 | 15 | LongND |
| 17 | | List up use cases | 1 day | Tue 6/5/18 | Wed 6/6/18 | 15 | ThangTQ,Quangl |
| 18 | | Create use cases description | 4 days | Wed 6/6/18 | Mon 6/11/18 | 17 | ThangTQ,Quangl |
| 19 | | Create report 3(Software) requirement specification) | 1 day | Mon 6/11/18 | Mon 6/11/18 | | LongND, NamBH,QuangN |
| 20 | | Team weekly meeting 3 | 1 day | Mon 6/11/18 | Tue 6/12/18 | 19 | |

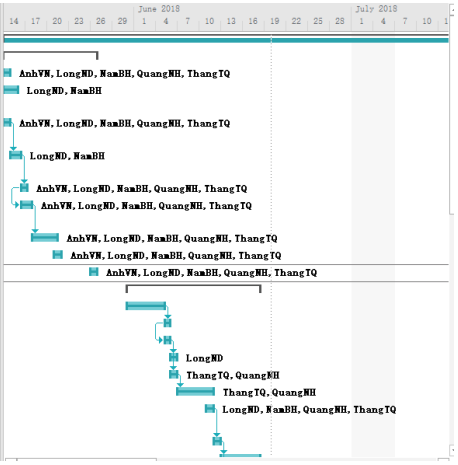



*Figure 2-1: Tasks*

Refer to "LongKProject.mpp" file.

## 2.2.2 Meeting Minutes

All Meeting Minutes follow the below template:

Fpt University
ĐẠI HỌC FPT

**MEETING MINUTES**

| | | | |
|---|---|---|---|
| **Project:** | GCS | **Project Code:** | GCS |
| **Project Manager:** | Bùi Hoàng Nam | **Conductor:** | Trần Quý Ban |
| | | **Secretary:** | |
| **Date:** | 12-May-2018 | **Time:** | 03:30 PM – 05:00 AM |
| **Venue:** | | | |
| **Topic of meeting:** | Project Team Meeting | | |

| **Attendees:** | | | | |
|---|---|---|---|---|
| **No** | **Full name** | **Unit/Group** | **Position** | **Attendance** |

| | | | | |
|---|---|---|---|---|
| 1 | Văn Ngọc Ánh | G1 | Technical leader | Present |
| 2 | Bùi Hoàng Nam | G1 | PM | Present |
| 3 | Văn Ngọc Ánh | G1 | DEV | Present |
| 4 | Nguyễn Huy Quang | G1 | DEV | Present |
| 5 | Tiêu Quang Thắng | G1 | Tester | Present |
| 6 | Tiêu Quang Thắng | G1 | Tester | Present |
| 7 | | | | |
| 8 | | | | |
| | | | | |
| | | | | |
| | | | | |

*Table 2-4: Meeting Minutes Template*

### 2.2.3 Document Convention

- Font family: Times New Roman
- Font size: 12 pt
- Page heading text: LongK
- Page header size: 12 pt

### 2.2.4 Coding Conventions

#### 2.2.4.1 Front -end Coding Convention

Folder/files naming:

- Written in “CamelCase” and “PascalCase”, for example: “Tests”, “Header.js”,…

JavaScript Style Guide:

We strictly follow JavaScript Standard Style Guide and ReactJS, with linter and automatic code fixer. Please refer to the official website at https://reactjs.org/ and https://standardjs.com/rules.html.

#### 2.2.4.2 Back -end Coding Convention

Folders/files naming:

- Wrritten in lowecase, with hyphen (“-“) between words. Example “auto-fun”.

TypeScript Style Guide:

- We strictly follow Airbnb JavaScript Style Guide, integrated with linter TSLint. Please refer to the offcicial website at https://github.com/airbnb/javascript.

#### 2.2.4.3 Database Coding Convention

Tables naming

- Names should be written entirely in lowercase, with underscrore("_") between words. Example:"car_gear","car_specification".
- Table should have their name written in singular form.

Columns naming:

- Names should be written entirely in lowercase, with underscore("_") between words. Exampl:"car_name","brand_id".
- Primary key fields shold be named id. Example:

CREATE TABLE car_gear(

Id BIGINT PRIMARY KEY NOT NULL,

Car_VARCHAR(255) NOT NULL);

### 2.2.5 Risk Management Plan

The project manager works with the project team and ensures that risks are actively identified, analyzed, and managed throughout the life of the project. Risks will be identified as early as possible in the project so as to minimize their impact. The steps for accomplishing this are outlined in the following sections. The PM will serve as the Risk Manager for this project.

### 2.2.6 Risk description

| No | Rank | Risk | Description | Category | Root Cause | Triggers | Probability | Impact |
|---|---|---|---|---|---|---|---|---|
| R1 | 1 | Technology and framework are difficult to use | Technology and framework are difficult to use, all team had not experience about this technology before | Technology | Team choosed wrong technology and framework | In any task or feature, the team must learn from the basics | High | High |
| R2 | 3 | Team member out of team | Some teammate out of team | People | The team has conflict or the members have another intention for the future | Some people are unhappy with others | High | High |
| R3 | 1 | Appearance of competitors | There is an other team has the same idea about this app | Market | May be the team is not secure internal | There is an other team has the same idea | Low | High |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | information | about this app | | |
| R4 | 2 | Some member work inefficiently | There are some member work inefficiently | People | May be the Technology is no suitable with them or is not their strength. Or may be they don't want to do it | A few people in the team do 80% of the work and the other do only 20% | High | Medium |
| R5 | 1 | Resources were lost or deleted | Resources were lost or deleted | Process | Does not have protection and back up | When project was delayed | Medium | High |
| R6 | 3 | Wrong design | Can't response the requirement of customer | Process | Because member design wrong | The customer plain | Medium | High |
| R7 | 2 | Delay task | The plan will not complete on time | People | The task is very long or difficult | when project was delayed | Medium | High |
| R8 | 1 | Customer do not provide enough resources for the project work | Customer do not provide enough resources for the project work. And team members can't complete system | process | Customer do not provide enough resources for the project work | when you work, and you don't have enough resources to do | Low | Low |
| R9 | 1 | Customer requests new features but team members can't response | Customer requests new features but team members can't response | People | Customer requests new features outside contracts | when customer request | Low | Low |
| R10 | 2 | Developer use a lot of time to learn new technology | The project have to use new technology and developer use a lot of time to learn it | People | The project use new technology | when project start | Medium | Medium |
| R11 | 3 | Conflict among development team member | Poor communication and misunderstanding | People | Arguments while doing project | Working attitude among members is not good | Medium | Medium |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | s | | | | | | |
| R12 | 1 | Requirements are not clear | The requirements and scope are not clear | Process | Team leader did not work efficiently | When a developer does not know what to do | Medium | Medium |
| R13 | 2 | Members intentionally hide errors during development | Team member in the development process encountered an error but did not want to report it to the team leader | People | Team member do not want to be responsible for their faults, poor personal sense | Software error during the demo | High | High |
| R14 | 3 | Many feature | Too many features and hard to predict was estimate | Structure / process | Team leader do not estimate correct workload | When the task cannot finish before deadline | Medium | Medium |
| R15 | 1 | Members have low skills | Team members do not reach the required technical standard to do some task. So that, we need a support from third-party organization | People | Lack of training | Members take more time to finish task or cannot do task requirement | Low | Low |

*Table 2-5: Risk description*

### 2.2.7 Probability – Impact matrix

| | | | | |
|---|---|---|---|---|
| Probability | High | | R4 | R1,R13,R2 |
| | Medium | | R10, R11,R12,R14 | R5,R6,R7 |
| | Low | R9,R15,R8 | | R3 |
| | | Low | Medium | High |
| | Impact | | | |

*Table 2-6: Probability – Impact matrix*

## 2.3 Communication Plan

### 2.3.1 Weekly Meeting Schedule

Every week, the project team will have a meeting with the supervisor to report progress, resolve issues and discuss problems together

### 2.3.2 Daily Discusing Schedule

Every day at 01:30 PM, the whole team will meet either in the FPT University dormitory or on Facebook in order to report daily progress. All team members must report what they have done during that day, which problems they encountered and what they will do the next day. The whole team will then discuss problems and work together to resolve issues (if any). By doing this, team members will achieve good understanding of the project's progress, what they have done and what they still have to do.

### 2.3.3 Unscheduled Meeting

In case there is an urgent problem, an unscheduled meeting may be held, usually on Facebook Messenger or Skype.

### 2.3.4 Communication Channel

Other than face-to-face meetings, our primary online communication channel is Facebook Messenger. Important messages are posted in the team's private Facebook group. By minimalizing the number of communication channels, we can more easily manage messages and reduce the risk of missing important information. However, in case there are problems, we also make phone calls and video calls.

## 2.4 User requirements

### 2.4.1 Admin functions

#### 1.1.1.1. Accept user’s post

Admin can accept post to sell car of user.

#### 1.1.1.2. Reject user’s post

Admin can reject user’s post.

#### 1.1.1.3. Send email or SMS to customer

Send notification to customer talking about new product, promotions …

#### 1.1.1.4. Manager customer’s information

Admin can manager customer’s information

#### 1.1.1.5. Display status of product

Display status of product which is in stock or sold out

#### 1.1.1.6. Processing order

Admin can agree or cancel order of user (guest)

#### 1.1.1.7. Order status

Admin can change order status

*Eg: Order Confirmed, Payment Confirmed, Order Shipped ...*

#### 1.1.1.8. Display related product

Display product with same price

### 2.4.2 User functions

#### 1.1.1.9. Post

User can create new post to sell car. But user must wait to admin accept it.

#### 1.1.1.10. Edit

User can edit their post like price, information of car…

#### 1.1.1.11. Delete

User can delete their post

#### 1.1.1.12. Signup

#### 1.1.1.13. Login

User can login by Facebook, google account or their account.

#### 1.1.1.14. Logout

#### 1.1.1.15. Change password

#### 1.1.1.16. Purchase history

User can review their purchase history.

#### 1.1.1.17. Upload picture

User can upload picture of car in their post

#### 1.1.1.18. Review

Everyone can write review on each product

### 2.4.3 Guest functions

#### 1.1.2.1. Add to bag

User can only add car's accessories to bag.

#### 1.1.2.2. Remove from bag

#### 1.1.2.3. Checkout

User can checkout by credit card or cash on delivery.

#### 1.1.2.4. Compare products

User (Guest) can choose 2 products and compare it.

#### 1.1.2.5. Chat with bot

User (Guest) can chat with bot.

#### 1.1.2.6. Basic search

Search by name, year of manufacture

#### 1.1.2.7. Advance search

Search by price from this amount to that amount, from this year to that year

#### 1.1.2.8. Sort

Sort by price, top seller, newest first

#### 1.1.2.9. View photo

View photos of car in slide format

#### 1.1.2.10. Zoom photo

Zoom photo when user view product

# Chapter 3 – Software Requirement Specification

## 3.1 Purpose

This chapter outlines functional and non-functional requirements of our website. It also provides some format constraints in common requirements and project success criteria. All member will work based on the information provided in this chapter.

## 3.2 Functional Requirements

### 3.2.1 Use Case Diagram

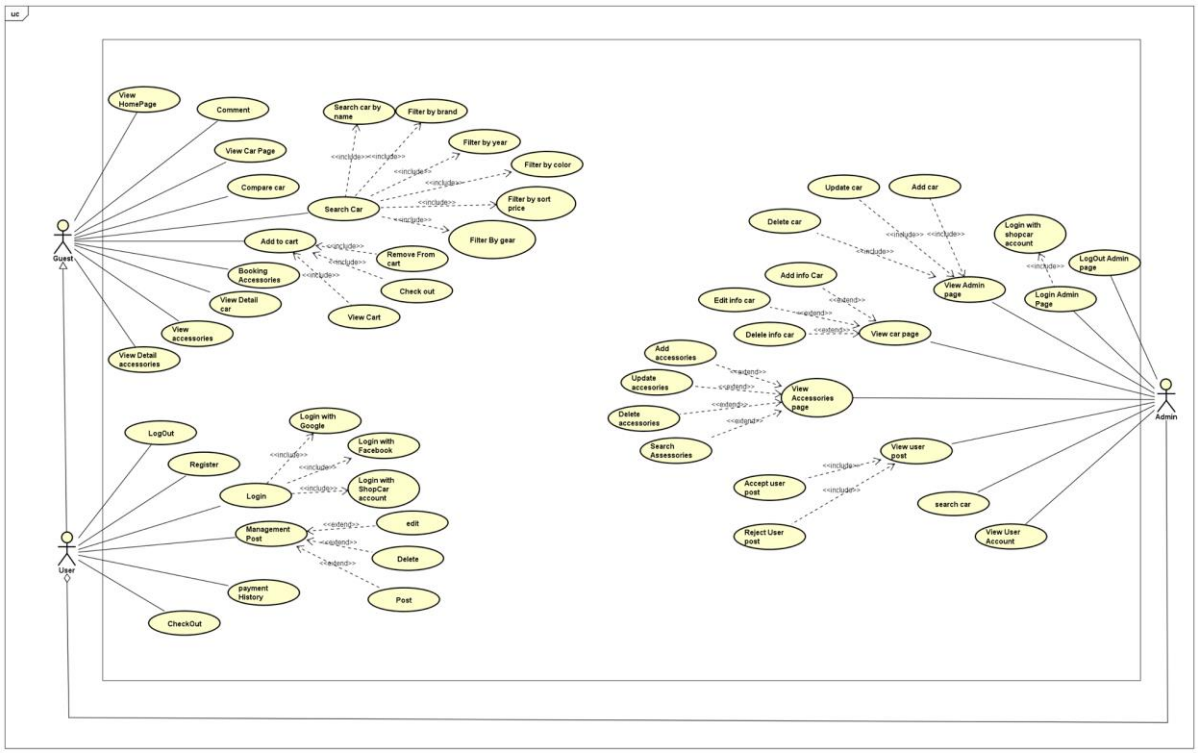



*Figure 3-1: Use Case Diagram*

### 3.2.2 Business Rules

| No | Description |
|---|---|
| B1 | Gmail account is valid |
| B2 | Facebook account is valid |

| | |
|---|---|
| B3 | Phone number must not be valid |
| B4 | Username must not be blank |
| B5 | Address must not be valid |
| B6 | Length of username must not exceed 100 characters |
| B7 | Length of user description must not exceed 2000 characters |
| B8 | Step name must not be blank |
| B9 | Name of car must not be blank |
| B10 | Price of car must not be blank |
| B11 | A comment to send must not be empty |
| B12 | Step phone number of user sale car must not be blank |

*Table 3-1: Business Rules*

## 3.3 Use Cases

| Actor | Description |
|---|---|
| Guest | Everyone |
| User | Everone who has logged into LongK by a Google or Facebook account or create with form of website |
| Admin | A person with the highest permission level, and is responsible for managing the website. |

*Table 3-2: Actor Description*

| ID | Primary Actor | Secondary Actors | Name | Assignee |
|---|---|---|---|---|
| UC-1 | Guest | User, Admin | View Homepage | QuangNH |
| UC-2 | Guest | User, Admin | View Car Page | QuangNH |
| UC-3 | Guest | User, Admin | Search by  Car name | QuangNH |
| UC-4 | Guest | User, Admin | Filter by brand | QuangNH |
| UC-5 | Guest | User, Admin | Filter by year | QuangNH |

| | | | | |
|---|---|---|---|---|
| UC-6 | Guest | User, Admin | Filter by price | QuangNH |
| UC-7 | Guest | User, Admin | Filter by color | QuangNH |
| UC-8 | Guest | User, Admin | Filter by sort price | QuangNH |
| UC-9 | Guest | User, Admin | Filter By gear | QuangNH |
| UC-10 | Guest | User, Admin | Compare Car | QuangNH |
| UC-11 | Guest | User, Admin | View Detail Car | QuangNH |
| UC-12 | Guest | User, Admin | View accessories | QuangNH |
| UC-13 | Guest | User, Admin | View detail accessories | QuangNH |
| UC-14 | Guest | User, Admin | Comment | QuangNH |
| UC-15 | Guest | User, Admin | Add to cart | QuangNH |
| UC-16 | Guest | User, Admin | View Cart | QuangNH |
| UC-17 | Guest | User,Admin | Checkout | ThangTQ |
| UC-18 | Guest | User, Admin | Remove from cart | QuangNH |
| UC-19 | User | Admin | Login with shopcar account | QuangNH |
| UC-20 | User | | Login with google | QuangNH |
| UC-21 | User | | Login with facebook | QuangNH |
| UC-22 | User | Admin | Register | AnhVN |
| UC-23 | User | Admin | Logout | AnhVN |
| UC-24 | User | Admin | Booking accessories | AnhVN |
| UC-25 | User | Admin | Post Car | AnhVN |
| UC-26 | User | Admin | Update Car | AnhVN |
| UC-27 | Admin | | Login Admin page | AnhVN |
| UC-28 | Admin | | LogOut Admin page | AnhVN |

| | | | | |
|---|---|---|---|---|
| UC-29 | Admin | | View admin page | AnhVN |
| UC-30 | Admin | | View accessories admin page | AnhVN |
| UC-31 | Admin | | View user account | AnhVN |
| UC-32 | Admin | | Add Car | ThangTQ |
| UC-33 | Admin | | Search Car | ThangTQ |
| UC-34 | Admin | | View user post | ThangTQ |
| UC-35 | Admin | | Up post of user | ThangTQ |
| UC-36 | Admin | | Check order status | ThangTQ |
| UC-37 | Admin | | Accept user post | ThangTQ |
| UC-38 | Admin | | Reject user post | ThangTQ |
| UC-39 | Admin | | Accept Order | ThangTQ |
| UC-40 | Admin | | Update Car | ThangTQ |
| UC-41 | Admin | | Delete Car | ThangTQ |
| UC-42 | Admin | | Add accessories | ThangTQ |
| UC-43 | Admin | | Update accessories | ThangTQ |
| UC-44 | Admin | | Delete accessories | ThangTQ |
| UC-45 | Admin | | Search accessories | ThangTQ |

*Table 3-3: Use Cases List*

### 3.3.1 Guest

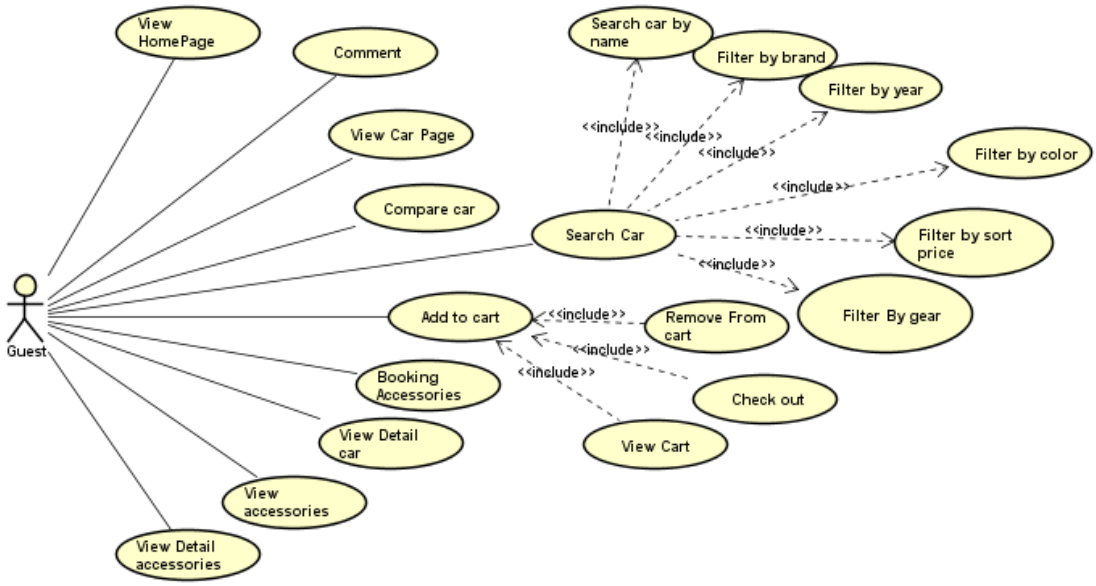



*Figure 3-2: Use Case Guest Diagram*

#### 3.3.1.1View HomePage

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-1 | | |
| **Use Case Name** | View Homepage | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest actor | **Secondary Actors** | Admin,User |
| **Description** | - View the homepage of the website. | | |
| **Pre-conditions** | - N/A | | |
| **Post-conditions** | - The homepage of the website is displayed | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage. | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.1.2View Car Page

| | |
|---|---|
| **Use Case ID** | UC-2 |

| | | | |
|---|---|---|---|
| **Use Case Name** | View page Car | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - View car page | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click menu “Bán xe” | | |
| **Post-conditions** | - Display the cars use already search with ascending price | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click menu “Ban xe” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.3Search by Car Name

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-3 | | |
| **Use Case Name** | Search by Car Name | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | Search list car by name | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click menu “Bán xe” | | |
| **Post-conditions** | - Display car by Carname | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click menu “Bán xe” .<br>3. Input carname in text search then click button “Search” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.4Filter by Brand

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-4 | | |
| **Use Case Name** | Filter by Car brand | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view car by brand | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click menu “Bán xe” | | |
| **Post-conditions** | -display car by brand | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Select Filter “ Brand”<br>4. Click on button “Filter” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.5Filter by Car year

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-5 | | |
| **Use Case Name** | Filter by Car year | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view car by year | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” | | |
| **Post-conditions** | - display car by year | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Select Filter “ Nam sx” | | |

| | |
|---|---|
| | 4. Click on button “Filter” |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.1.6Filter by Price

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-6 | | |
| **Use Case Name** | Filter by Car price | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view car by price | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” | | |
| **Post-conditions** | -display car by price | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Input price in text “ pirce “<br>4. Click on button “Filter” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.7Filter by Car Color

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-7 | | |
| **Use Case Name** | Filter by Car color | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |

| | |
|---|---|
| **Description** | - view car by color |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” |
| **Post-conditions** | -display car by color |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Select Filter “ color”<br>4. Click on Filter |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.1.8 Filter by sort price

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-8 | | |
| **Use Case Name** | Filter by Sort Price | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view car by sort | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” | | |
| **Post-conditions** | -display car by sort | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Select Filter “ sort ”<br>4. Click on “Filter” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.9 Filter by Gear

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-9 | | |
| **Use Case Name** | Filter by Car gear | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view car by gear | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” | | |
| **Post-conditions** | -display car by gear | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Select Filter “ gear”<br>4. Click on button “Filter” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.10 Compare Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-10 | | |
| **Use Case Name** | Filter by Car brand | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - view Compare car | | |
| **Pre-conditions** | Pre 1.0 Guest must access the website<br>Pre 1.1 Click on menu “Bán xe” | | |
| **Post-conditions** | -display Compare car | | |
| **Normal Flow** | 1. User enters the URL of the website to visit the homepage.<br>2. Click on menu “Ban xe”<br>3. Click on radio button “Compare car” | | |
| **Alternative Flows** | N/A | | |

| | |
|---|---|
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.1.11 View Detail Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-11 | | |
| **Use Case Name** | View Detail Car | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest actor | **Secondary Actors** | Admin,User |
| **Description** | - View page detail car of website. | | |
| **Pre-conditions** | - 1 .View “Xe” page<br>- 2 .Click on product of “Xe” page | | |
| **Post-conditions** | - 1.The page Detail car of the website is displayed<br>- 2.Show image and infomation of car in page car detail | | |
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest click “Xe” on menu<br>3. Guest click image product of “Xe” page | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.1.12 View Accessories Page

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-12 | | |
| **Use Case Name** | View accessories page | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - View page accessories of website | | |
| **Pre-conditions** | - View Accessories page | | |
| **Post-conditions** | - The page Accessories of the website is displayed | | |

| | |
|---|---|
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest click “Phụ kiện” on menu |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.1.13 View Detail Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-13 | | |
| **Use Case Name** | View detail accessories | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - View page accessories detail of website | | |
| **Pre-conditions** | - 1 .View Accessories page<br>- 2 .Click on product of “Phụ kiện” page | | |
| **Post-conditions** | - 1.The page Detail accessories of the website is displayed<br>- 2.Show image and infomation of car in page accessories detail | | |
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest click “Phụ kiện” on menu<br>3. Guest click image product of “Phụ kiện” page | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.1.14 Comment

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-14 | | |
| **Use Case Name** | Comment | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |

| | | | |
|---|---|---|---|
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Comment with account facebook in website | | |
| **Pre-conditions** | - 1. Login website with account FB<br>- 2 .View “Xe” page<br>- 3 .Click on product of “Xe” page<br>- 4 .Comment | | |
| **Post-conditions** | - 1.The page Detail car of the website is displayed<br>- 2.Show comment of user on “Bình luận” | | |
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest login with Facebook<br>3. Guest click “Xe” on menu<br>4. Guest click image product of “Xe” page<br>5. Guest comment on textbox “Bình luận” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | website displays error message | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.1.15 Add to Cart

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-15 | | |
| **Use Case Name** | Add to cart | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Add product of accessories page on cart | | |
| **Pre-conditions** | - 1 View Accessories page<br>- 2 Click icon “ Giỏ hang “ on product of page “Phụ kiện” | | |
| **Post-conditions** | - Product added to cart | | |
| **Normal Flow** | 1. Guest r enters the URL of the website to visit the homepage.<br>2. Guest click “Phụ kiện” on menu<br>3. Guest click icon “ Giỏ hàng “ on product of “Phụ kiện” page | | |
| **Alternative Flows** | - User login on website | | |

| | |
|---|---|
| | - go to step 3 in normal flow |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.1.16 View Cart

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-16 | | |
| **Use Case Name** | View Cart | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - View product added to cart | | |
| **Pre-conditions** | - Click icon " Giỏ hàng " on menu | | |
| **Post-conditions** | - Cart page displayed<br>- View product added to cart | | |
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest click icon " Giỏ hàng " on menu | | |
| **Alternative Flows** | - User login on website<br>- go to step 2 in normal flow | | |
| **Exceptions** | - Product is not added to cart | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.1.17 Check out

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-17 | | |
| **Use Case Name** | Checkout | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Remove product has been added to cart | | |
| **Pre-conditions** | - Click icon " Giỏ hàng " on menu | | |

| | |
|---|---|
| | - Click icon “ xác nhận thanh toán” of product on cart |
| **Post-conditions** | - Cart page displayed<br>- Check out product has been added to cart |
| **Normal Flow** | 1. Guest enters the URL of the website to visit the homepage.<br>2. Guest click icon “ Giỏ hàng “ on menu<br>3. Guest click icon “xác nhận thanh toán” of product on cart |
| **Alternative Flows** | - User login on website<br>- go to step 3 in normal flow |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.1.18 Remove from Cart

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-18 | | |
| **Use Case Name** | Remove from cart | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Remove product has been added to cart | | |
| **Pre-conditions** | - Click icon “ Giỏ hàng “ on menu<br>- Click icon “ X” of product on cart | | |
| **Post-conditions** | - Cart page displayed<br>- Remove product has been added to cart | | |
| **Normal Flow** | 4. Guest enters the URL of the website to visit the homepage.<br>5. Guest click icon “ Giỏ hàng “ on menu<br>6. Guest click icon “X” of product on cart | | |
| **Alternative Flows** | - User login on website<br>- go to step 3 in normal flow | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.2 User

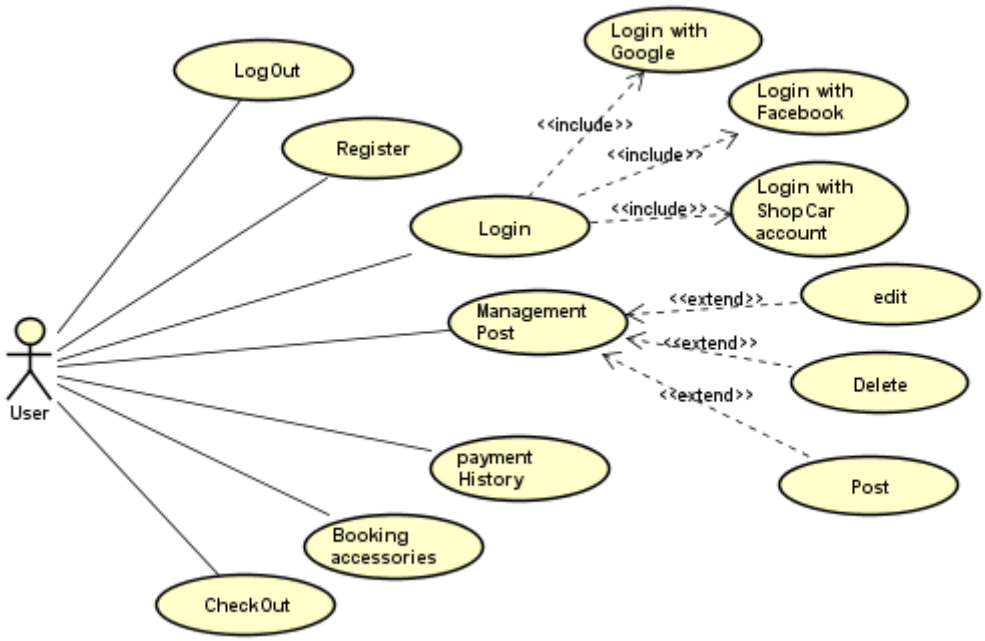



*Figure 3-3: Use Case User Diagram*

#### 3.3.2.1 Login with shopcar account

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-19 | | |
| **Use Case Name** | Login with shopcar account | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Allow User who has LongK website account login | | |
| **Pre-conditions** | - User has a LongK website account | | |
| **Post-conditions** | - User is logged into website successfully<br>- Website displays Home page<br>- Sever saves the new user data into database of this is the first time login by this LongK website account | | |
| **Normal Flow** | 1. From the website header, user clicks "Đăng nhập".<br>2. System loads the Login page.<br>3. User enters login credentials.<br>4. User clicks "Đăng nhập". | | |
| **Alternative Flows** | N/A | | |

| | |
|---|---|
| **Exceptions** | 1: At step 3 of normal flow, user leaves email and/or password blank, then proceeds to step 4.<br>The system displays the error message "Tài khoản và mật khẩu không được để trống". User is not logged in.<br>2: At step 3 of normal flow, user enters invalid login credentials, then proceeds to step 4.<br>The system displays the error message "Sai tài khoản hoặc mật khẩu". User is not logged in. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.2.2 Login with google

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-20 | | |
| **Use Case Name** | Login with shopcar account | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Guest | **Secondary Actors** | Admin,User |
| **Description** | - Allow User who has LongK website account login | | |
| **Pre-conditions** | - User has a LongK website account | | |
| **Post-conditions** | - User is logged into website successfully<br>- Website displays Home page<br>- Sever saves the new user data into database of this is the first time login by this LongK website account | | |
| **Normal Flow** | 1. From the website header, user clicks "Đăng nhập".<br>2. System loads the Login page.<br>3. User enters login credentials.<br>4. User clicks "Đăng nhập". | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | 1: At step 3 of normal flow, user leaves email and/or password blank, then proceeds to step 4.<br>The system displays the error message "Tài khoản và mật khẩu không được để trống". User is not logged in.<br>2: At step 3 of normal flow, user enters invalid login credentials, then proceeds to step 4.<br>The system displays the error message "Sai tài khoản hoặc mật khẩu". | | |

| | |
|---|---|
| | User is not logged in. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.2.3 Login with facebook

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-21 | | |
| **Use Case Name** | Login with Facebook | | |
| **Creator** | ThangTQ | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | - Allow User who has Facebook account login | | |
| **Pre-conditions** | - User has a Facebook account | | |
| **Post-conditions** | - Guest is logged into website successfully<br>- Website displays Home page<br>- Sever saves the new user data into database of this is the first time login by this facebook account | | |
| **Normal Flow** | 1. User opens website and click on login with facebook button in login screen<br>2. Website displays login with facebook screen<br>3. User fills valid email or phone number and password then taps on login button<br>4. Website display home page | | |
| **Alternative Flows** | 1.sever creates new account with that facebook account for use<br>2.go to step 4 in normal flow | | |
| **Exceptions** | website displays error message | | |
| **Priority** | High | | |
| **Frequency of Use** | Medium | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.2.4 Register

| | |
|---|---|
| **Use Case ID** | UC-22 |
| **Use Case Name** | User Register |

| | | | |
|---|---|---|---|
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | - Guest register account to become a user | | |
| **Pre-conditions** | Pre 1.0 : Guest access to login page | | |
| **Post-conditions** | Post 1.0 : Guest register a account sussesfuly<br>Post 2.0 : the website display homepage | | |
| **Normal Flow** | 1.0 Guest access to the Homepage<br>2.0 Guest click on “Đăng nhập ”<br>3.0 Guest clik on “Đăng ký”<br>4.0 User enters required information<br>5.0 User clicks "Đăng ký". | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | **EXC1: At step 4 of normal flow, user leaves a required field blank**<br>The system displays the error message "("Họ”) không được bỏ trống".<br>**EXC2: At step 4 of normal flow, user leaves a required field blank**<br>The system displays the error message "("Tên”) không được bỏ trống<br>**EXC3: At step 4 of normal flow, user leaves a required field blank**<br>The system displays the error message "("Tài khoản”) không được bỏ trống<br>**EXC4: At step 4 of normal flow, user enters a password with 8 or more characters and a number but does not have an uppercase letter.**<br>The system displays the error message "Mật khẩu phải có 1 ký tự chữ in hoa". If user proceeds to step 4, nothing happens.<br>**EXC4: At step 4 of normal flow, user enters different passwords in "Mật khẩu " and "Nhập lại mật khẩu.**<br>The system displays the error message "Mật khẩu không khớp". If user proceeds to step 4, nothing happens.<br>**EXC5: At step 4 of normal flow, user enters an email that already exists in the system.**<br>The system displays the error message "Địa chỉ email đã tồn tại!". If user proceeds to step 4, nothing happens.<br>**EXC6: At step 3 of normal flow, user leaves a required field** | | |

| | |
|---|---|
| | **blank, then proceeds to step 4.**<br>The system displays the error message "("Số điện thoại") không được bỏ trống".<br>**EXC7: At step 3 of normal flow, user leaves a required field blank, then proceeds to step 4.**<br>The system displays the error message "("Địa chỉ") không được bỏ trống".<br><br>. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.2.5 Logout

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-23 | | |
| **Use Case Name** | logOut | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | user login from the website | | |
| **Pre-conditions** | -pre 1.0: user Login sussessful | | |
| **Post-conditions** | Post 1.0 : User logout from the website<br>Post 2.0 : the website display homepage | | |
| **Normal Flow** | 1. From the website, user click on "Đăng xuất" | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |

| | |
|---|---|
| **Other Information** | N/A |

### 3.3.2.6 Booking Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-24 | | |
| **Use Case Name** | Booking accessories | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | User book a accessories item | | |
| **Pre-conditions** | Pre 1.0: User login sussessfull to the system | | |
| **Post-conditions** | Post 1.0 : The system display the information of the item that user has book | | |
| **Normal Flow** | 1. User access to the website<br>2. User login to the system<br>3.User click on “Phụ kiện” page<br>4. User click on “cart” icon | | |
| **Alternative Flows** | 1. Guest click on “phụ kiện” page<br>2. The system display login page<br>3. Guest login to system<br>4. Click on “cart” icon | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.2.7 Post Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-25 | | |
| **Use Case Name** | Post Car | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | User post the car that they want to sell | | |
| **Pre-conditions** | Pre 1. 0 : User login sussessfly to the system | | |

| | |
|---|---|
| **Post-conditions** | Post 1.0 : User post the information of the car<br>Post 2.0 : Display information of that car has posted in website |
| **Normal Flow** | 1. User access the website<br>2. Login to the system<br>3. Click on “Đăng xe”<br>4. User enters required information<br>5. User click on “Đăng” button |
| **Alternative Flows** | N/A |
| **Exceptions** | **EXC1: At step 4 of normal flow, user leaves a required field blank at the dropdown list information.**<br>The system displays the error message need choose information from dropdown list".<br>**EXC2 : At step 4 of normal flow, user not uploat at least a car image**<br>The system displays the error message : need upload at least car image |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.2.8 Update Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-26 | | |
| **Use Case Name** | Update car | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | User | **Secondary Actors** | Admin |
| **Description** | User update information about the their post | | |
| **Pre-conditions** | -pre 1.0: User has at least a post before | | |
| **Post-conditions** | Post 1.0 : The screen display update successful<br>Post 2.0 : The information of the car posted has change | | |
| **Normal Flow** | 1. User assess the website<br>2. User login susscessful<br>3. User click on “Đăng xe” | | |

| | |
|---|---|
| | 4. The website display post information<br>5. User click on “Chỉnh sửa” button<br>6. User click on “Lưu” button |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.3 Admin

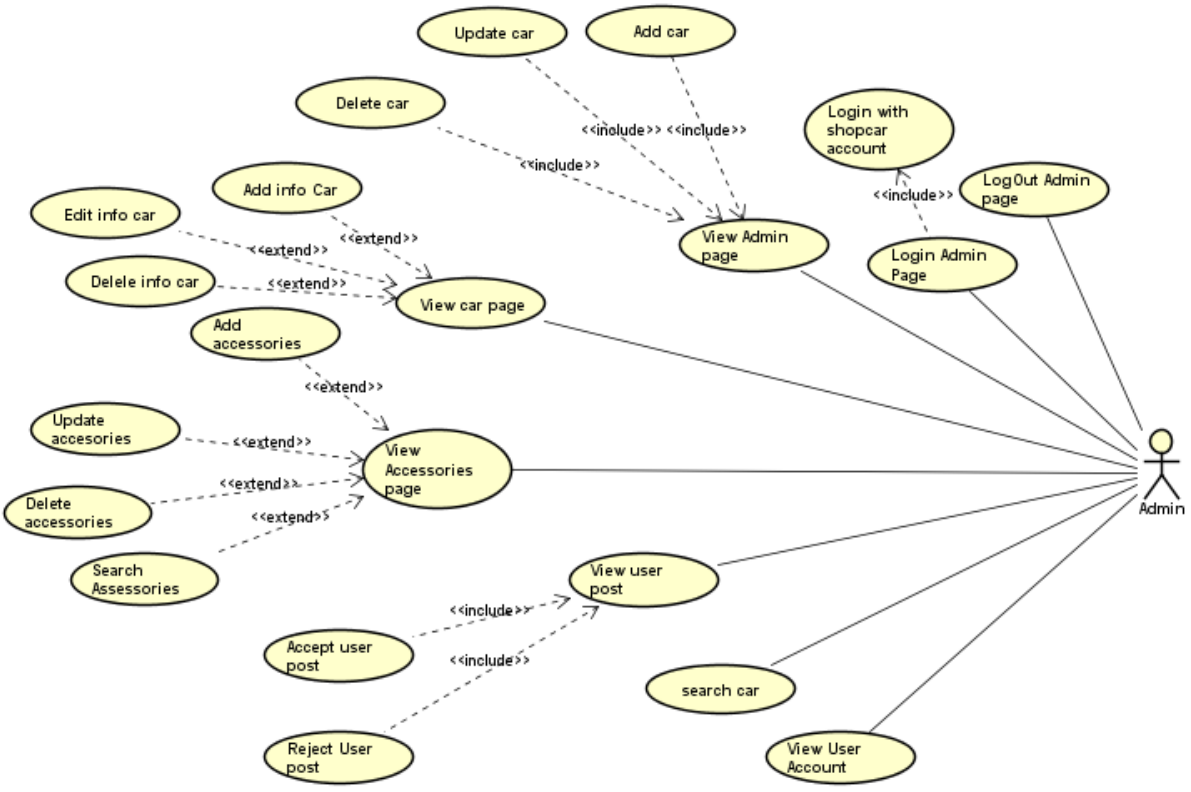



*Figure 3-4: Use Case Admin Diagram*

#### 3.3.3.1 Login admin page

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-27 | | |
| **Use Case Name** | Login Admin page | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |

| | | | |
|---|---|---|---|
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Admin | **Secondary Actors** | |
| **Description** | User access to the system with administrator privileges | | |
| **Pre-conditions** | Pre 27.0: User has an admin account<br>Pre 27.1 : User access to the website for admin | | |
| **Post-conditions** | Post 27.0 : logged the website<br>Post 27.1 : Screen display admin page | | |
| **Normal Flow** | 1. User access to the admin website by URL<br>2. Click on “Đăng nhập” button<br>3. Enters required information<br>4. Click on “Đăng nhập” | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | **EXC1: At step 3 of normal flow,leave a require filed UserName or filed Password**<br>The system displays the error message :"Tài khoản hoặc mật khẩu sai”<br>**EXC2: At step 3 of normal flow, user enter incorrect userName or Password**<br>The system displays the error message : “Tài khoản hoặc mật khẩu sai” | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.2 Logout admin page

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-28 | | |
| **Use Case Name** | Logout Admin Page | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Admin | **Secondary Actors** | |
| **Description** | user logout from the admin website | | |
| **Pre-conditions** | -PRE 28.0: User has login susscessful admin website | | |

| | |
|---|---|
| **Post-conditions** | Post 28.0 : The screen dispplay Login page |
| **Normal Flow** | 1. From the admin website, user click on “Đăng xuất” |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.3.3 View admin page

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-29 | | |
| **Use Case Name** | View admin page | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | Admin | **Secondary Actors** | |
| **Description** | User as a admin can view admin website | | |
| **Pre-conditions** | -pre 29.0: user Login sussessful | | |
| **Post-conditions** | Post 29.0 : The screen display Admin home page in Admin website | | |
| **Normal Flow** | 1. User assess the admin website by URL<br>2. Login to the system | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.4 View Accessories page

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-30 | | |
| **Use Case Name** | View accessories admin page | | |
| **Creator** | QuangNH | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |

| | | | |
|---|---|---|---|
| **Primary Actor** | Admin | **Secondary Actors** | |
| **Description** | User as a admin can view accessories in admin page | | |
| **Pre-conditions** | -pre 30.0: user Login sussessful | | |
| **Post-conditions** | Post 30.0 : The screen display accessories in admin website | | |
| **Normal Flow** | 1. User assess the admin website by URL<br>2. Login to the system<br>3. Click on “Phụ kiện ” button<br>4. The screen display “phụ kiện” page in admin website | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.5 View User account

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-31 | | |
| **Use Case Name** | View user account | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all user account of system | | |
| **Pre-conditions** | - PRE-31.0 Log in with admin account | | |
| **Post-conditions** | - POST-31.0 List all account of user | | |
| **Normal Flow** | 4. Admin open admin page.<br>5. Admin log in with admin account then click log in button<br>6. Admin page display all account of of user | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | - Alert wrong username or password | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.6 Add Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-32 | | |
| **Use Case Name** | Add Car | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - Add Car to LongK System<br>- Add a car object to database | | |
| **Pre-conditions** | - PRE-32.0 Log in successfully with admin account | | |
| **Post-conditions** | - POST-32.0 Show a template add car<br>- POST-32.1 Upload a car to LongK System | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin click " thêm xe" button<br>4. Admin write data into template<br>5. Click " save " button to accept add car to LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | - E-32.0 Missing upload image<br>- E-32.1 Error add car to database | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.7 Search Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-33 | | |
| **Use Case Name** | Search Car | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - Search a car in LongK System<br>- Display item in " Search" button | | |
| **Pre-conditions** | - PRE-33.0 Log in with admin account<br>- PRE-33.1 Insert text to text box Search | | |
| **Post-conditions** | - POST-33.0 Show car want to search | | |

| | |
|---|---|
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Click " xe " link and wirte text to search text box<br>4. Click " TÌm kiếm" button<br>5. Display all car same of keyword in search text box |
| **Alternative Flows** | N/A |
| **Exceptions** | - E-33.0 Show all car if text box is empty |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.3.8 View user post

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-34 | | |
| **Use Case Name** | View user post | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all user post | | |
| **Pre-conditions** | - PRE-34.0 Log in successfully with admin account | | |
| **Post-conditions** | - POST-34.0 List all user post | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all user account | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | - E-34.0 Error query data off user in DB | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.9 Up post of user

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-35 | | |
| **Use Case Name** | Up post of user | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |

| | | | |
|---|---|---|---|
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Accept post to up in LongK Webstie | | |
| **Pre-conditions** | - PRE-35.0 Admin must log in with admin account<br>- PRE-35.1 In system user has already up a post | | |
| **Post-conditions** | - POST-35.0 List all post of user | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all post of user<br>4. View detail post<br>5. Confirm then click " Duyệt " button<br>6. The post are pushed to LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.10 Check order status

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-36 | | |
| **Use Case Name** | Check order status | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Accept post to up in LongK Webstie | | |
| **Pre-conditions** | - PRE-36.0 Log in with admin account<br>- PRE-36.1 User up a post | | |
| **Post-conditions** | - POST-36.0 List all post of user | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all post of user<br>4. View detail post<br>5. Confirm then click " Duyệt " button | | |

| | |
|---|---|
| | 6. The post are pushed to LongK System |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.3.11 Accept user post

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-37 | | |
| **Use Case Name** | Accept user post | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Accept post to up in LongK Webstie | | |
| **Pre-conditions** | - PRE-37.0 Log in with admin account<br>- PRE-37.1 User up a post | | |
| **Post-conditions** | - POST-37.0 List all post of user | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all post of user<br>4. View detail post<br>5. Confirm then click " Duyệt " button<br>6. The post are pushed to LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.3.12 Reject user post

| | |
|---|---|
| **Use Case ID** | UC-38 |
| **Use Case Name** | Up post of user |

| | | | |
|---|---|---|---|
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Accept post to up in LongK Webstie | | |
| **Pre-conditions** | - PRE-38.0 Admin must log in with admin account<br>- PRE-38.1 In system user has already up a post | | |
| **Post-conditions** | - POST-38.0 List all post of user | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all post of user<br>4. View detail post<br>5. Confirm then click " Hủy " button<br>6. The post has remove from LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | E-38.0 At the step 4 when " Hủy" button this post will not delete in LongK System | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

#### 3.3.3.13 Accept order

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-39 | | |
| **Use Case Name** | Accept user post | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Accept post to up in LongK Webstie | | |
| **Pre-conditions** | - PRE-37.0 Admin must log in with admin account<br>- PRE-37.1 In system user has already up a post | | |
| **Post-conditions** | - POST-37.0 List all post of user | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all post of user | | |

| | |
|---|---|
| | 4. View detail post<br>5. Confirm then click " Duyệt " button<br>6. The post are pushed to LongK System |
| **Alternative Flows** | N/A |
| **Exceptions** | N/A |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.3.14 Update Car

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-40 | | |
| **Use Case Name** | Up post of user | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Update information of car in LongK System | | |
| **Pre-conditions** | - PRE-40.0 Admin must log in with admin account<br>- PRE-40.1 Admin is performing *View detail car* use case | | |
| **Post-conditions** | - POST-40.0 View Detail Car<br>- POST-40.1 System show detail car in admin page | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all car in LongK System<br>4. Click "Detail" icon<br>5. View detail of each car<br>6. Confirm then click " Lưu " button<br>7. The post has updated from LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | N/A | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.15 DeleteCar

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-41 | | |
| **Use Case Name** | Delete Car | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all post of user<br>- Delete information of car in LongK System | | |
| **Pre-conditions** | - PRE-41.0 Admin must log in with admin account<br>- PRE-41.1 Admin is performing *View detail car* use case | | |
| **Post-conditions** | - POST-41.0 View Detail Car<br>- POST-41.1 System delete Car select | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin page display all car in LongK System<br>4. Click “Delete” icon<br>5. System show log “ bạn có muốn xóa không”<br>6. Confirm then click “ OK “ button<br>7. The car has delete from LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | E-41.0 At the step 4 when “ Cancel” button this post will delete in LongK System | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.16 Add Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-42 | | |
| **Use Case Name** | Add Car | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - Add Accessories to LongK System<br>- Add a accessories object to database | | |

| | |
|---|---|
| **Pre-conditions** | - PRE-42.0 Log in successfully with admin account |
| **Post-conditions** | - POST-42.0 Show a template add accessories<br>- POST-42.1 Upload accessories to LongK System |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Admin click " thêm phụ kiện" button<br>4. Admin write data into template<br>5. Click " save " button to accept add accessories to LongK System |
| **Alternative Flows** | N/A |
| **Exceptions** | - E-42.0 Missing upload image<br>- E-42.1 Error add accessories to database |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

#### 3.3.3.17 Update Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-43 | | |
| **Use Case Name** | Add Car | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - Update Accessories to LongK System<br>- Update a accessories object to database | | |
| **Pre-conditions** | - PRE-43.0 Log in successfully with admin account<br>- PRE-43.1 View detail of accessories | | |
| **Post-conditions** | - POST-43.0 View Detail Car<br>- POST-43.1 System show detail car in admin page | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Click " Phụ kiện " link<br>4. Admin page display all accessories in LongK System<br>5. Click "Detail" icon<br>6. View detail of each accessories | | |

| | |
|---|---|
| | 7. Confirm then click " Lưu " button<br>8. This accessories has updated from LongK System |
| **Alternative Flows** | N/A |
| **Exceptions** | - E-43.0 Error update accessories to database |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | N/A |
| **Other Information** | N/A |

### 3.3.3.18 Delete Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-44 | | |
| **Use Case Name** | Delete Accessories | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - View all accessories of user<br>- Update information of accessories in LongK System | | |
| **Pre-conditions** | - PRE-44.0 Admin must log in with admin account<br>- PRE-44.1 Admin is performing *View detail car* use case | | |
| **Post-conditions** | - POST-44.0 View Detail Accessories<br>- POST-44.1 System delete Accessories select | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Click " Phụ kiện " link<br>4. Admin page display all accessories in LongK System<br>5. Click "Delete" icon<br>6. View detail of each accessories<br>7. Confirm then click " Có " button<br>8. This accessories has delete from LongK System | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | E-44.0 At the step 4 when " Cancel" button this accessories will delete in LongK System | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

### 3.3.3.19 Search Accessories

| | | | |
|---|---|---|---|
| **Use Case ID** | UC-45 | | |
| **Use Case Name** | 1. Search Accessories | | |
| **Creator** | AnhVN | **Date Created** | 2018/06/18 |
| **Version** | 1.0 | **Last Updated** | |
| **Primary Actor** | | **Secondary Actors** | Admin |
| **Description** | - Search a car in LongK System<br>- Display item in " Search" button | | |
| **Pre-conditions** | - PRE-45.0 Log in with admin account<br>- PRE-45.1 Insert text to text box Search | | |
| **Post-conditions** | - POST-45.0 Show accessories want to search | | |
| **Normal Flow** | 1. Admin open admin page.<br>2. Admin log in with admin account then click log in button<br>3. Click " phụ " link and wirte text to search text box<br>4. Click " TÌm kiếm" button<br>5. Display all accessories same of keyword in search text box | | |
| **Alternative Flows** | N/A | | |
| **Exceptions** | - E-45.0 Show all accessories if text box is empty | | |
| **Priority** | High | | |
| **Frequency of Use** | High | | |
| **Business Rules** | N/A | | |
| **Other Information** | N/A | | |

# 3.4 Entity relationship diagram

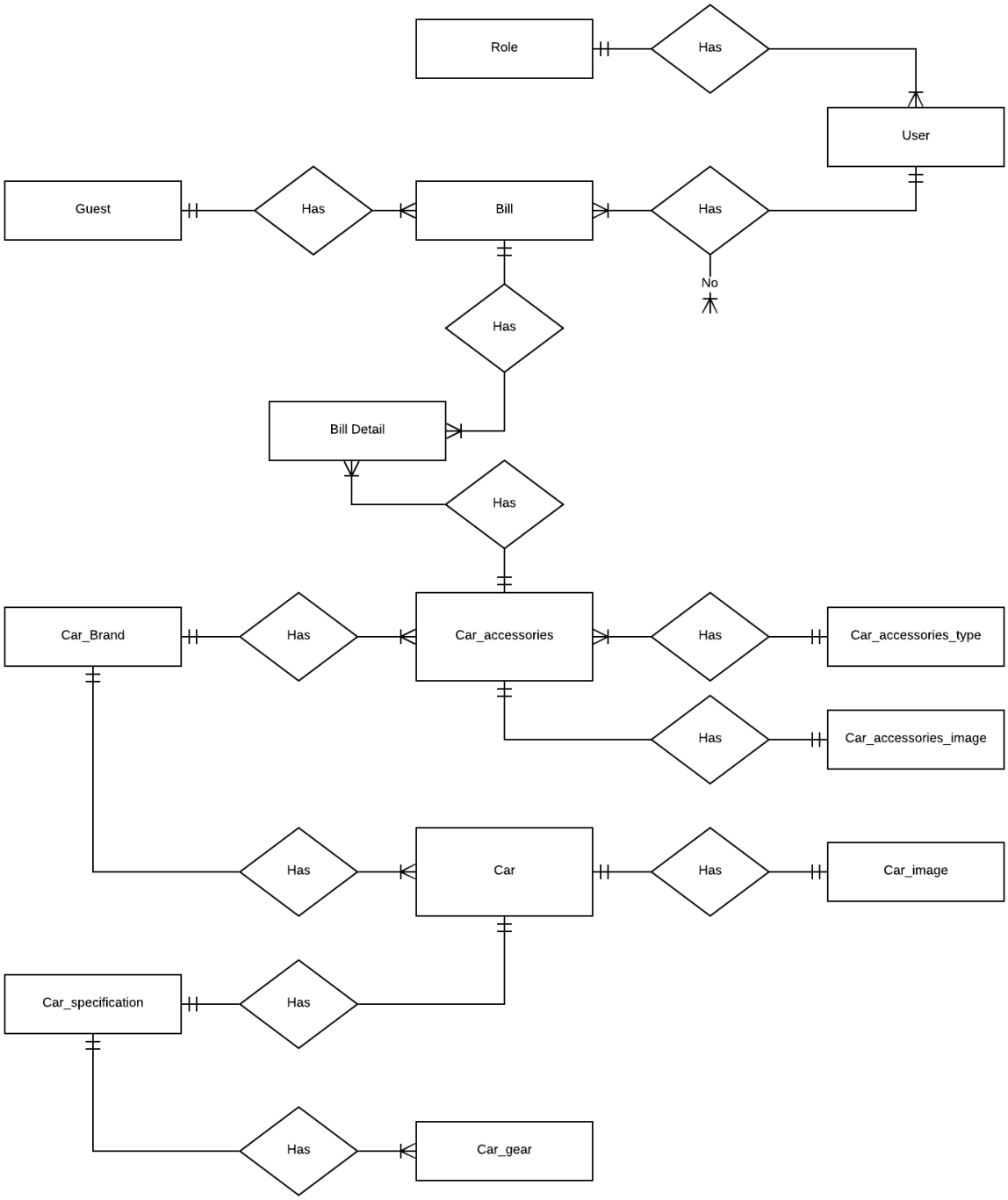



*Figure 3-5: Entity relationship diagram*

# Chapter 4 : Software Design

## 4.1 Purpose

This chapter provides an overview of software design, helping developers understand the system architecture, database design, common design and detail design.

## 4.2 Design of LongK System

### 4.2.1 System Architecture

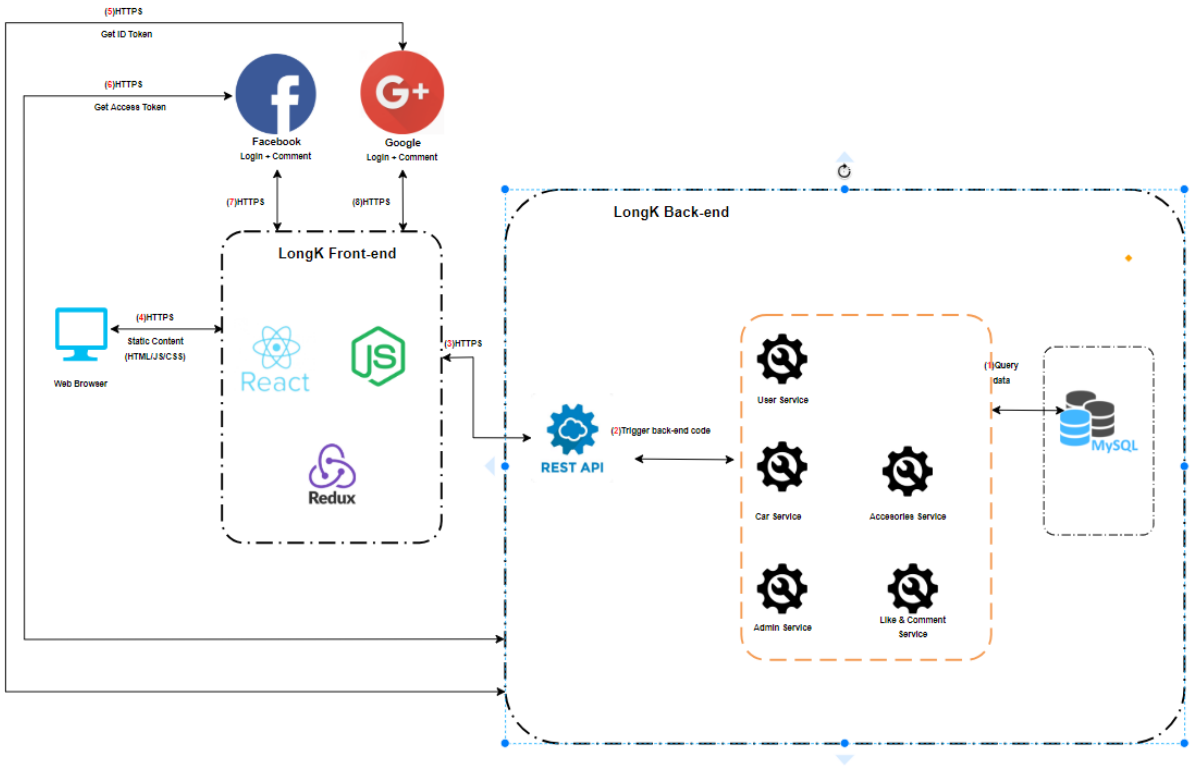



*Figure 4-1: System Architecture*

### 4.2.2 System architecture Explanation

#### 4.2.2.1 React

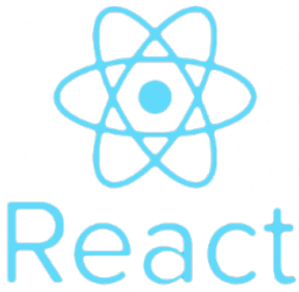


**Figure 4-2: React**

React is an open-source JavaScript library for building user interfaces from declarative components. It is used in combination with other Javascript libraries or framworks to builds web or mobile applications**. We use it to build entire view layer fpr Single Page Application in LongK Front-end.**

#### 4.2.2.2 Redux




**Figure 4-3: Redux**

Redux is an open-source JavaScript librabry designed for managing application state. Redux introduces new front-end architecture based on Flux. Redux architecture revolves around a strict unidirectional data flow, which makes logic of the front-end app more predictable and easier to understand**. Redux is used together with React to build user interfaces.**

#### 4.2.2.3 NodeJs




**Figure 4-4: NodeJs**

**NodeJs** is an open-source, cross-platform JavaScript run-time environment that executes JavaScript code server-side. It uses an event-driven, non blocking I/O model that makes it lightweight and efficient**. We use it to develop API services in LongK System.**

#### 4.2.2.4 MySQL




**Figure 4-5: MySQL**

**MySQL** is an open-source relational database management system (RDBMS). **We use it to manage LongK operational data.**

#### 4.2.2.5 RestAPI




**Figure 4-6: Rest API**

**A RESTful API** is an application program interface (API) that uses HTTP requests to GET, PUT, POST and DELETE data. We use it to development a API to LongK System.

### 4.2.3 Architecture Layers Design

### 4.2.4 Database Design

### 4.2.5 Database Diagram

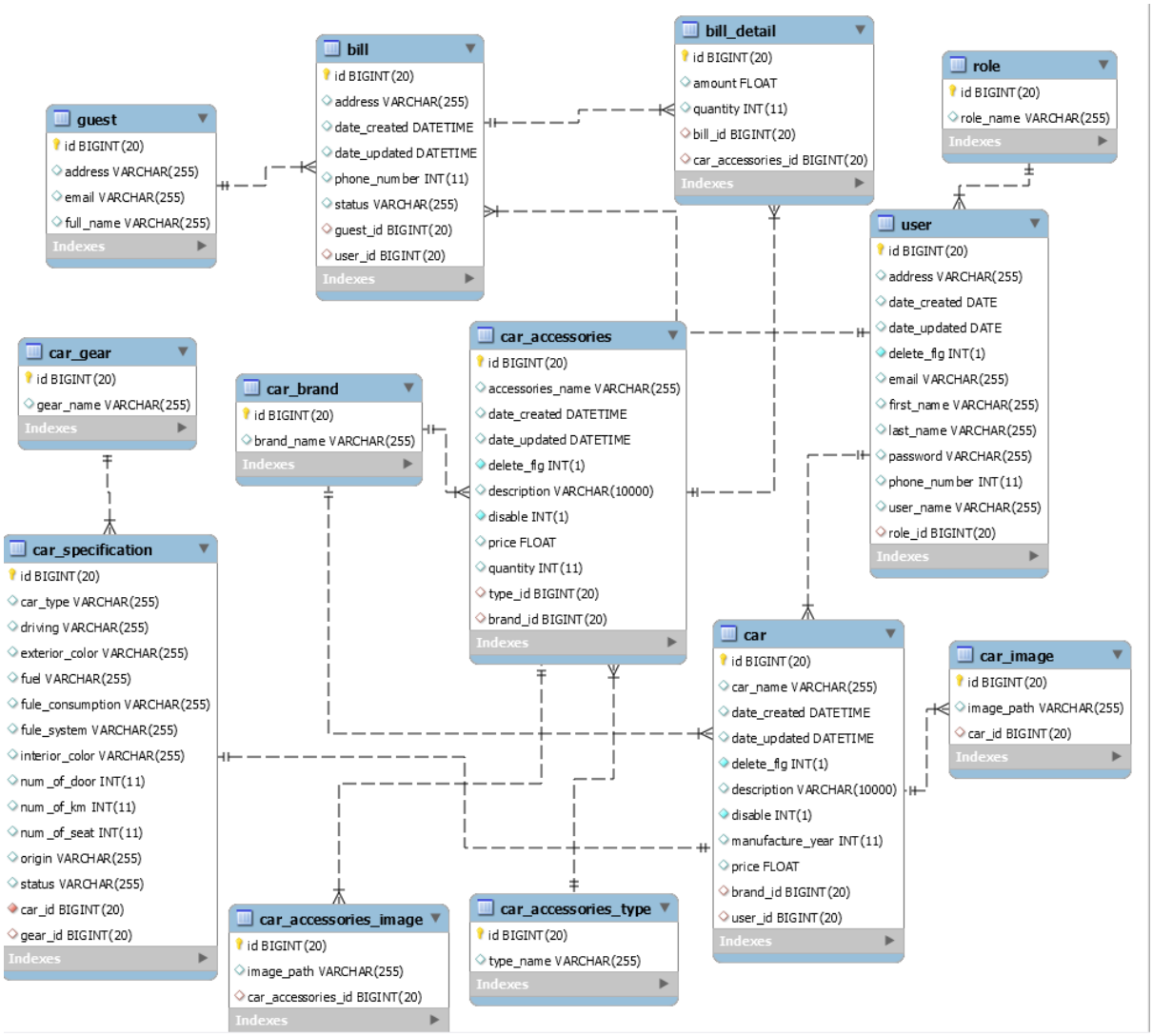



*Figure 4-2: Database Diagram*

### 4.2.6 Data Dictionary

| **user** | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| first_name | VARCHAR(255) | First name of user |
| last_name | VARCHAR(255) | Last name of user |
| email | VARCHAR(255) | Email of user |

| | | |
|---|---|---|
| phone_number | INT(11) | Phone number of user |
| delete_flg | INT(1) | |
| address | VARCHAR(255) | Address of user |
| user_name | VARCHAR(255) | User name |
| password | VARCHAR(255) | Pass word |
| date_created | DATE | |
| date_updated | DATE | |
| role_id | BIGINT(20) | |
| | | |

| **role** | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| role_name | VARCHAR(30) | |

| **Bill_detail** | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| quantity | INT(11) | Quantity of product |
| amount | FLOAT | |
| bill_id | BIGINT(20) | |
| car_accessories_id | BIGINT(20) | |
| | | |
| | | |
| | | |
| | | |

| bill | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| date_created | DATETIME | |
| date_updated | DATETIME | |
| phone_number | INT(11) | Phone number of user |
| status | VARCHAR(30) | Status of order |
| guest_id | BIGINT(20) | id of guest |
| user_id | BIGINT(20) | id of user |

| guest | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| address | VARCHAR(255) | Address of guest |
| full_name | VARCHAR(255) | Full name of guest |
| email | VARCHAR(255) | Email of guest |

| car_accessories | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| description | VARCHAR(10000) | Description about each accessories |
| accessories_name | VARCHAR(255) | Name of accessories |
| price | FLOAT | Price of accessories |
| quantity | INT(11) | Quantity of accessories |
| brand_id | BIGINT(20) | |
| type_id | BIGINT(20) | |
| Delete_flg | INT(1) | |
| date_created | DATETIME | date_created of accessories |
| date_updated | DATETIME | date_updated of accesories |

*Table 4-1: Data Dictionary*

| car_accessories_type | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| type_name | VARCHAR(255) | type of accessories |

| car_accessories_image | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| image_path | VARCHAR(255) | |
| car_accessories_id | BIGINT(20) | Identifier |

| car_brand | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| brand_name | VARCHAR(100) | Brand of car |

| car | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| car_name | VARCHAR(255) | |
| description | VARCHAR(10000) | description about each car |
| manufacture_year | INT(4) | manufacture year of car |
| price | FLOAT | price of car |
| brand_id | BIGINT(20) | Identifier |
| Delete_flg | INT(1) | |

| | | |
|---|---|---|
| disable | INT(1) | |
| date_created | DATETIME | |
| date_updated | DATETIME | |
| user_id | BIGINT(20) | |

| **car_specification** | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| | | |
| car_type | VARCHAR(255) | |
| driving | VARCHAR(255) | |
| exterior_color | VARCHAR(255) | |
| fule_system | VARCHAR(255) | |
| fule_consumption | VARCHAR(255) | |
| fule_system | VARCHAR(255) | |
| interior_color | VARCHAR(255) | |
| number_of_door | INT(11) | |
| num_of_km | INT(11) | |
| number_of_seat | INT(11) | |
| origin | VARCHAR(255) | |
| status | VARCHAR(255) | |
| car_id | BIGINT(20) | |
| gear_id | BIGINT(20) | |

| **car_gear** | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |

| | | |
|---|---|---|
| gear_name | VARCHAR(100) | |

| car_image | | |
|---|---|---|
| **Field Name** | **Data type** | **Description** |
| id | BIGINT(20) | Identifier |
| image_path | VARCHAR(255) | |
| car_id | BIGINT(20) | |

*Table 4-1: Data Dictionary*

# 4.3 Detail Design

## 4.3.1 View HomePage

Screen Design

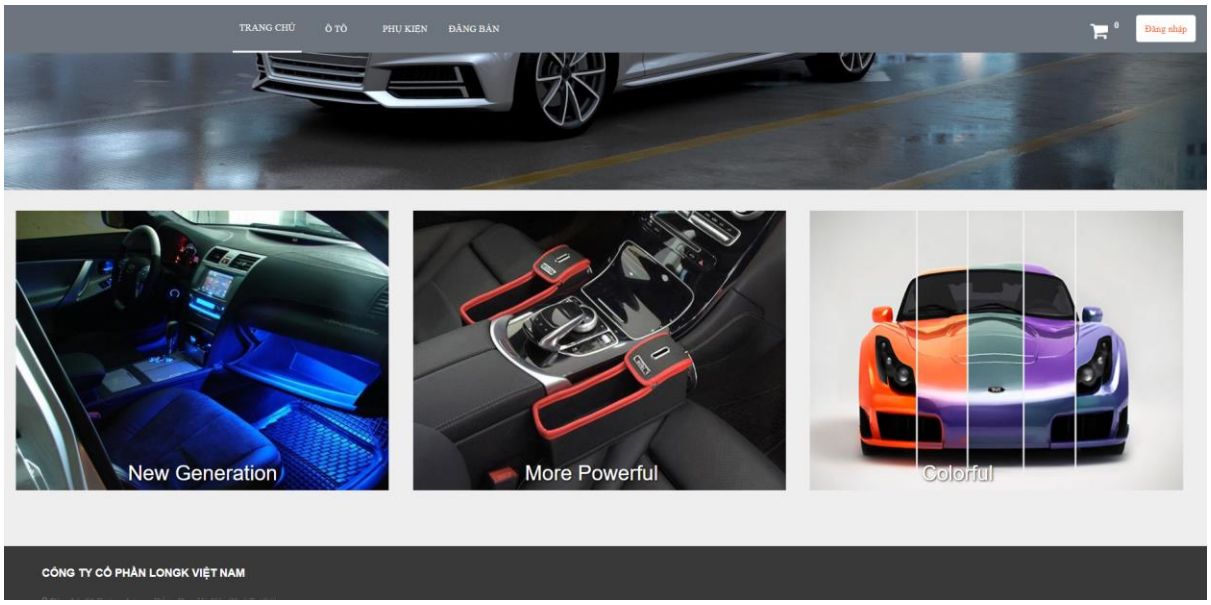



Figure 4-3: View HomePage

Class Diagram

Class Specification

| View Homepagae | |
|---|---|
| Physical address | **/car** |
| Base class | |

| *Attributes* | | | |
|---|---|---|---|
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | Void | | |
| Parameters | Name | Type | Description |
| | | | |

Sequence Diagram

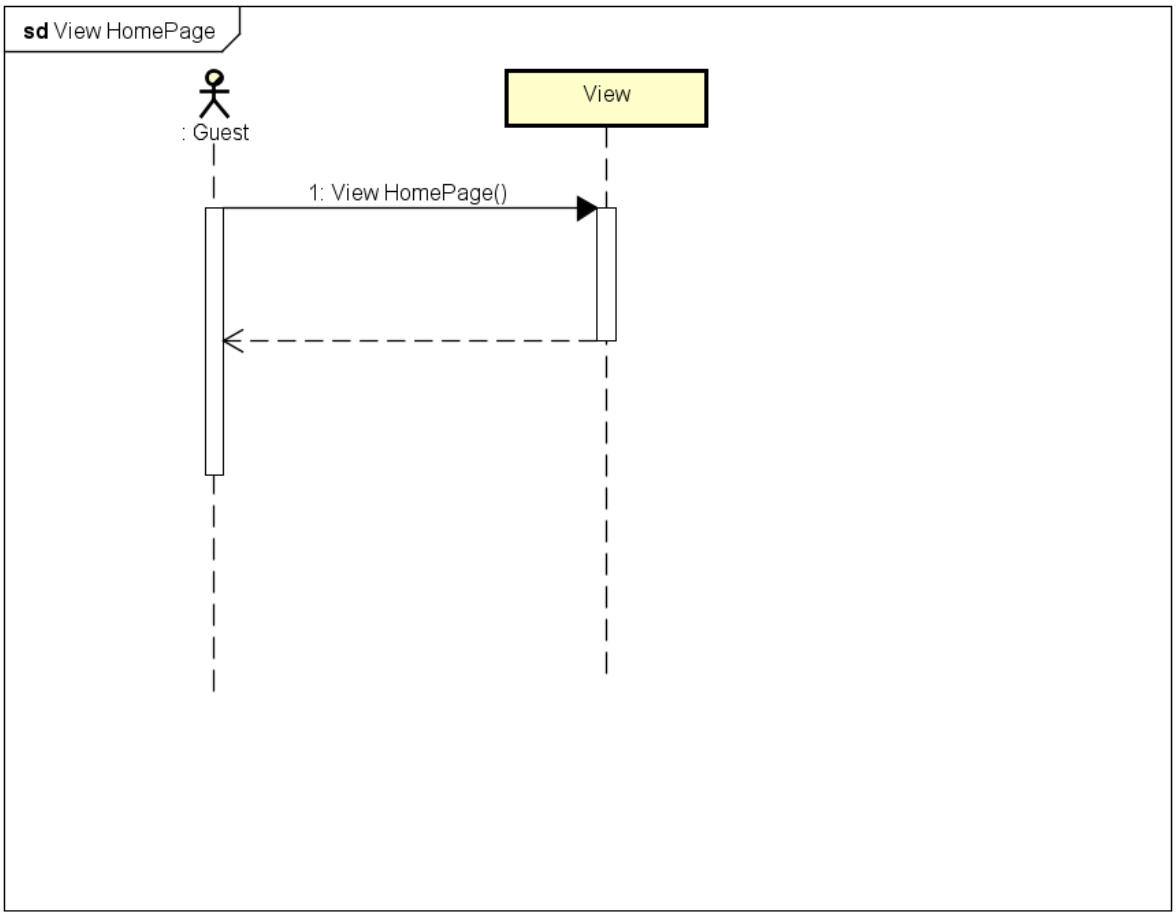



Figure 4-5: View HomePage Sequence

### 4.3.2 Filter by brand

Screen Design

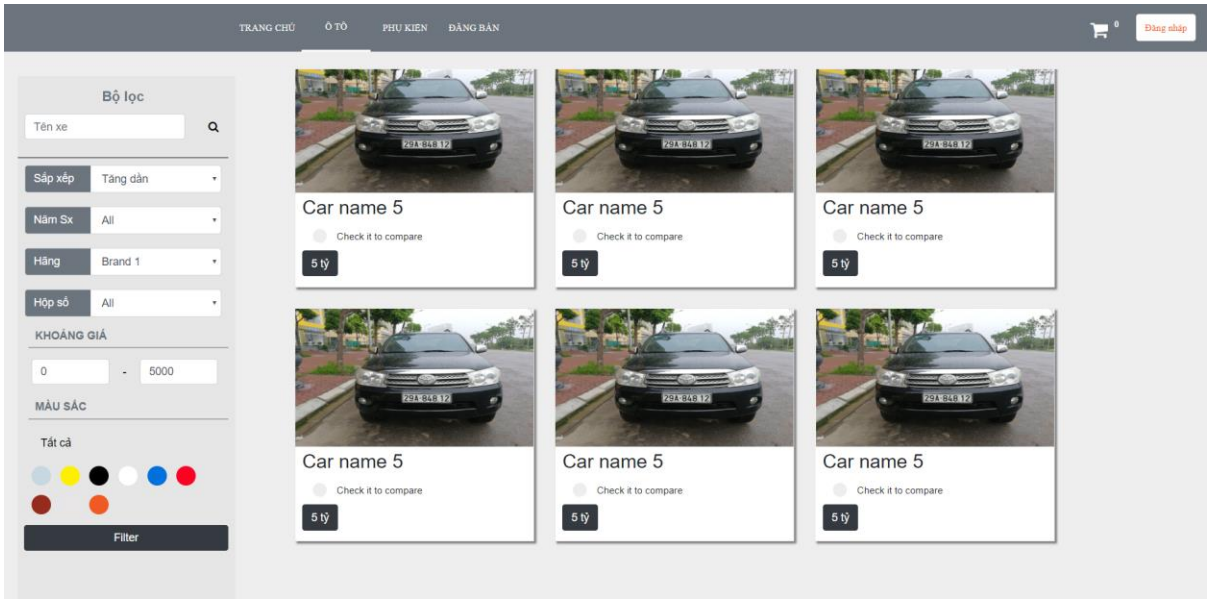



*Figure 4-6: Filter by brand*

## Class Diagram

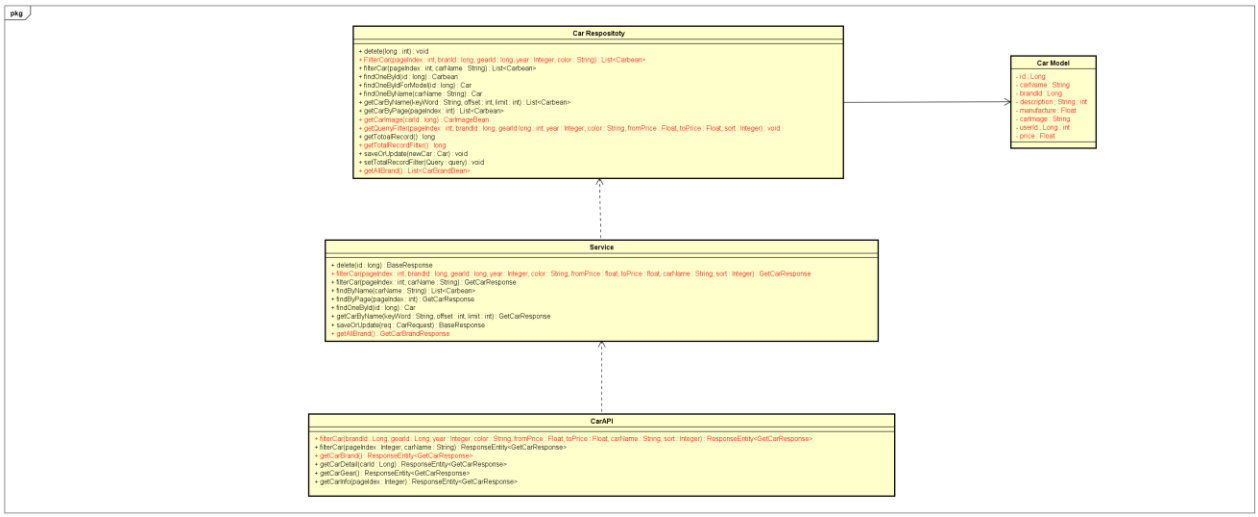



*Figure 4-7: Class Diagram of filter by brand*

## Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | Id | Long | |
| 2 | carName | String | |

| | | | |
|---|---|---|---|
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

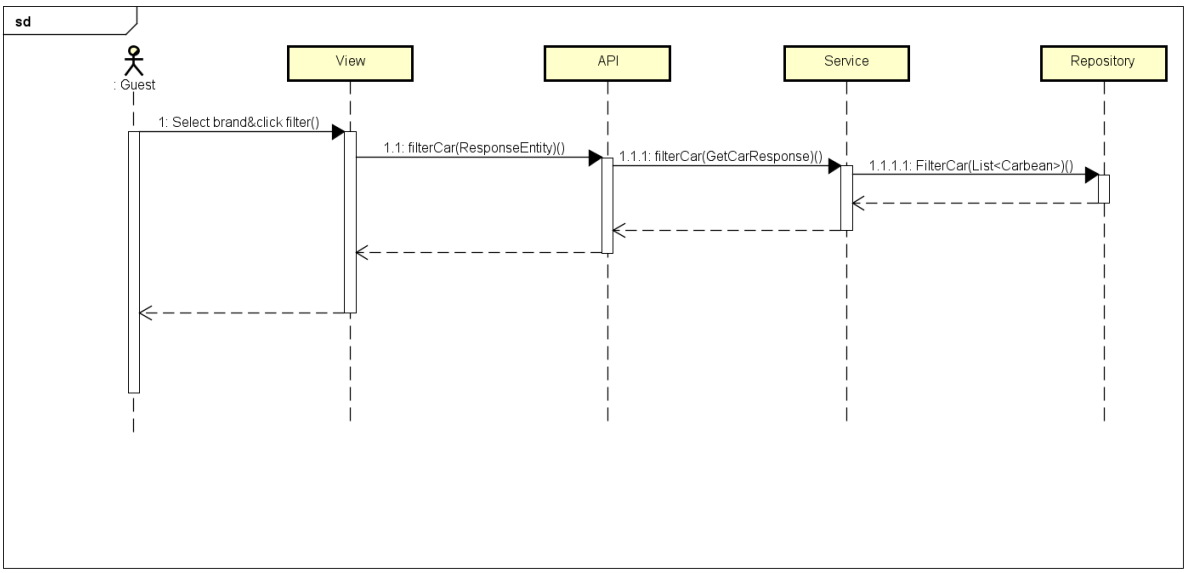


*Figure 4-8: Filter by brand Sequence*

### 4.3.3 Filter by color

Screen Design

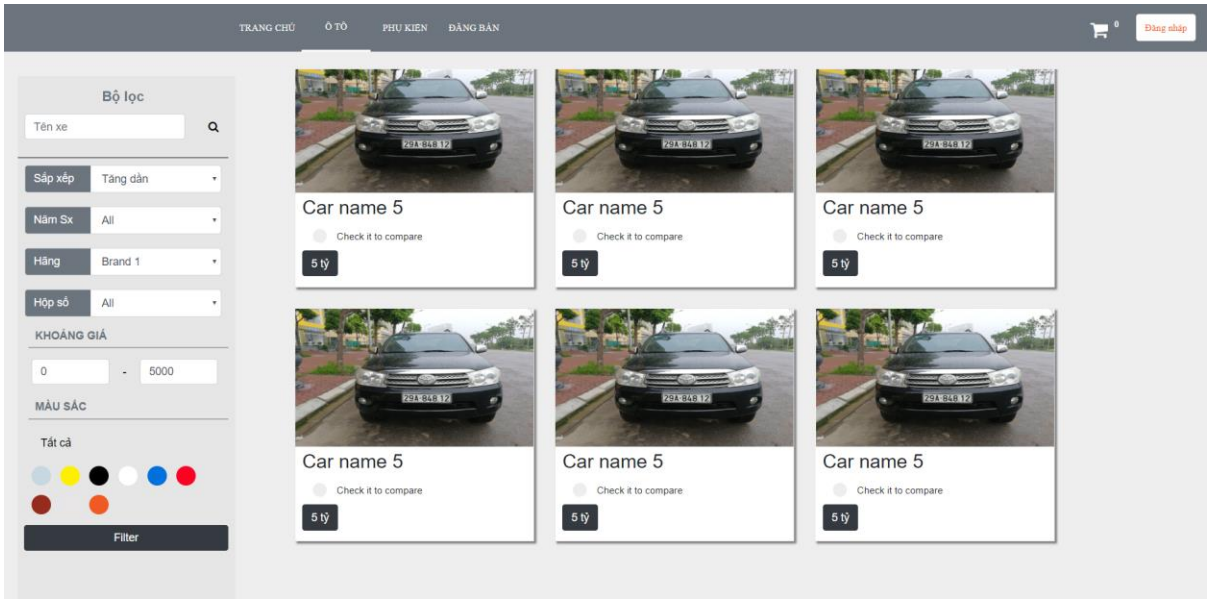



*Figure 4-9: Filter by Color page*

ClassDiagram

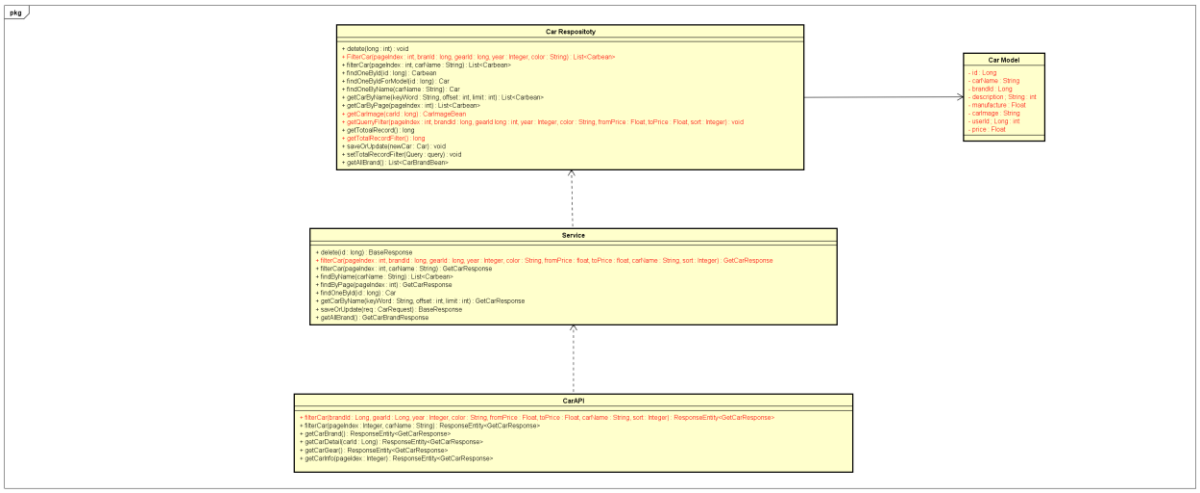



*Figure 4-10: Filter by Color classdiagram*

Class Specification

| Car Service | |
|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** |
| Base class | |
| *Attributes* | |

| No | Name | Type | Description |
|---|---|---|---|
| 1 | Id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

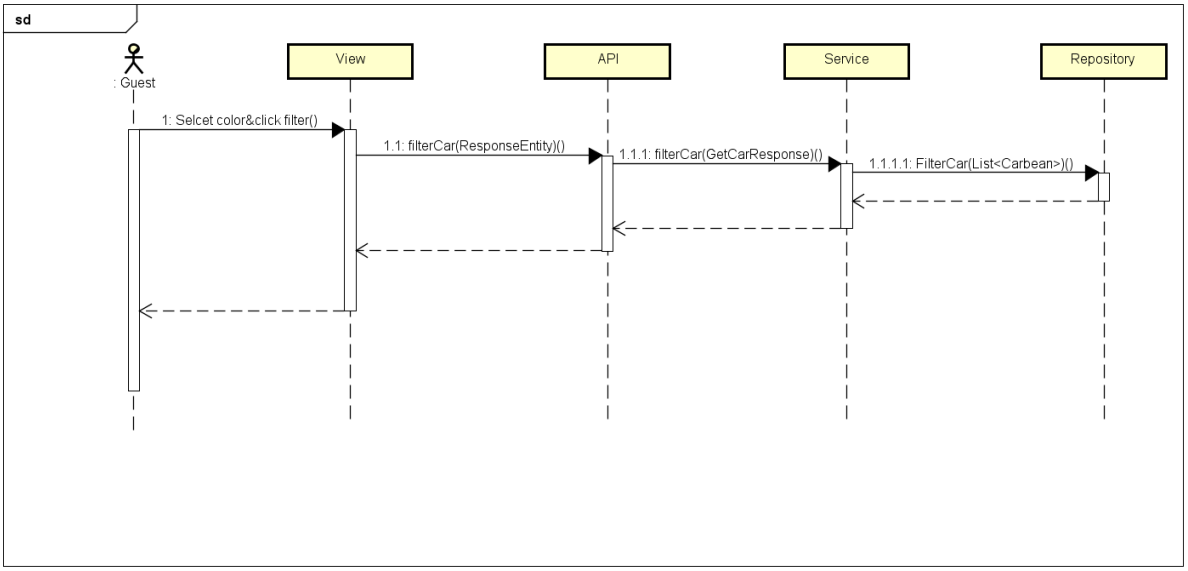



*Figure 4-11:Filter by Color Sequence diagram*

### 4.3.4 Filter by gear

Screen Design

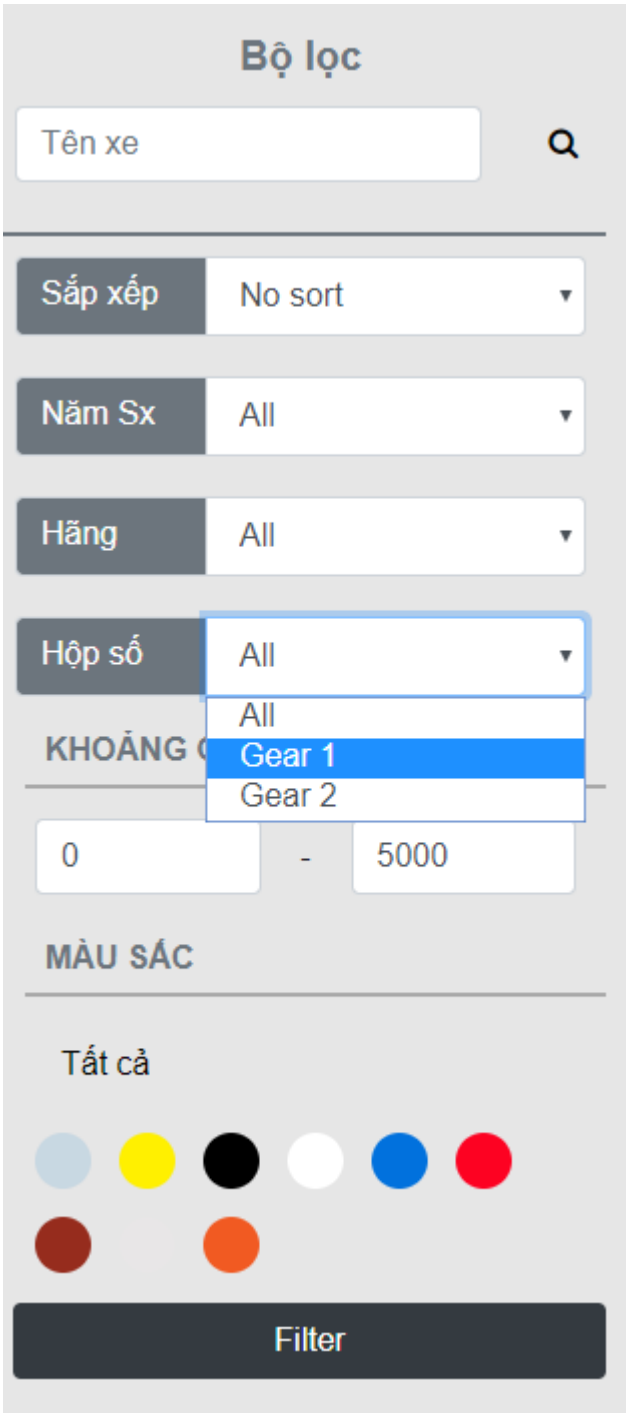



*Figure 4-12:Filter by Gear*

Class Diagram

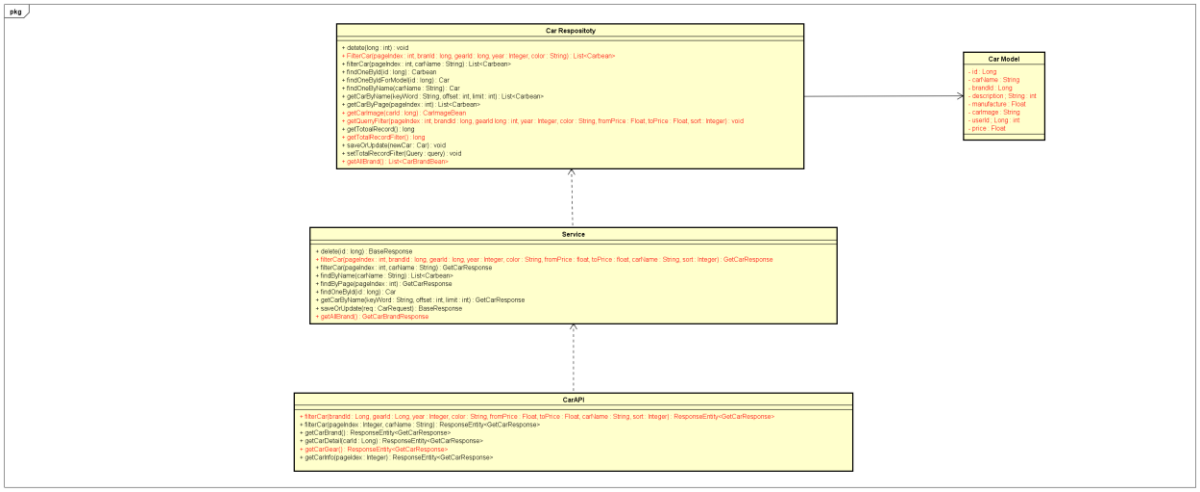



*Figure 4-13:Filter by gear class diagram*

Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | Id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |

| | | | |
|---|---|---|---|
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

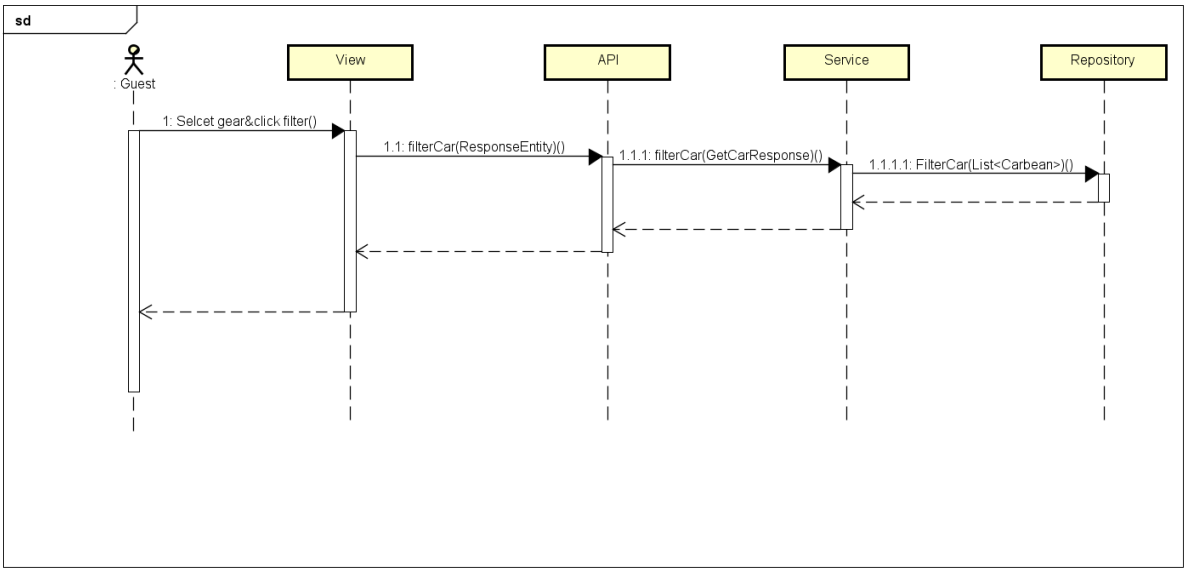



*Figure 4-14:Filter by gear sequence diagram*

### 4.3.5 Filter by price

Screen Design

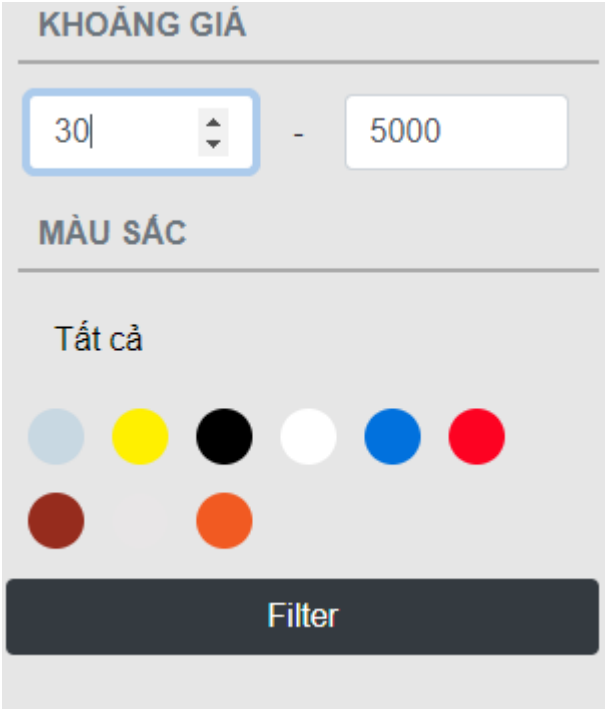



*Figure 4-15:Filter by price*

Class Diagram

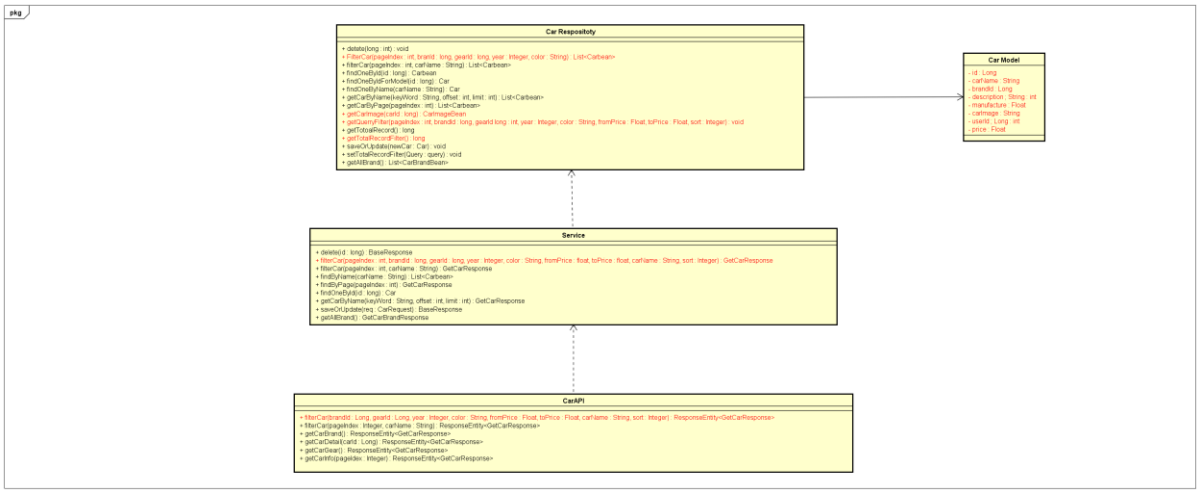



*Figure 4-16:Filter by price class diagram*

Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |

| | | | |
|---|---|---|---|
| 1 | Id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

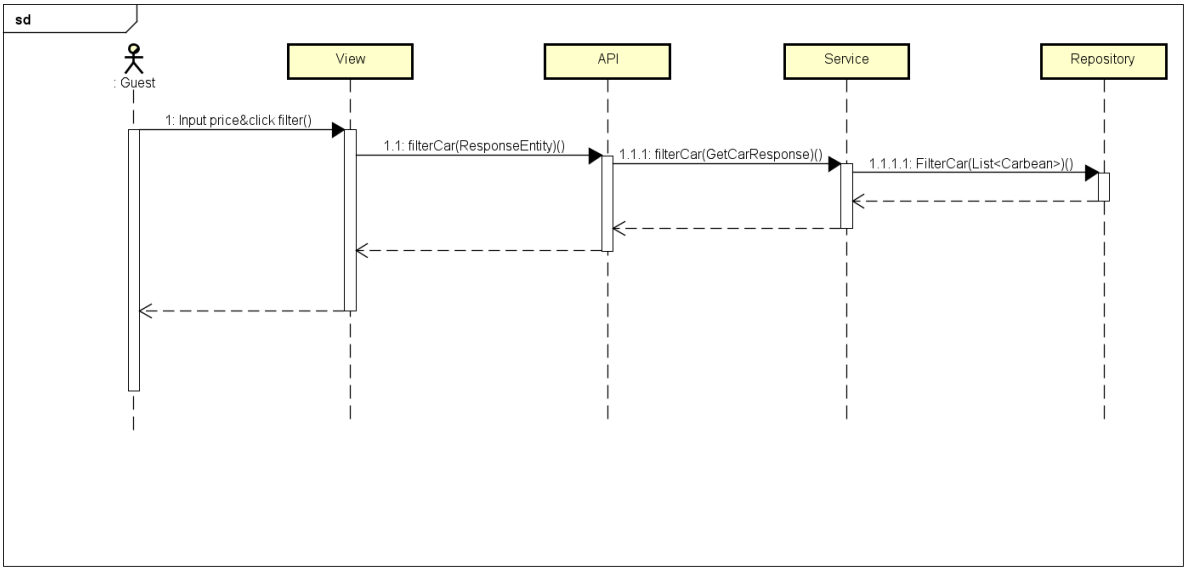



*Figure 4-17:Filter by price sequence diagram*

### 4.3.6 Filter by sort price

Screen Design

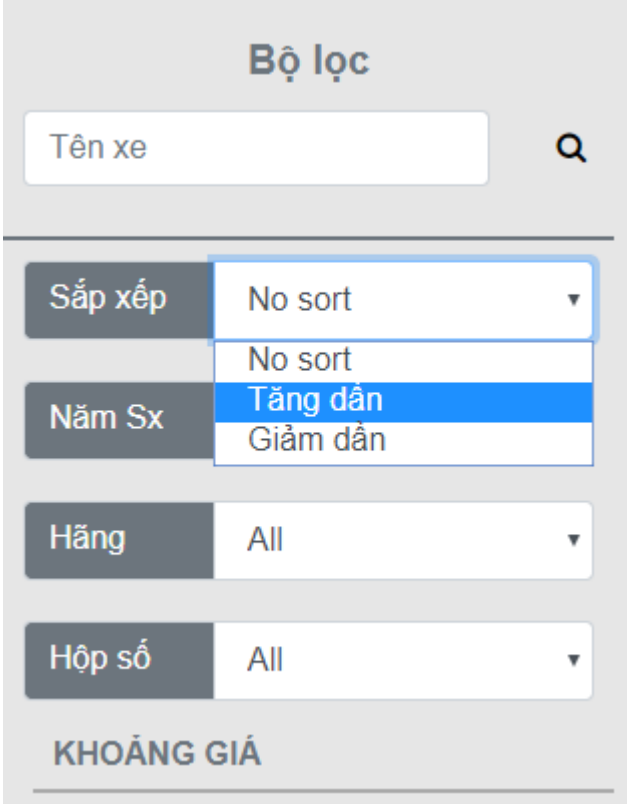



*Figure 4-18:Filter by sort price*

Class Diagram

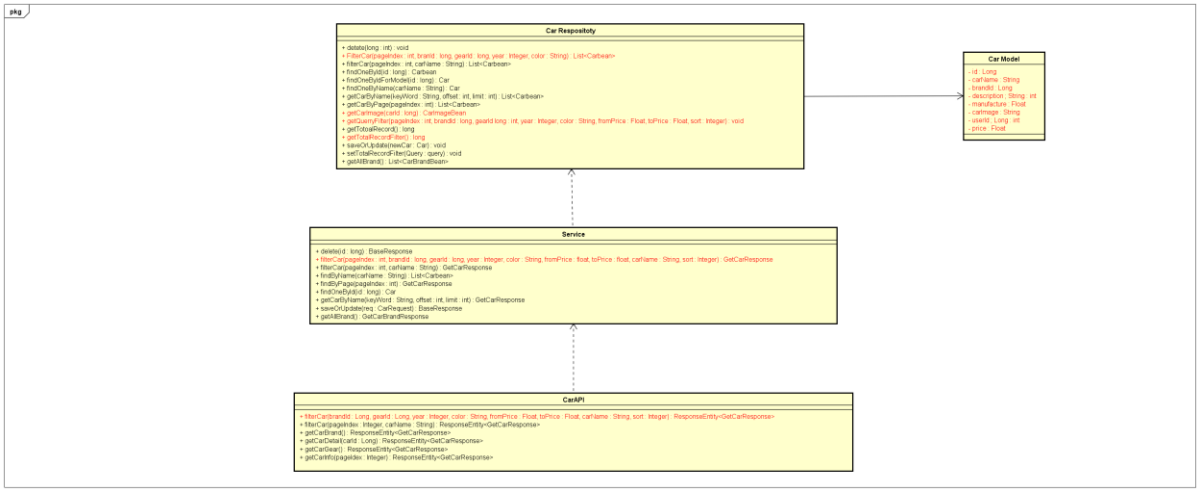



*Figure 4-19:Filter by sort price class diagram*

Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | Id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |

| | | | |
|---|---|---|---|
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

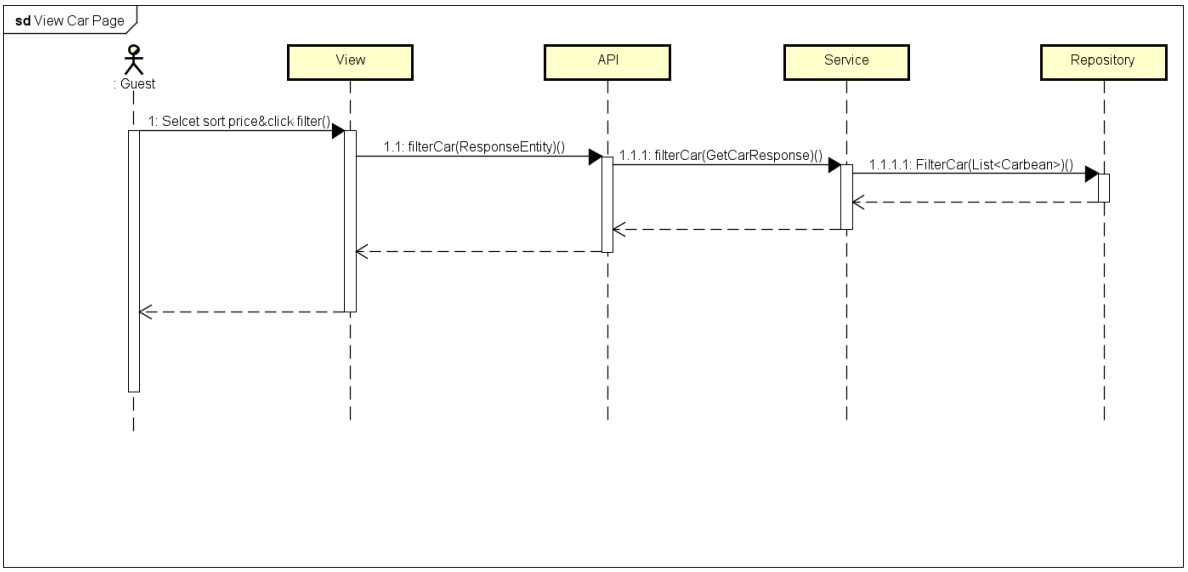



*Figure 4-20:Filter by sort price sequence diagram*

### 4.3.7 Filter by Year

Screen Design

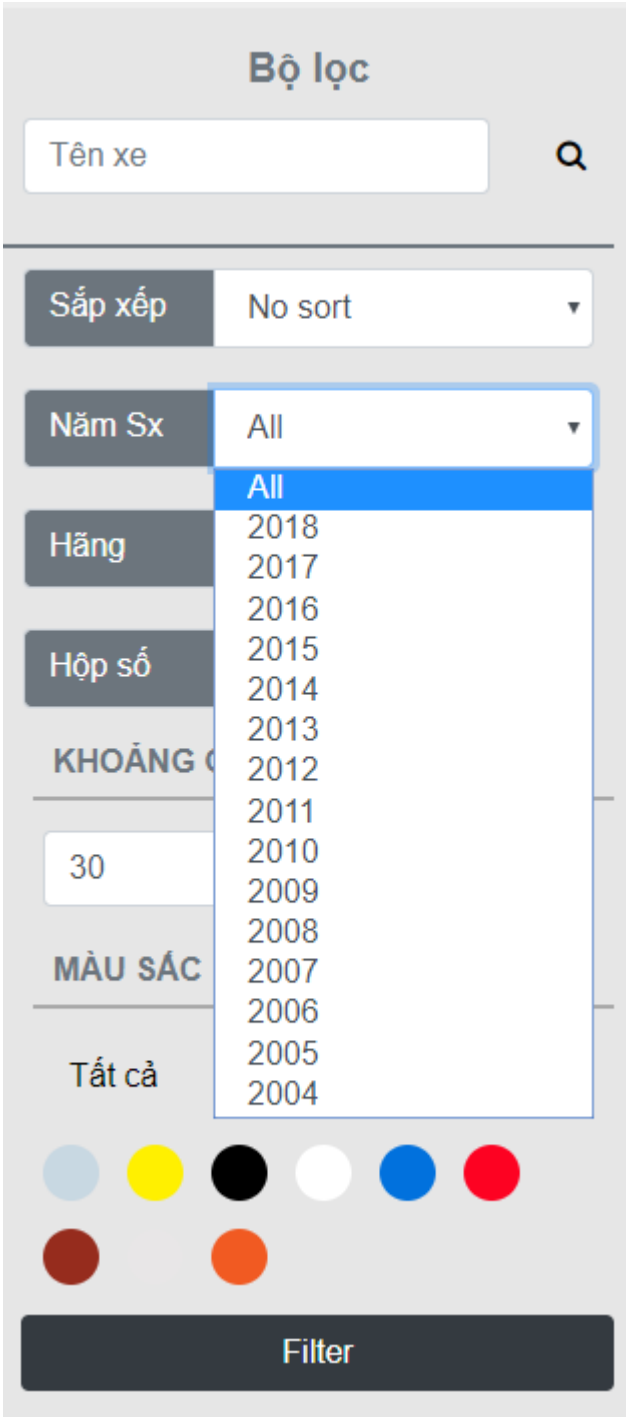



*Figure 4-21:Filter by year*

Class Diagram

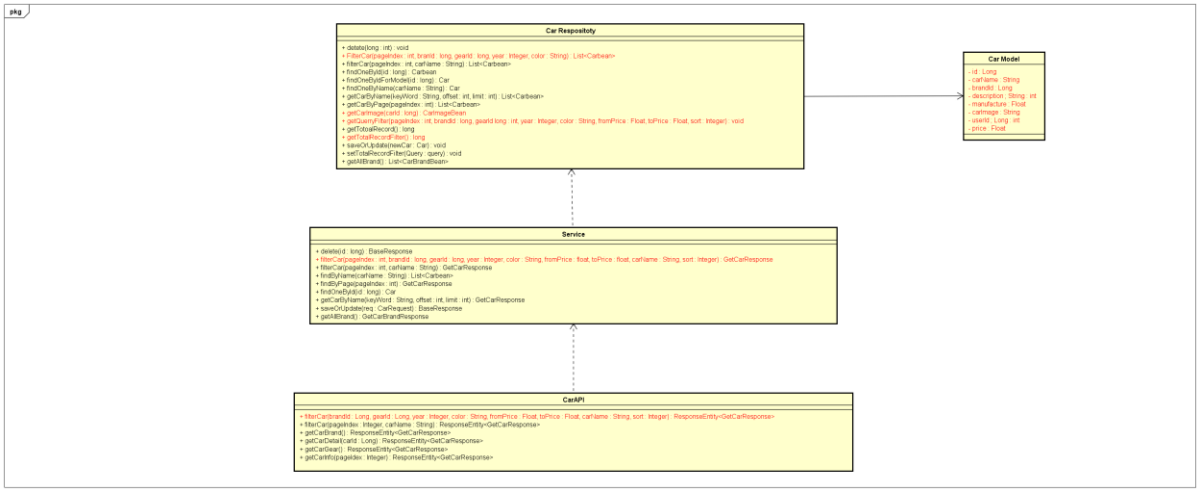



*Figure 4-22:Filter by year class diagram*

Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | Id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| FilterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | brandId | long | |

| | | | |
|---|---|---|---|
| | gearId | long | |
| | year | interger | |
| | **color** | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

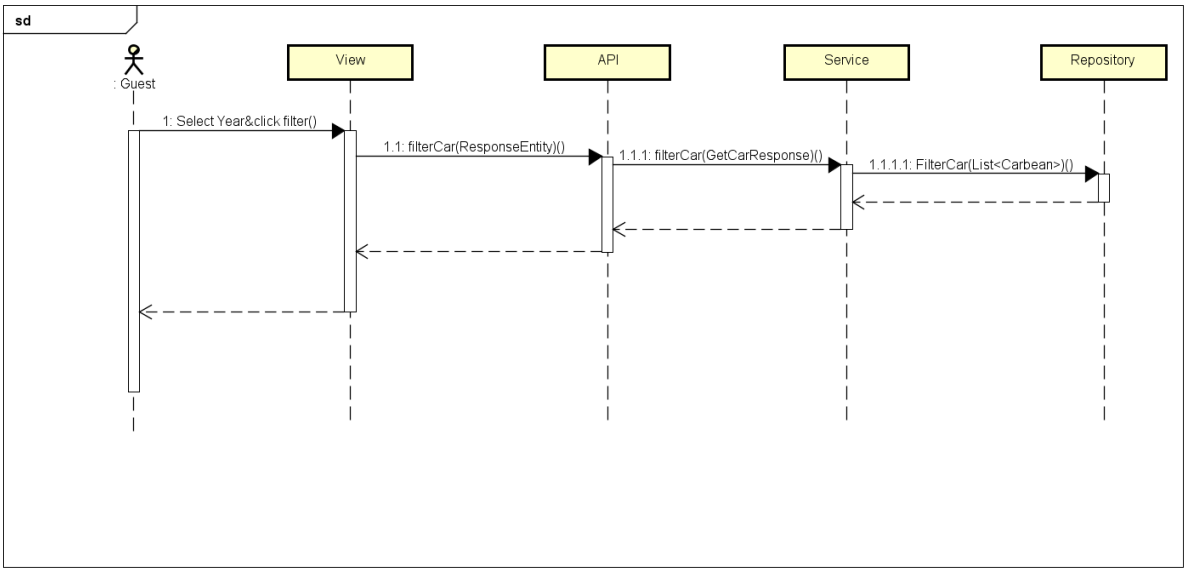



*Figure 4-23:Filter by year sequence diagram*

### 4.3.8 Guest View Car Page

Screen Design

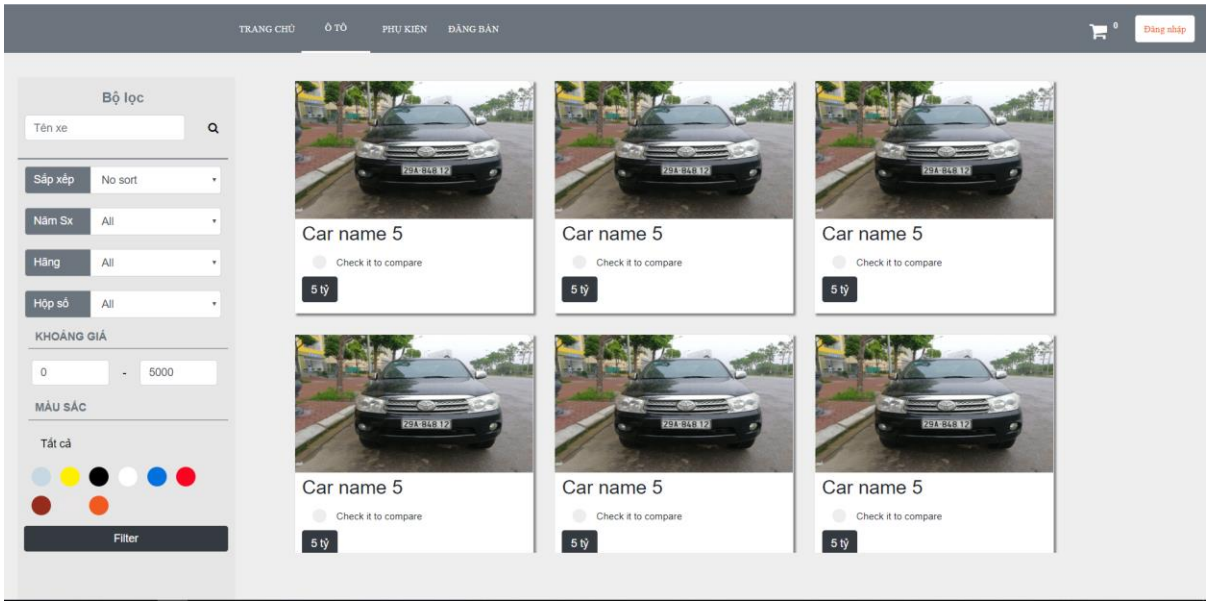



*Figure 4-24:Guest view car page*

## Class Diagram

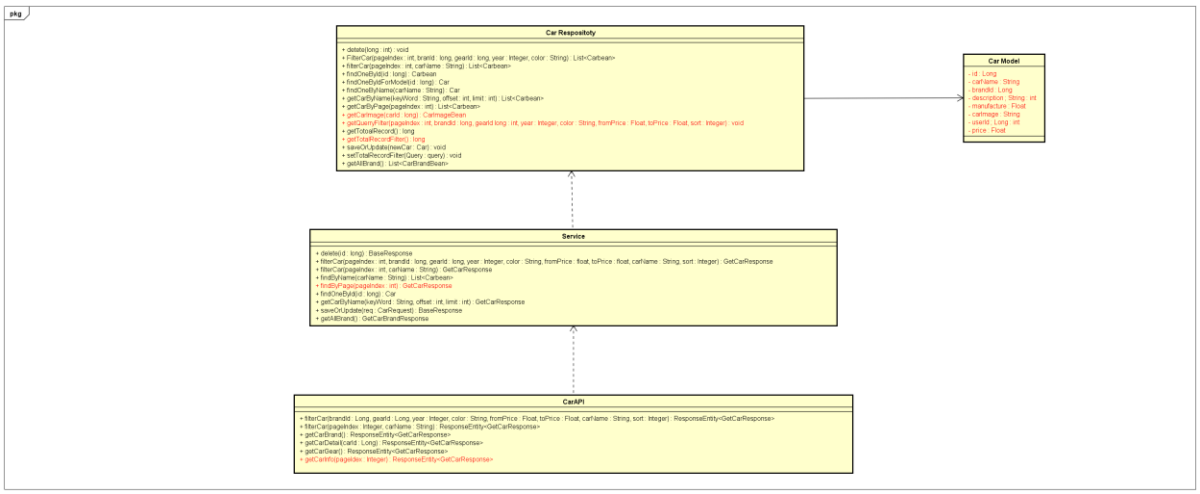



*Figure 4-25:Guest view car page class diagram*

## Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | id | Long | |
| 2 | carName | String | |

| | | | |
|---|---|---|---|
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| getCarspecificationByCar | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |
| | **carId** | long | |

Sequence Diagram

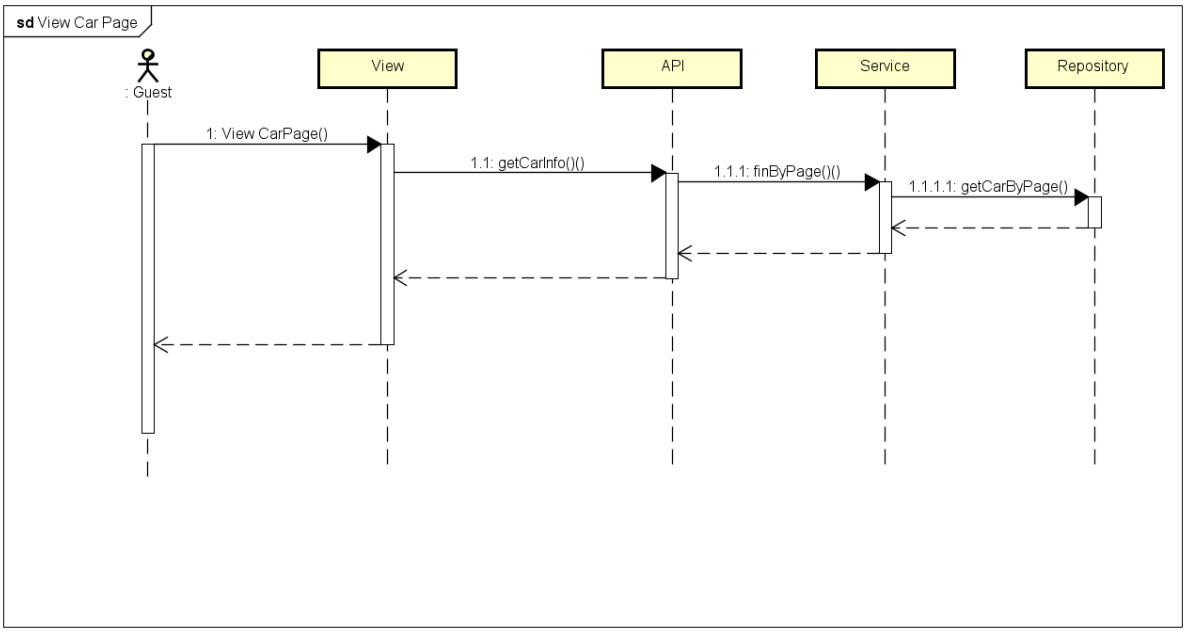



*Figure 4-26:Guest view car page sequence diagram*

### 4.3.9 Guest View HomePage

Screen Design

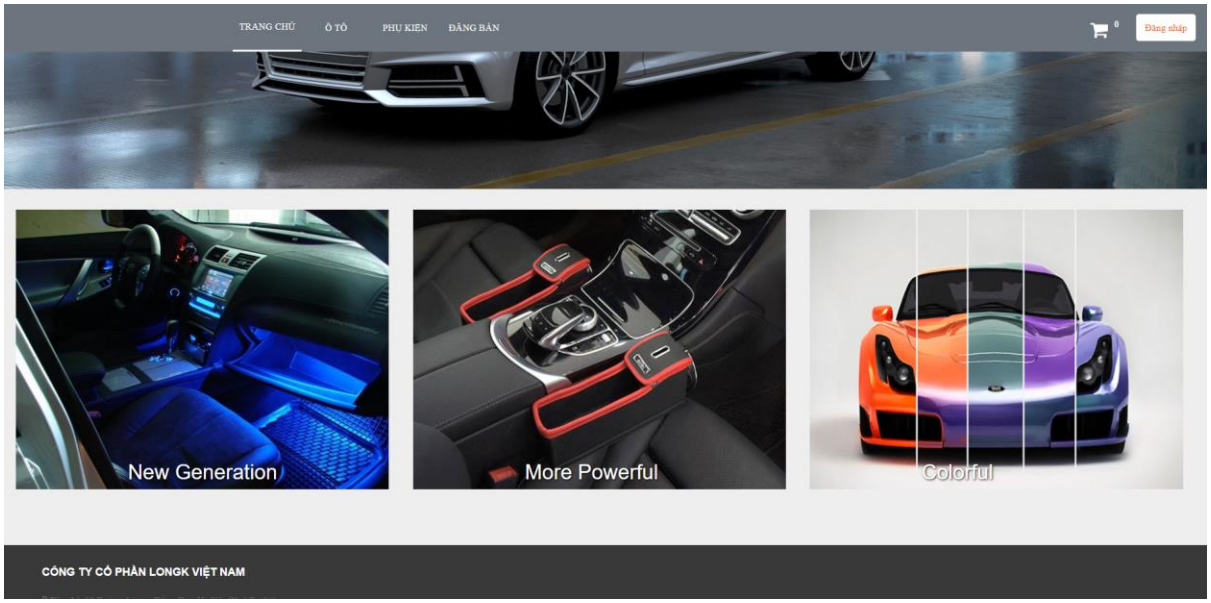



*Figure 4-27:Guest view home page*

Class Diagram

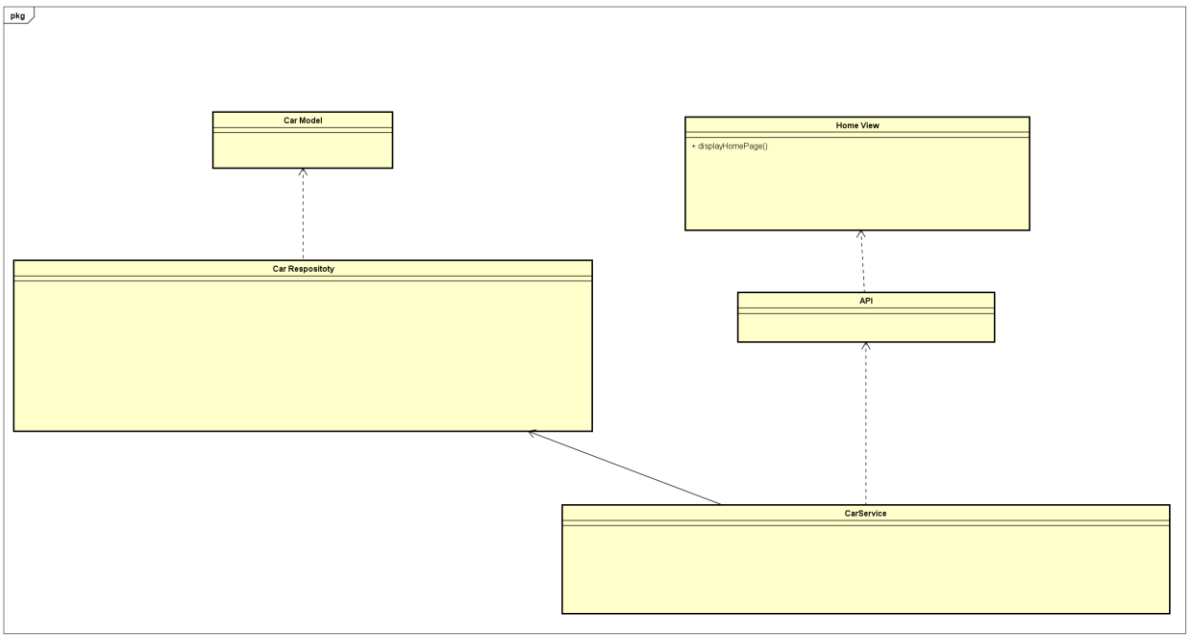



*Figure 4-28:Guest view home page class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |

| | | | |
|---|---|---|---|
| | | | |
| *Operation* | | | |
| | | | |
| Return Type | | | |
| Parameters | Name | Type | Description |
| | | | |

Sequence Diagram

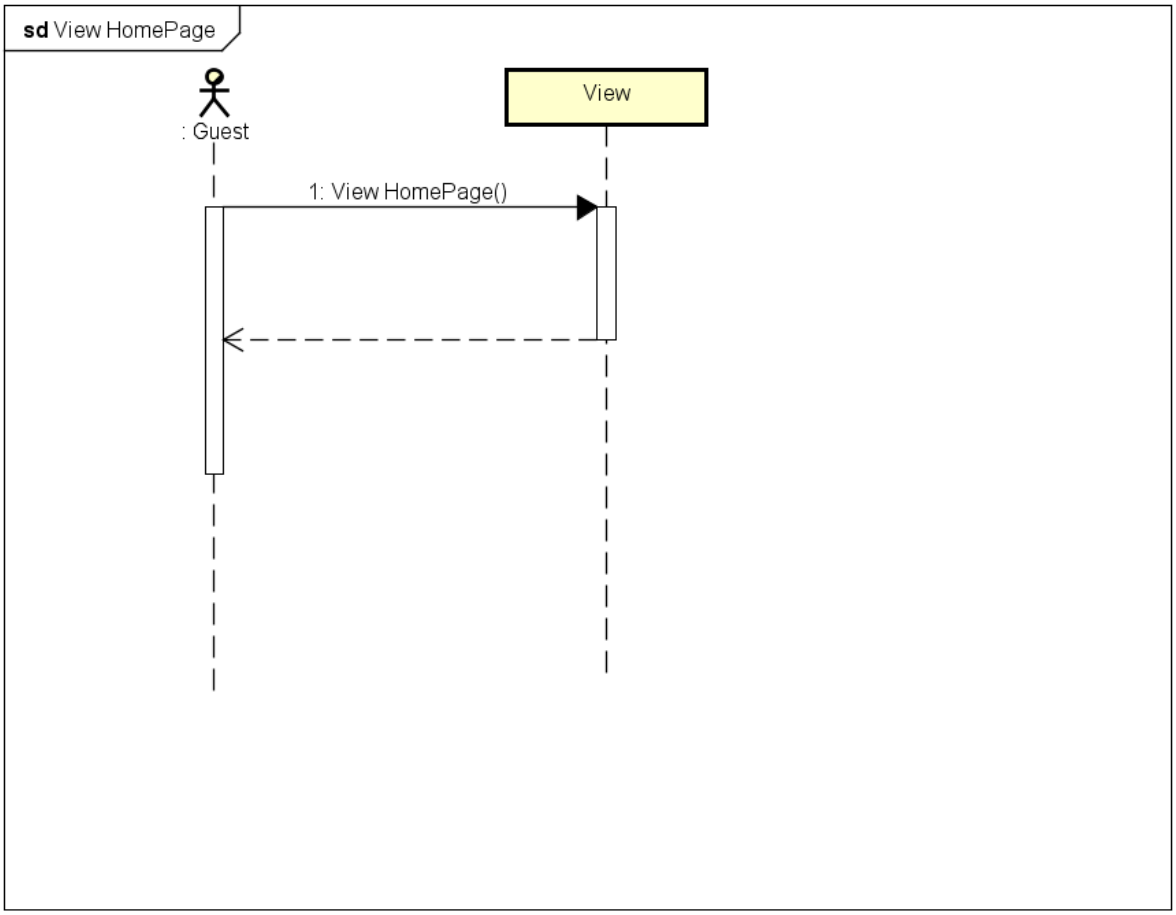



*Figure 4-29:Guest view home page sequence diagram*

### 4.3.10 Guest View Cart

Screen Design

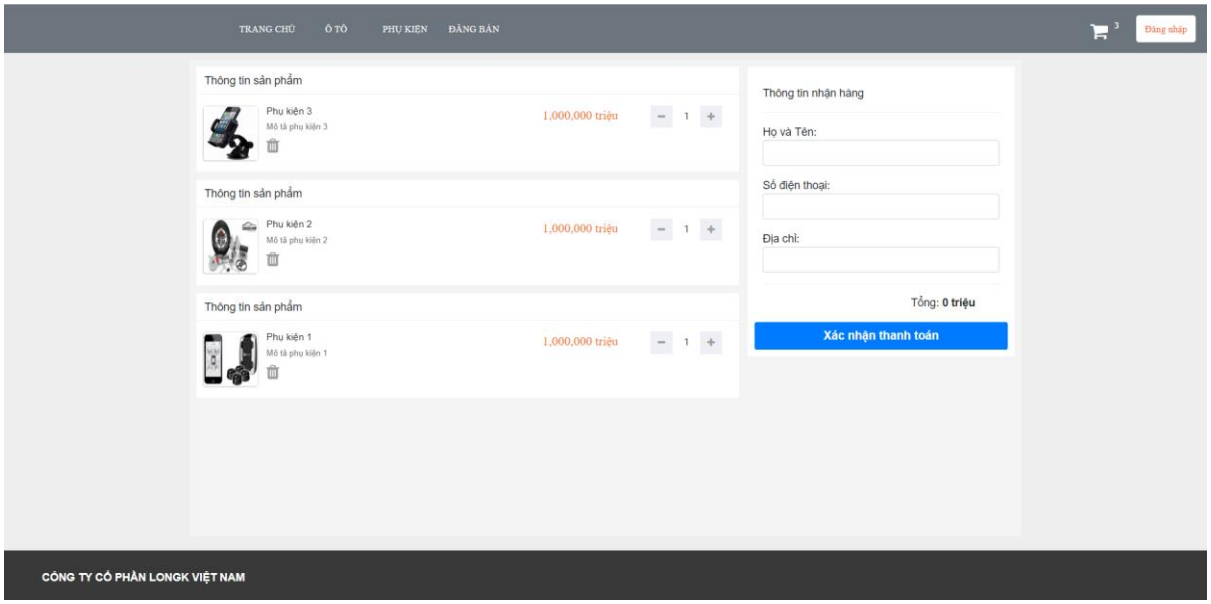



*Figure 4-30:Guest view cart*

## Class Diagram

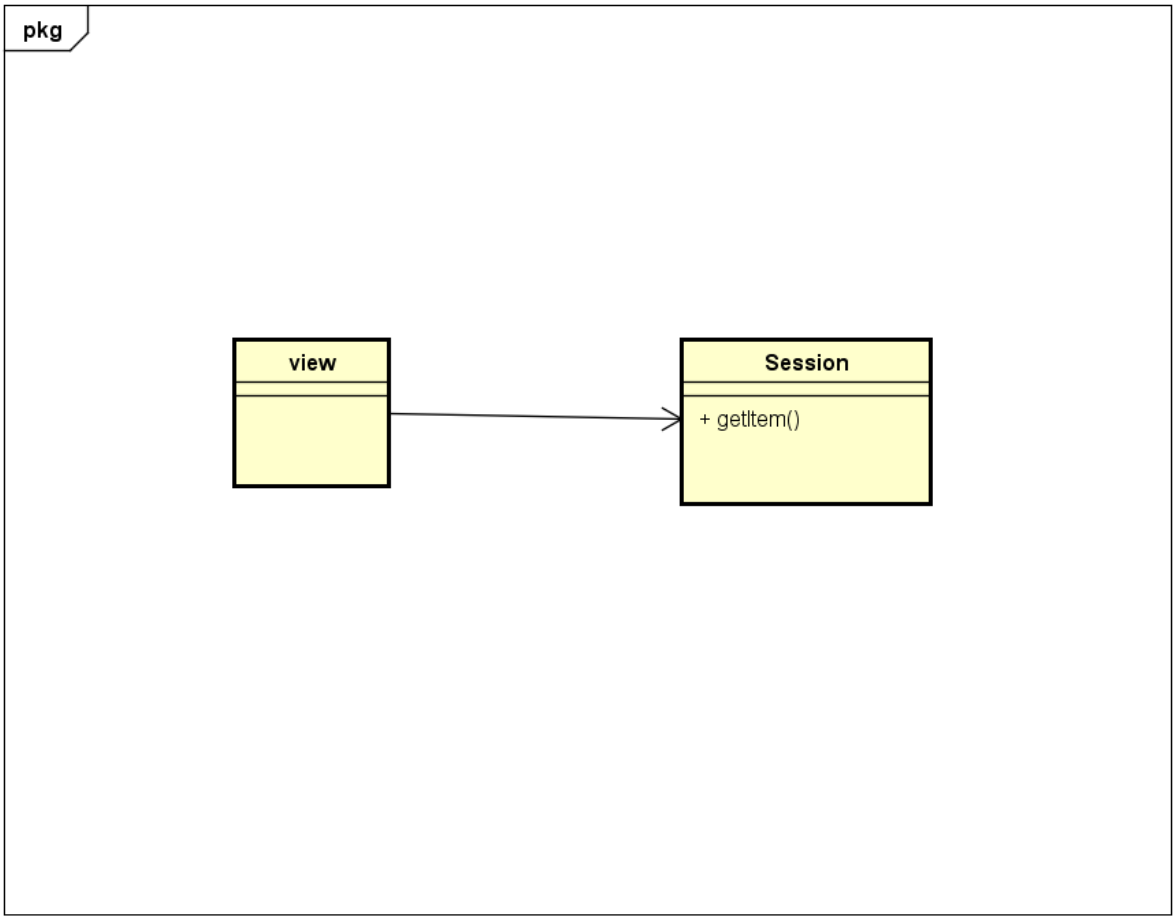



*Figure 4-31:Guest view cart page class diagram*

## Class Specification

| Service | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| | | | |
| *Operation* | | | |
| getItem | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |
| | accessoriesId | int | |

Sequence Diagram

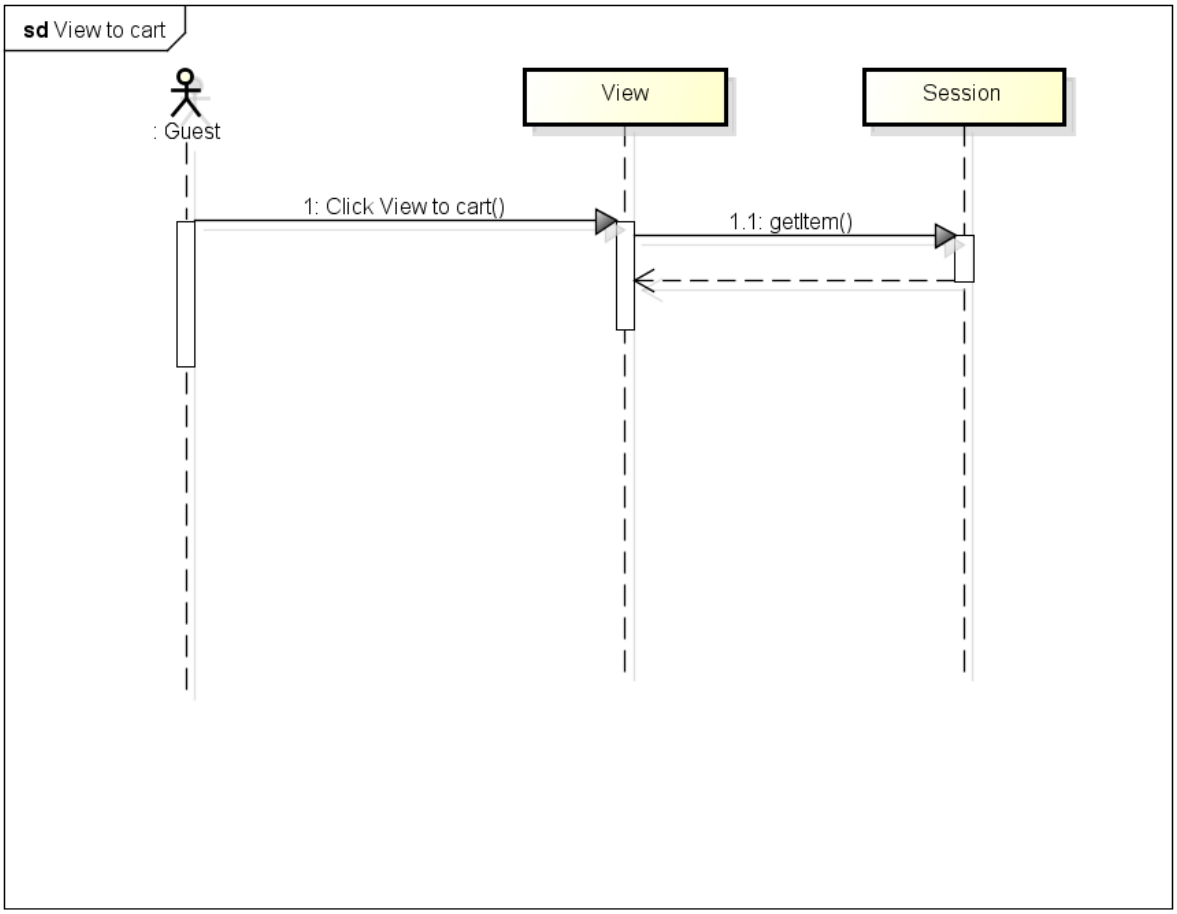


*Figure 4-32:Guest view cart page sequence diagram*

### 4.3.11 Guest Add To Cart

Screen Design



*Figure 4-33:Guest add to cart*

Class Diagram

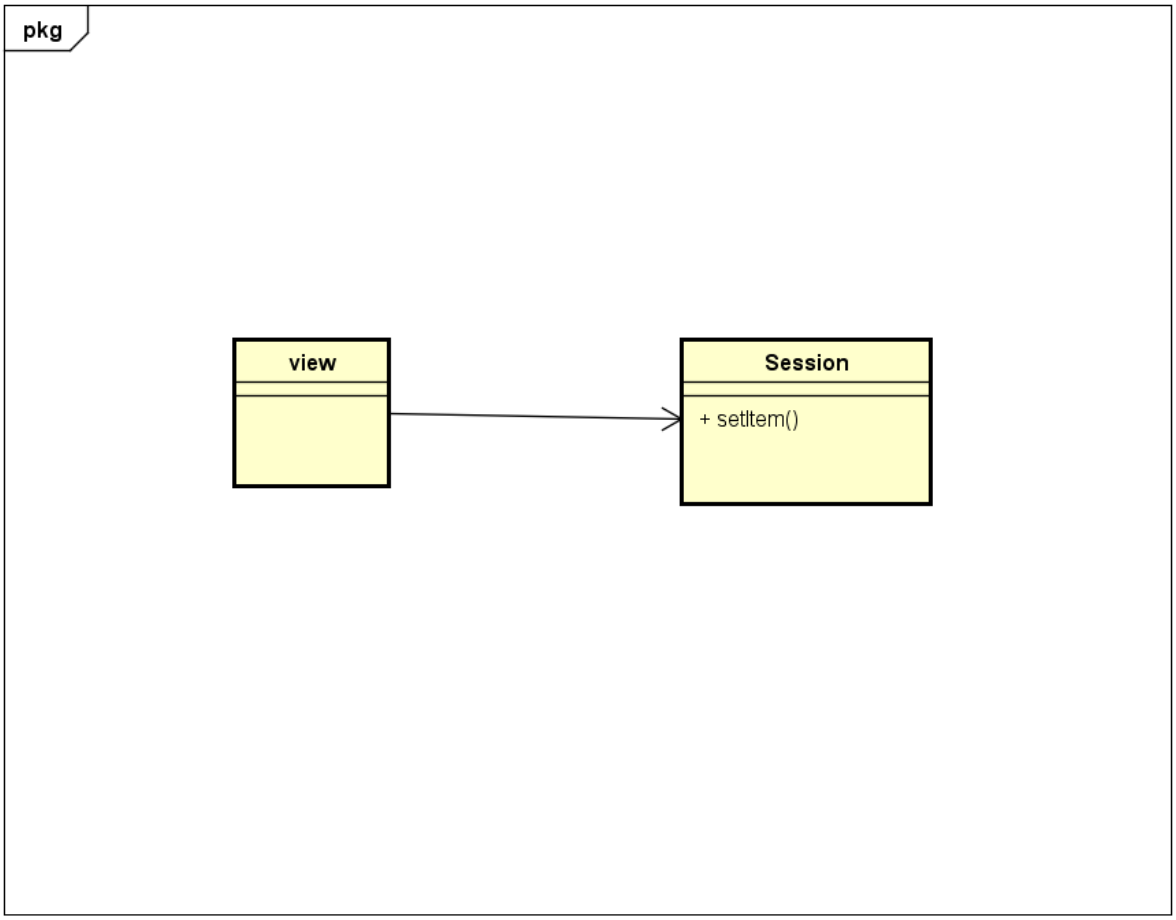



*Figure 4-34:Guest add to cart class diagram*

Class Specification

| **Session** | |
|---|---|
| Physical address | **/auto-fun** |
| Base class | |
| *Attributes* | |

| No | Name | Type | Description |
|---|---|---|---|
| | | | |
| *Operation* | | | |
| setItem | | | |
| Return Type | | | |
| Parameters | Name | Type | Description |
| | accessoriesId | int | |

Sequence Diagram

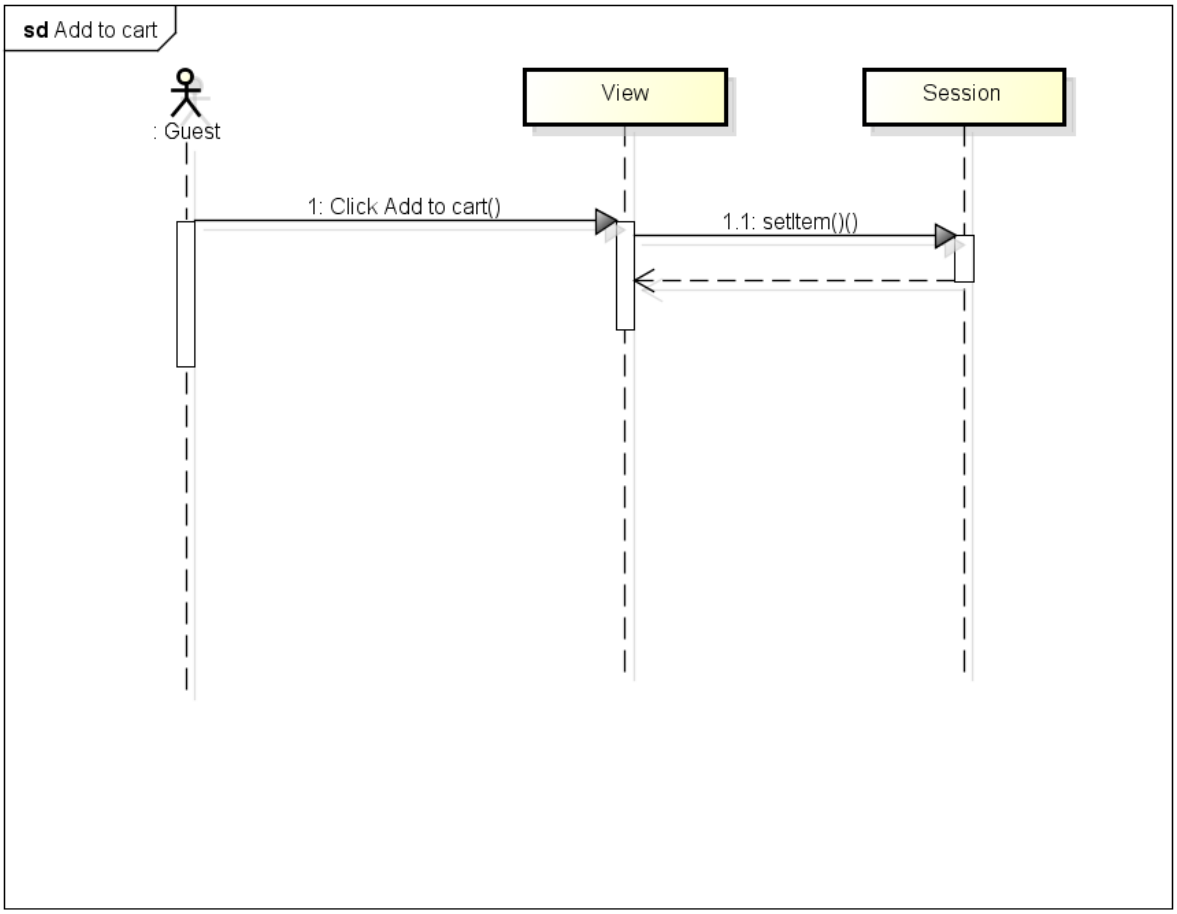



*Figure 4-35:Guest add to cart sequence diagram*

### 4.3.12 Guest Remove Cart

Screen Design




*Figure 4-36:Guest remove cart*

Class Diagram

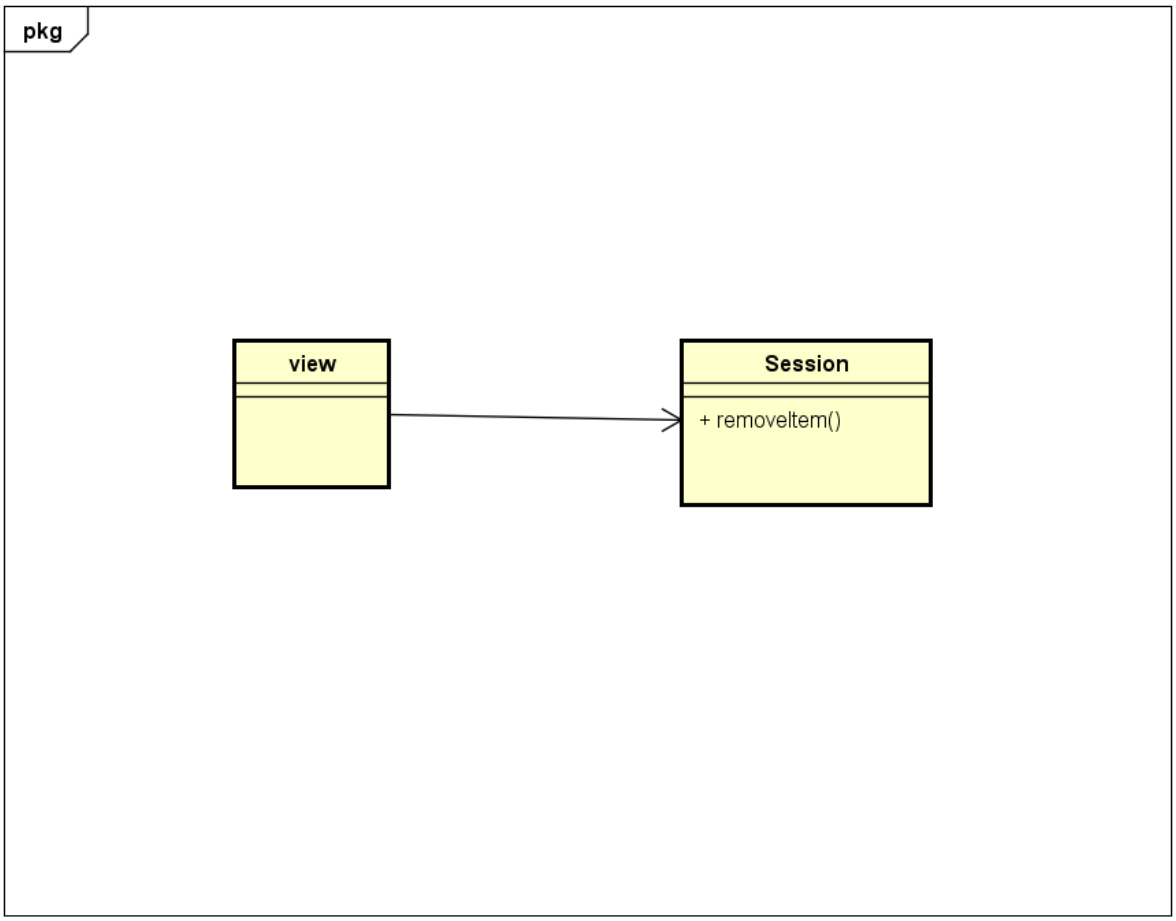



*Figure 4-37:Guest remove cart class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |

| | | | |
|---|---|---|---|
| *Operation* | | | |
| getCarAccessories | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |
| | accessoriesId | int | |

Sequence Diagram

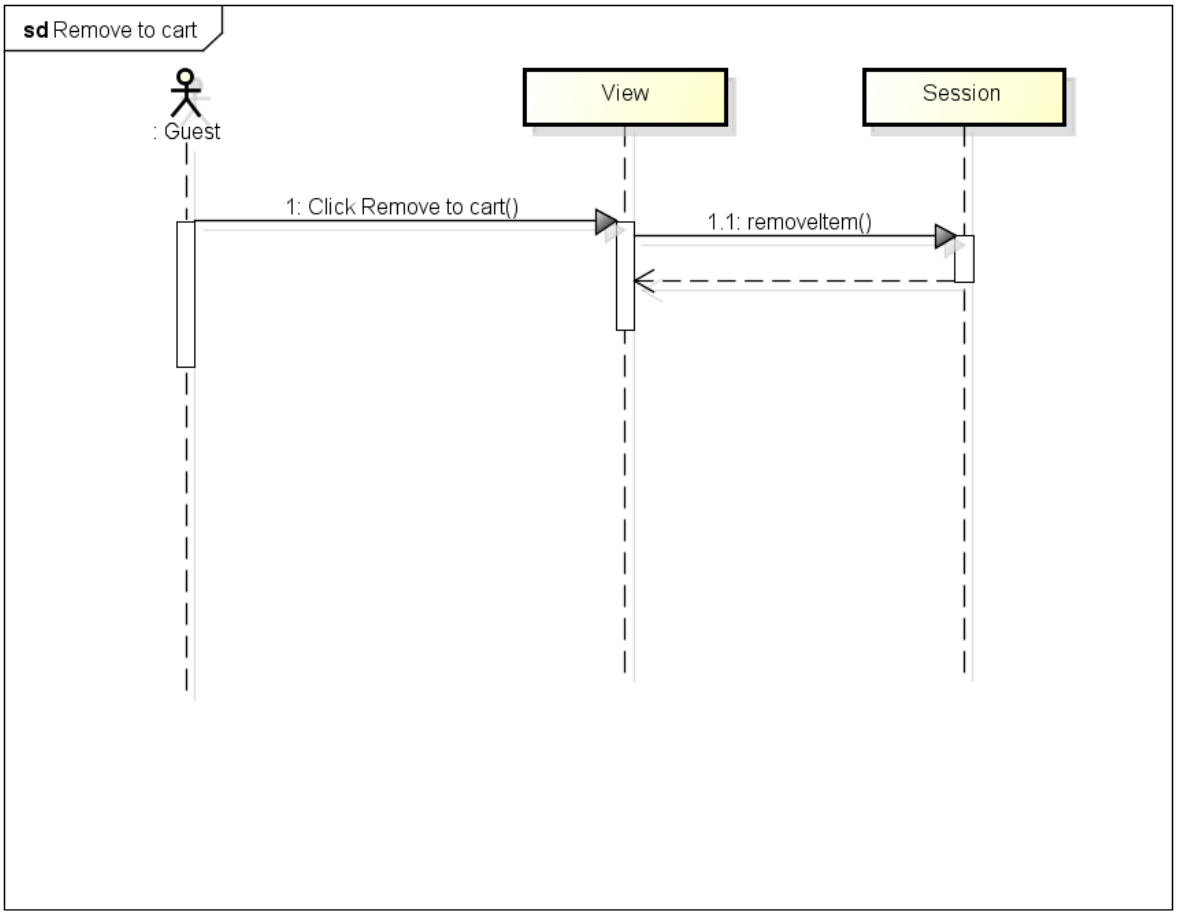



*Figure 4-38:Guest remove cart sequence diagram*

### 4.3.13 Guest View Accessories

Screen Design

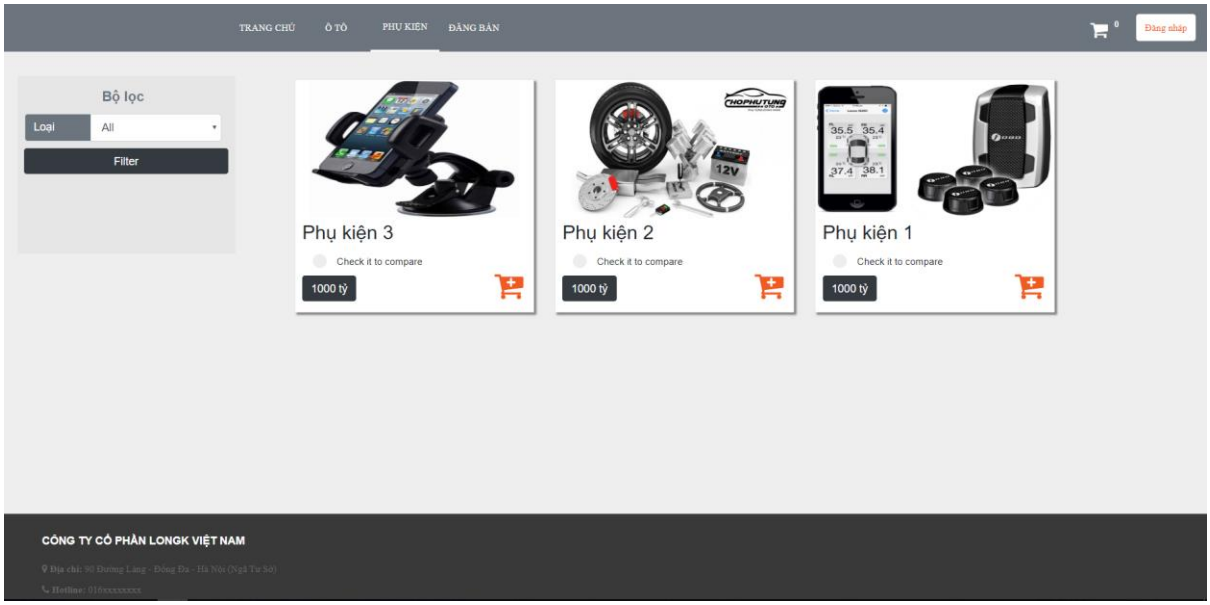



*Figure 4-39:Guest view accessories*

## Class Diagram

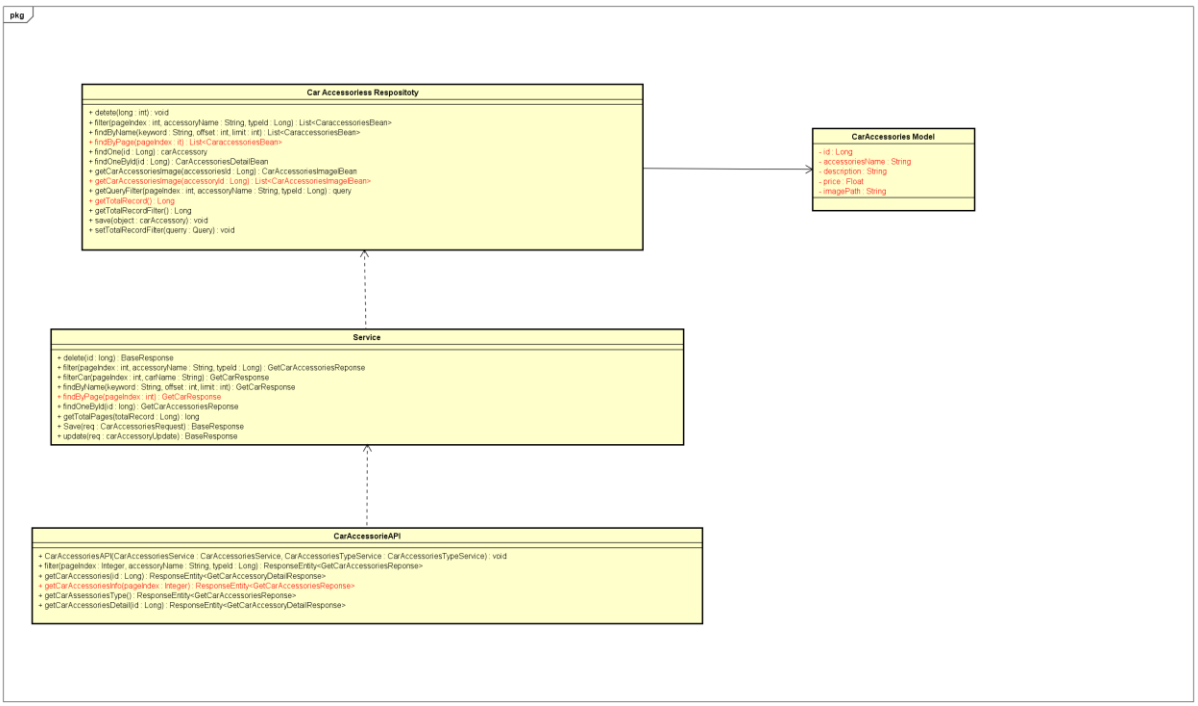



*Figure 4-39:Guest view accessories class diagram*

## Class Specification

| Service | |
|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** |
| Base class | |
| *Attributes* | |

| No | Name | Type | Description |
|---|---|---|---|
| 1 | id | long | |
| 2 | accessoriesName | String | |
| 3 | Description | String | |
| 4 | Price | Float | |
| 5 | imagePath | String | |
| *Operation* | | | |
| getCarAccessories | | | |
| Return Type | getCarAccessories<br>Response | | |
| Parameters | Name | Type | Description |
| | pageIndex | int | |

Sequence Diagram

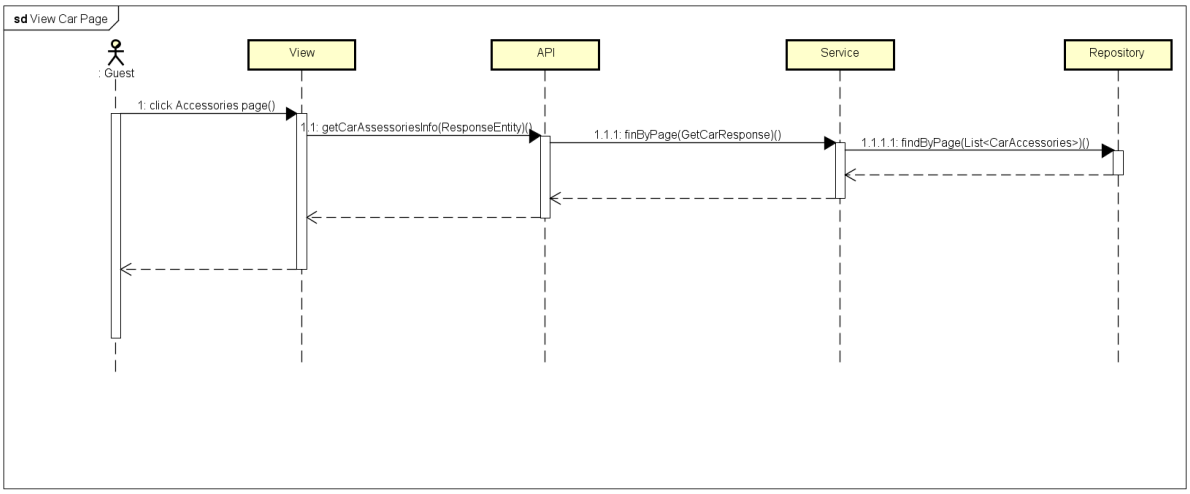



*Figure 4-39:Guest view accessories sequence diagram*

## 4.3.14 Guest View Car Detail

Screen Design

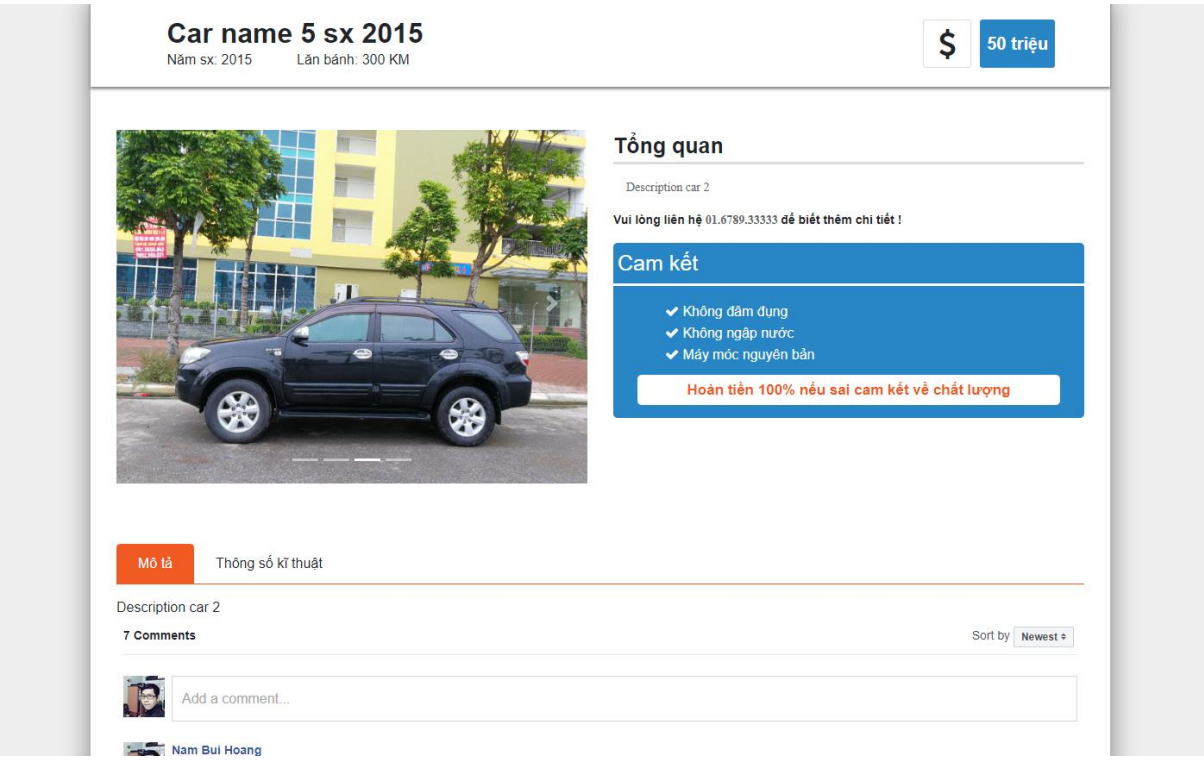



*Figure 4-40:Guest view Car Detail*

## Class Diagram

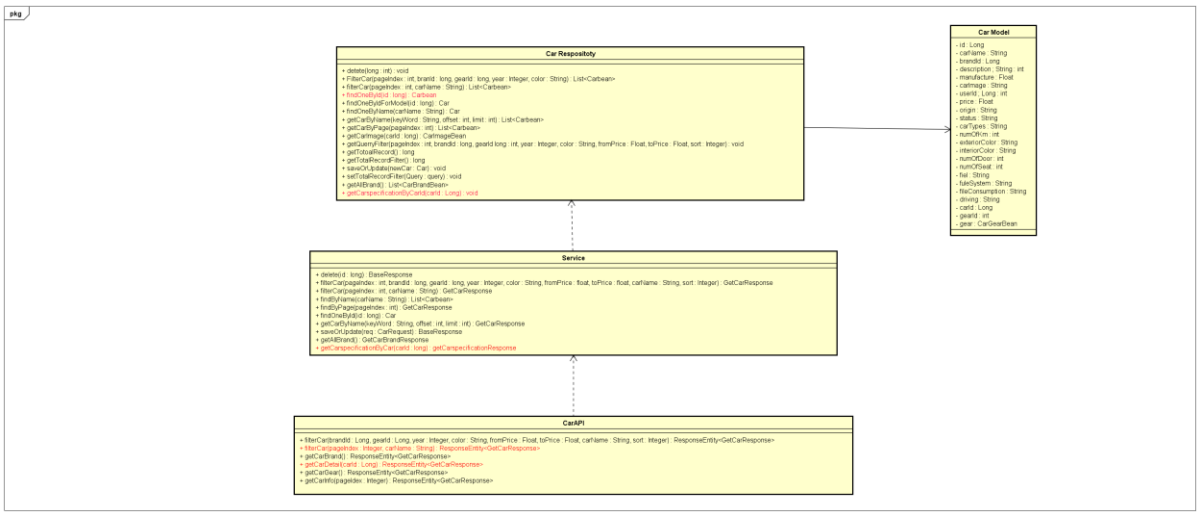



*Figure 4-41:Guest view Car Detail class diagram*

## Class Specification

| Car Service | |
|---|---|
| Physical address | **/car** |
| Base class | |
| *Attributes* | |

| No | Name | Type | Description |
|---|---|---|---|
| 1 | id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| getCarspecificationByCar | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |
| | **carId** | long | |

Sequence Diagram

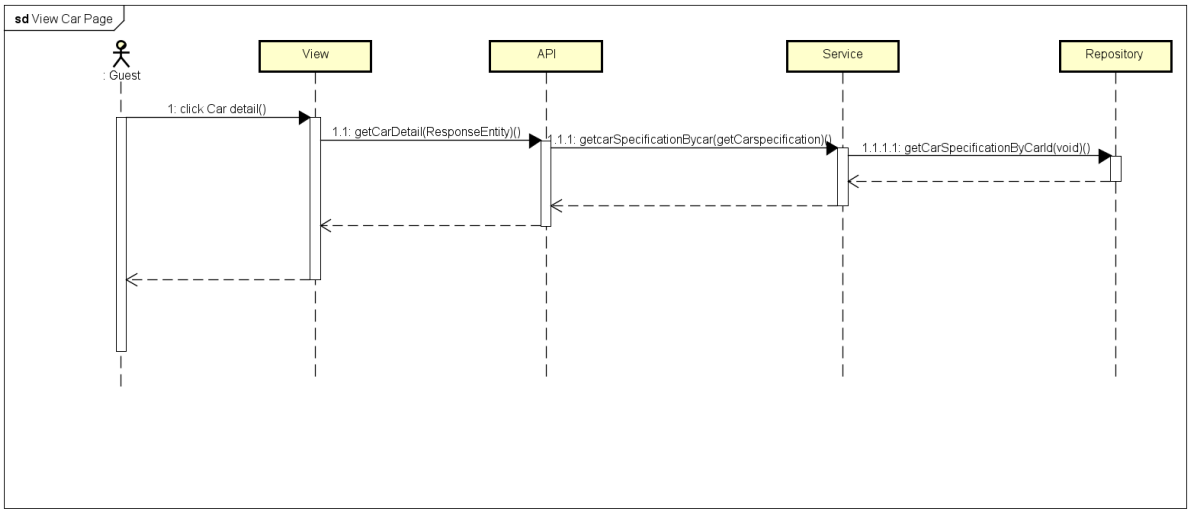



*Figure 4-42:Guest view Car Detail sequence diagram*

## 4.3.15 Guest View Detail Accessories

Screen Design

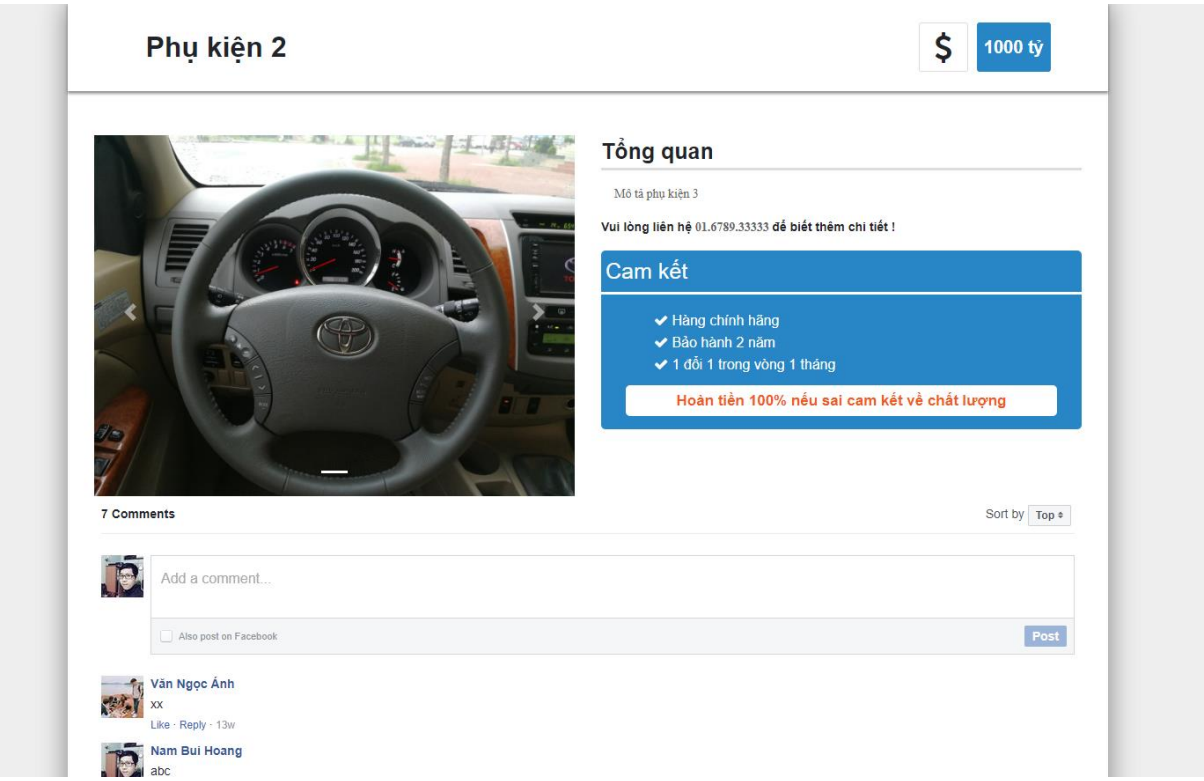



*Figure 4-43:Guest view Accessories Detail*

## Class Diagram

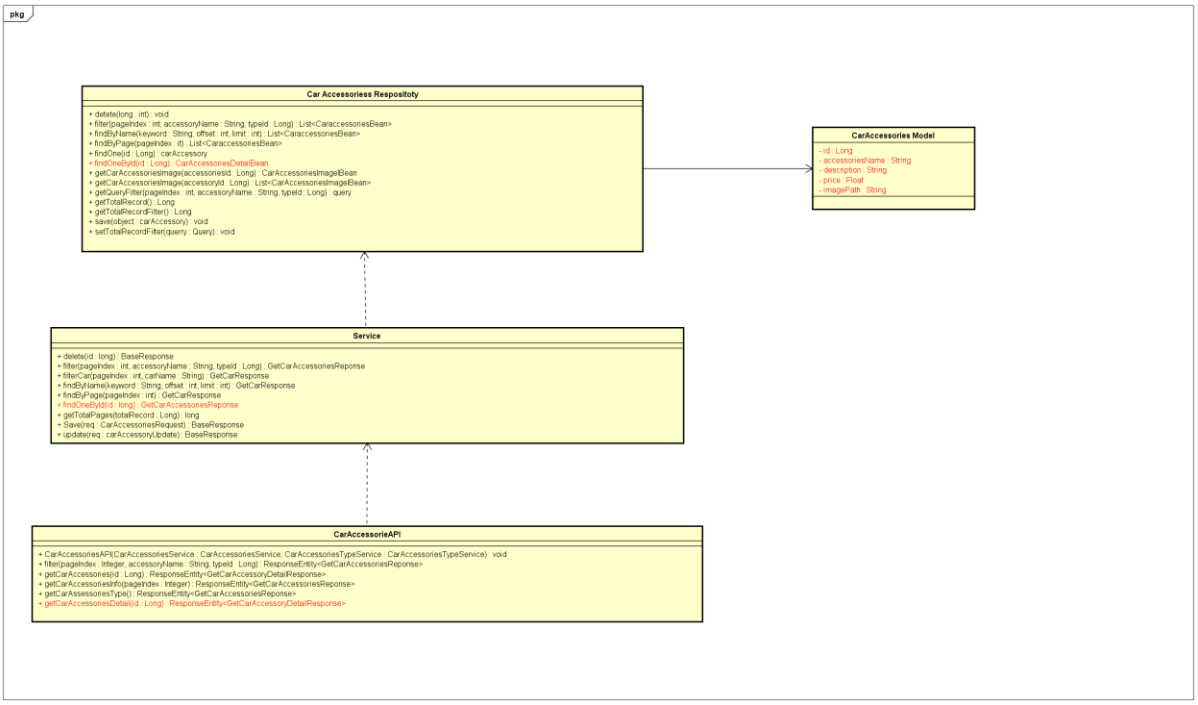



*Figure 4-44:Guest view Accessories Detail class diagram*

## Class Specification

| Car Service |
|---|

| | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | id | Long | |
| 2 | description | String | |
| 6 | price | Float | |
| 7 | imagePath | String | |
| *Operation* | | | |
| getCarAccessories | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |

Sequence Diagram

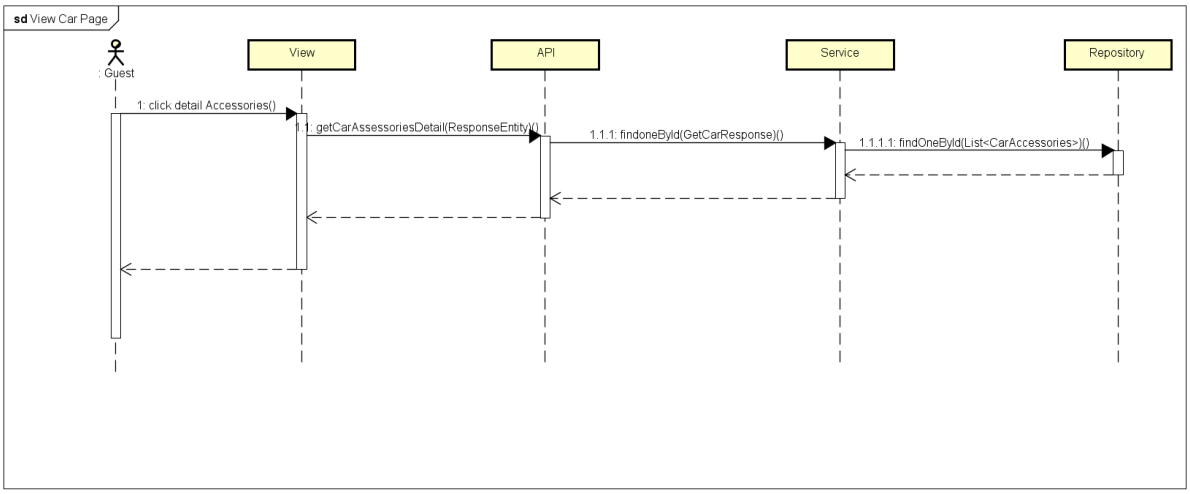



*Figure 4-45:Guest view Accessories Detail sequence diagram*

## 4.3.16 Search Accessories

Screen Design




*Figure 4-46:Search Accessories Detail*

Class Diagram

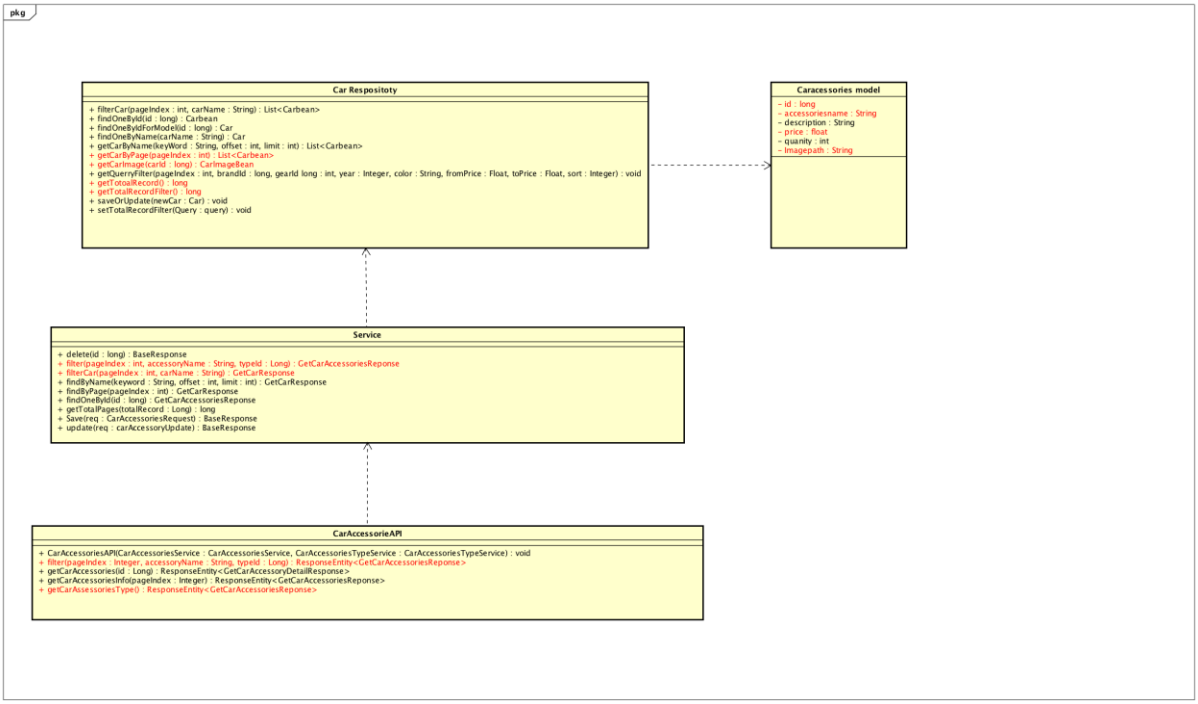



*Figure 4-47:Search Accessories Detail class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |

| | | | |
|---|---|---|---|
| 1 | id | long | |
| 2 | accessoriesName | String | |
| 3 | Description | String | |
| 4 | Price | Float | |
| 5 | imagePath | String | |
| *Operation* | | | |
| findByPage | | | |
| Return Type | getCarAccessories Response | | |
| Parameters | Name | Type | Description |
| | pageIndex | int | |

Sequence Diagram

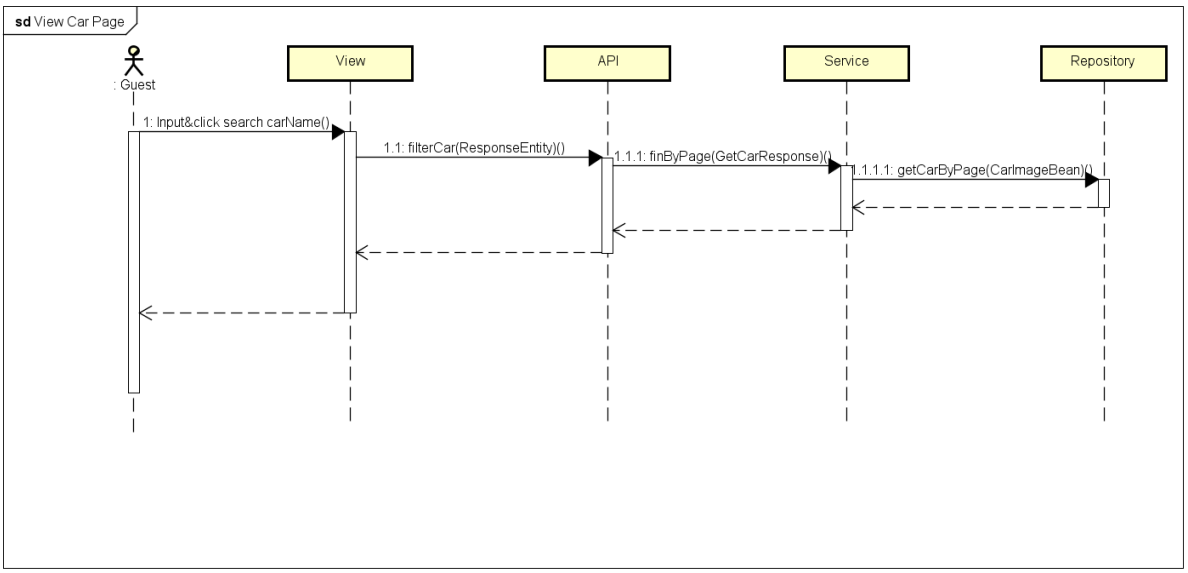



*Figure 4-48:Search Accessories Detail sequence diagram*

### 4.3.17 Guest search by car name

Screen Design

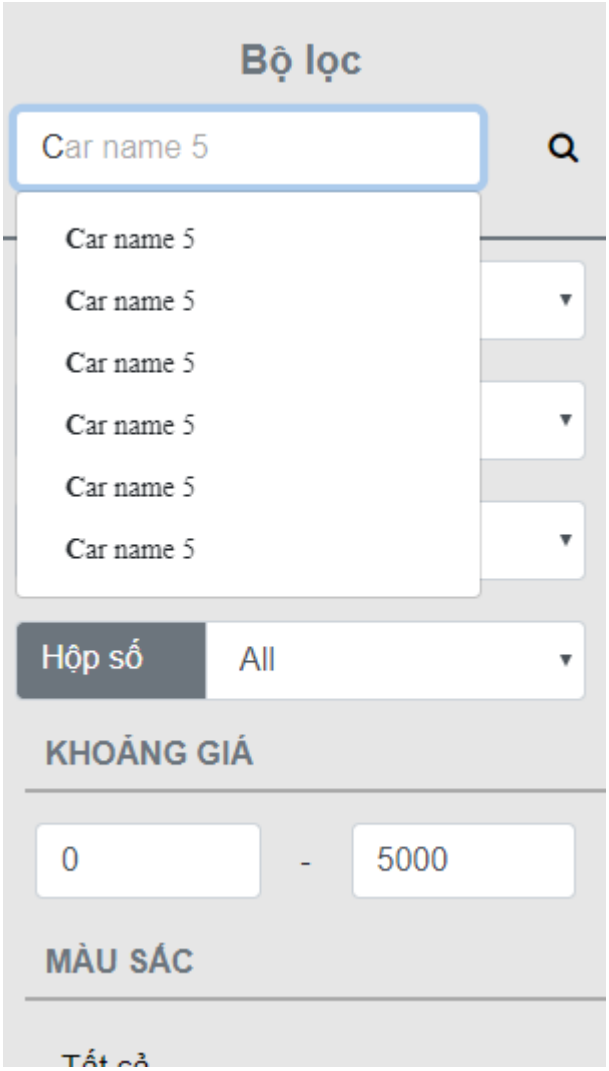



*Figure 4-49:Search by car name*

Class Diagram

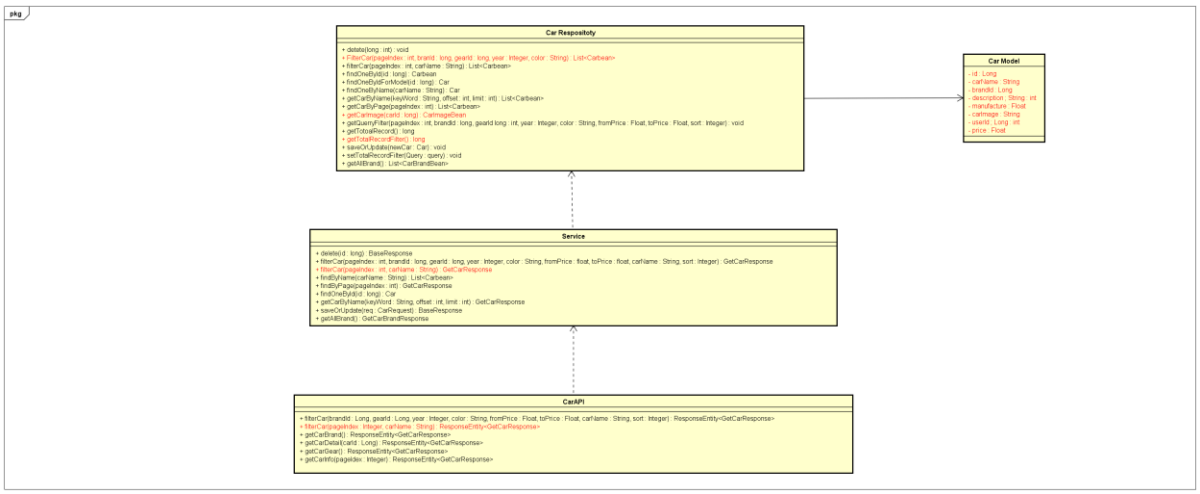



*Figure 4-50:Search by car name class diagram*

Class Specification

| Car Service | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| filterCar | | | |
| Return Type | GetCarReponse | | |
| Parameters | Name | Type | Description |
| | pageIndex | Int | |
| | carName | String | |
| **getAllBrand** | | | |
| **Return Type** | **Object** | | |
| **Parameters** | **Name** | Type | Description |

Sequence Diagram

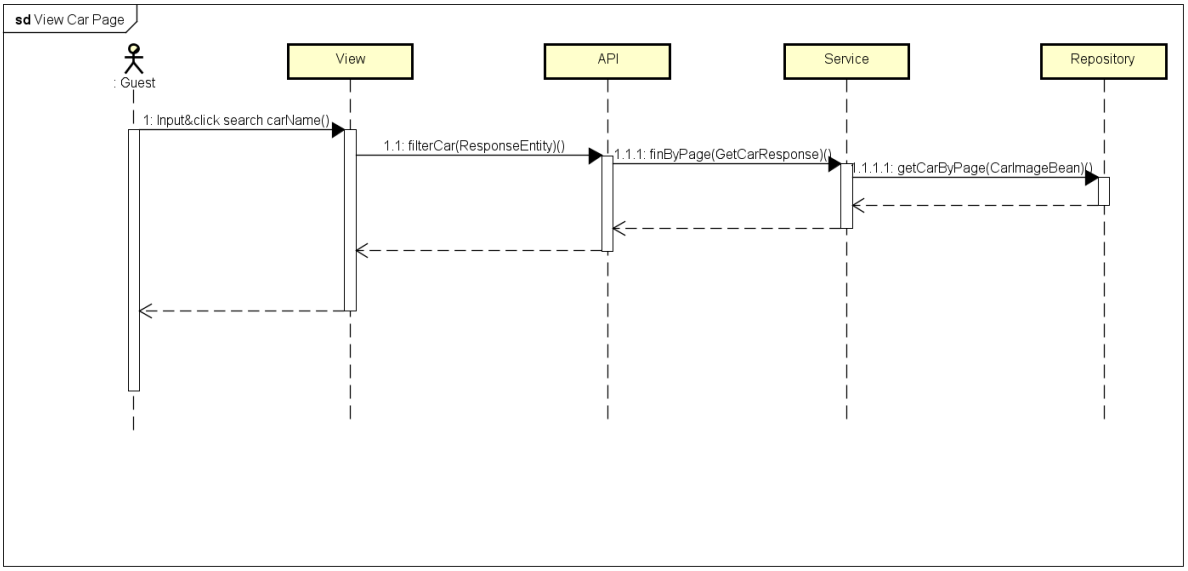



*Figure 4-51:Search by car name sequence diagram*

### 4.3.18 User Login

Screen Design

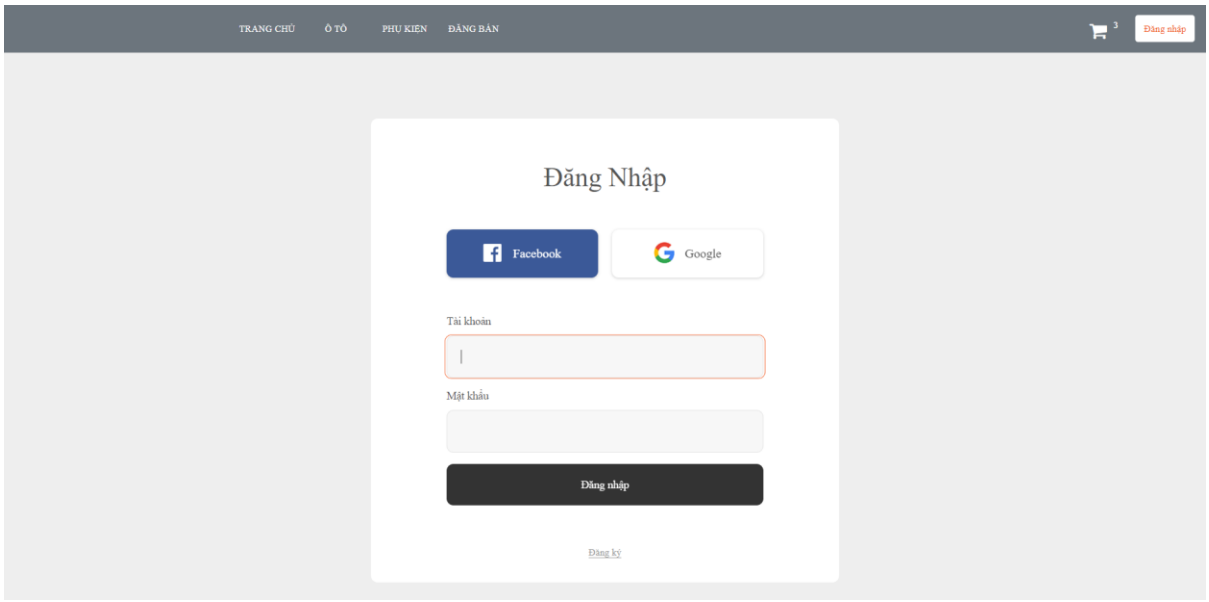



*Figure 4-52:User login*

Class Diagram

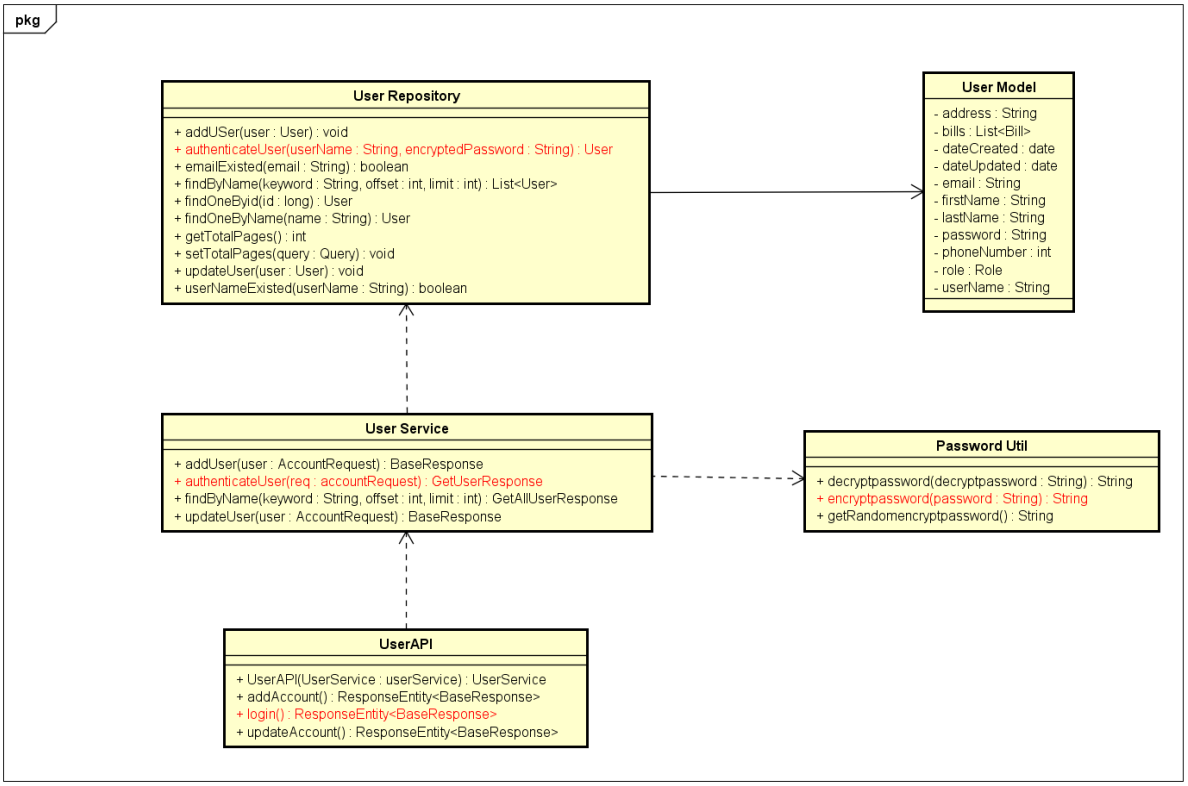



*Figure 4-53:User login class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| | | | |
| *Operation* | | | |
| | | | |
| Return Type | | | |
| Parameters | Name | Type | Description |
| | **carId** | long | |

Sequence Diagram

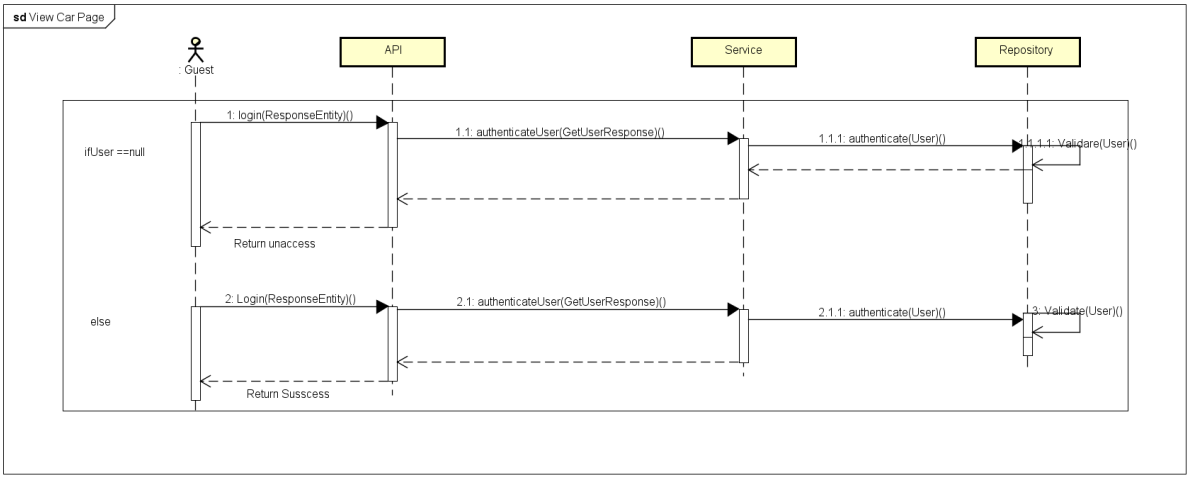



*Figure 4-54:User login sequence diagram*

### 4.3.19 User Logout

Screen Design

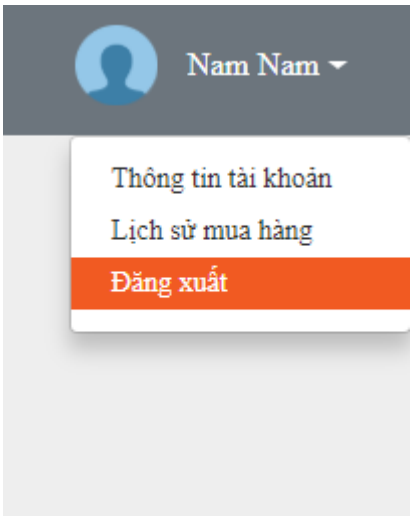



*Figure 4-55:User logout*

Class Diagram

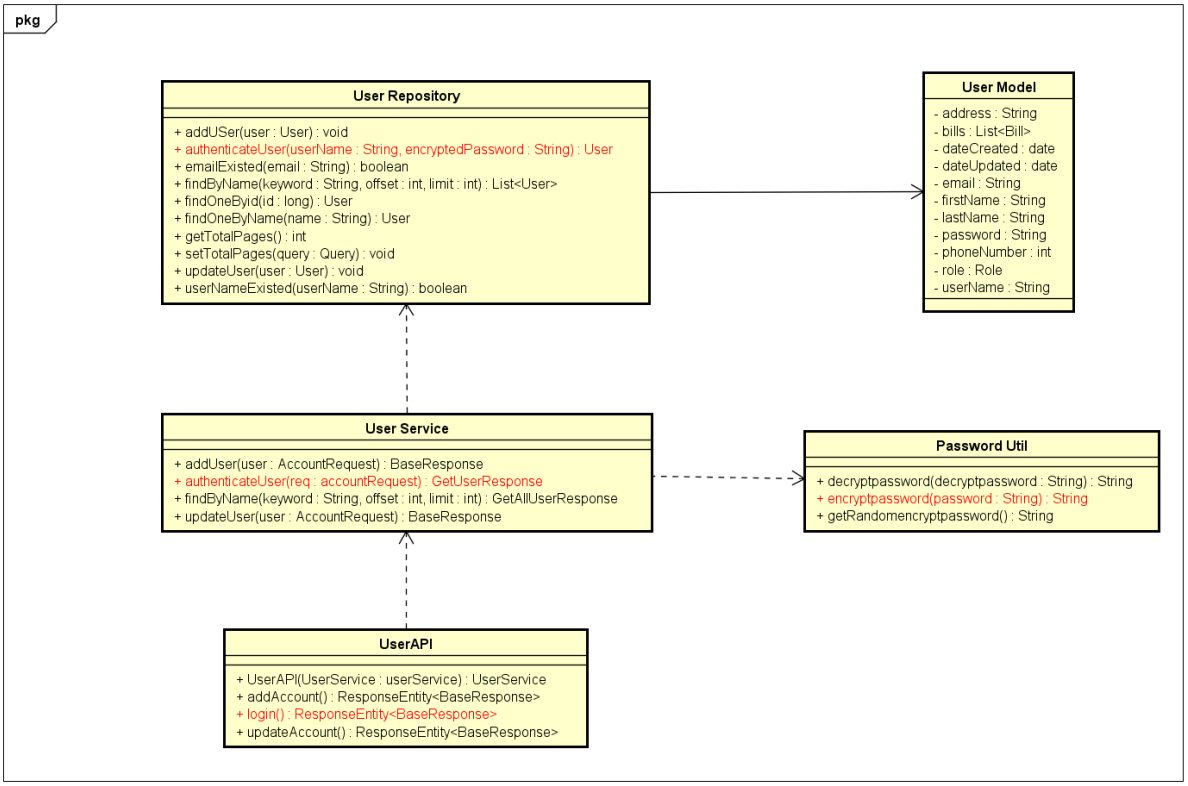



*Figure 4-56:User logout class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| | | | |
| *Operation* | | | |
| | | | |
| Return Type | | | |
| Parameters | Name | Type | Description |
| | **carId** | long | |

Sequence Diagram

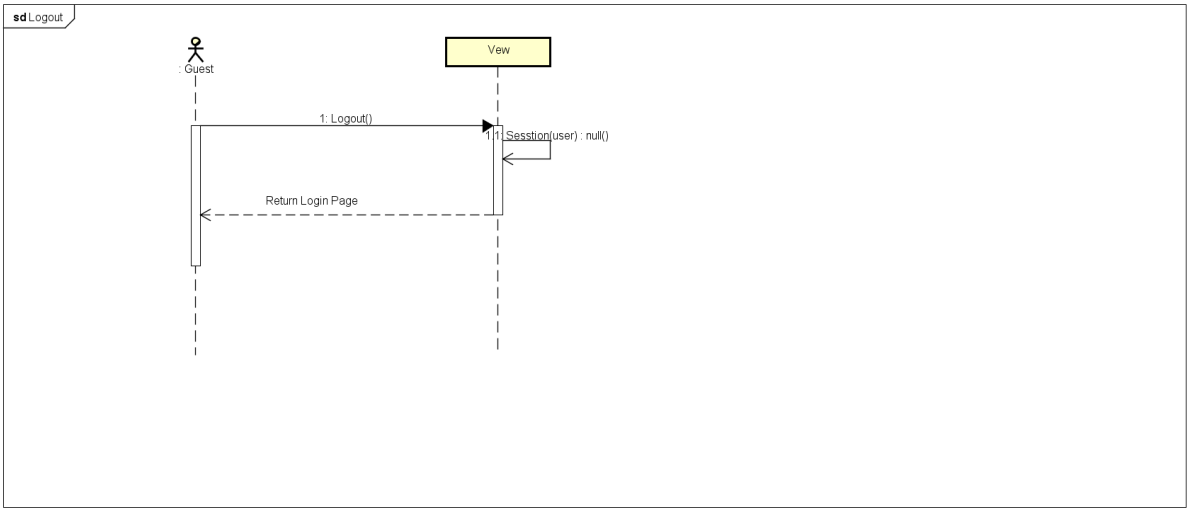



*Figure 4-57:User logout sequence diagram*

## 4.3.20 Register

Screen Design

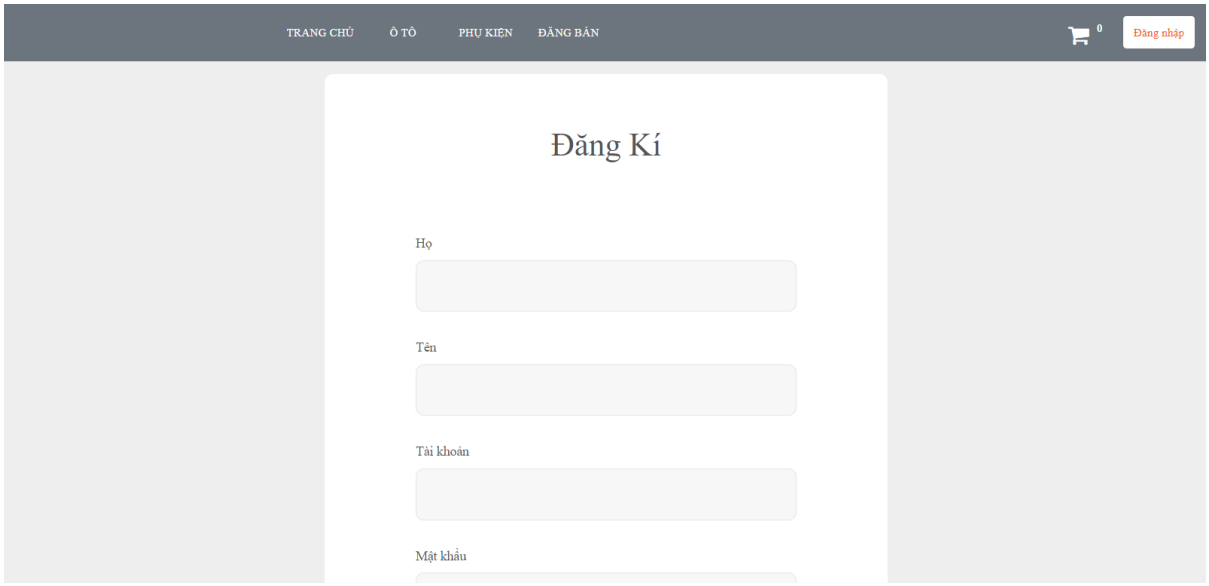



*Figure 4-58:Register*

Class Diagram

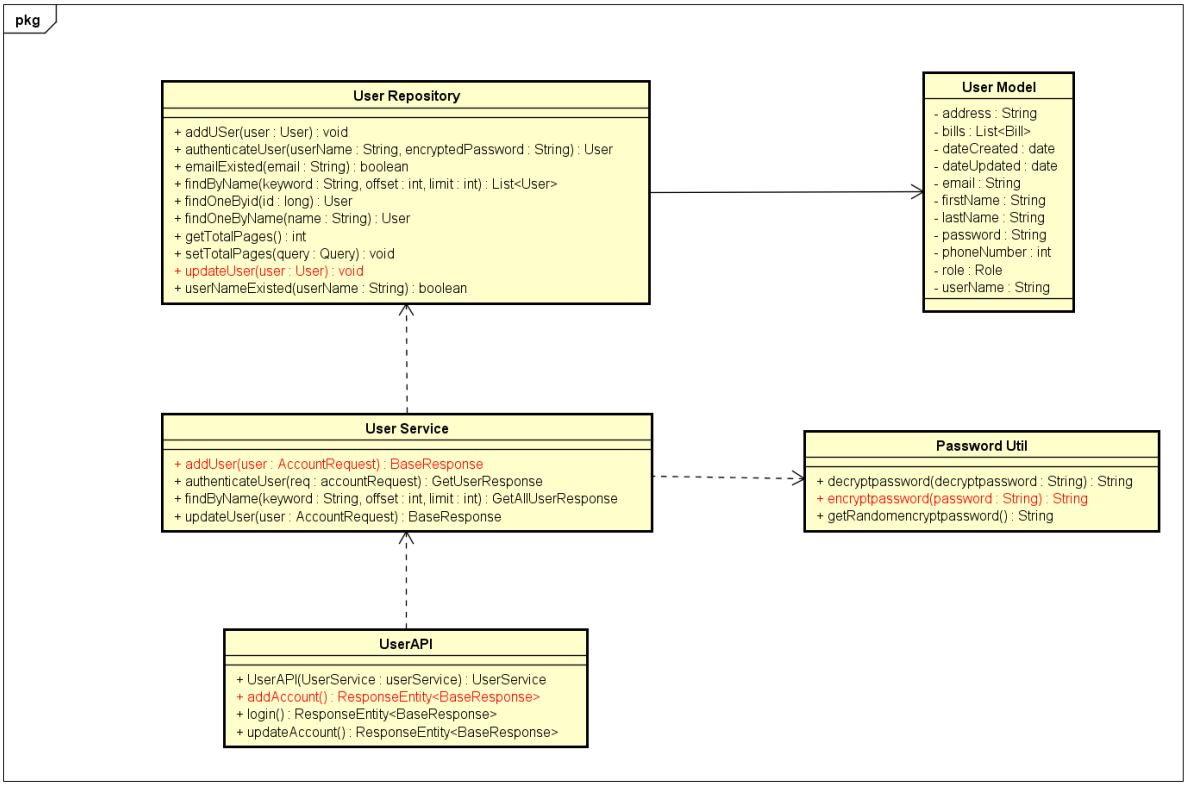



*Figure 4-59:Register class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| | | | |
| Return Type | | | |
| Parameters | Name | Type | Description |
| | User | accountRequest | |

Sequence Diagram

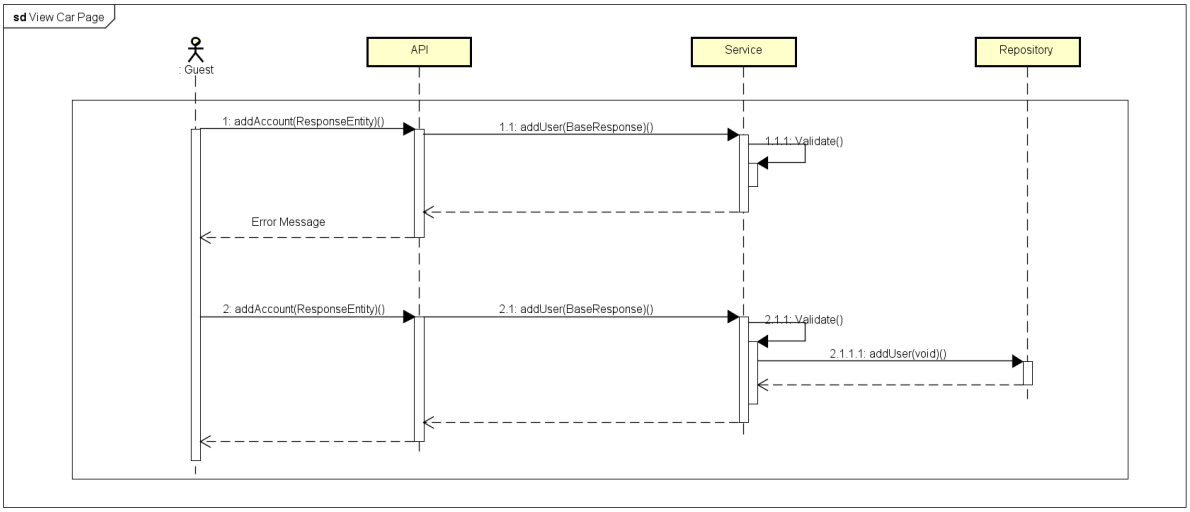



*Figure 4-60:Register sequence diagram*

## 4.3.21 User Booking Accessories

Screen Design

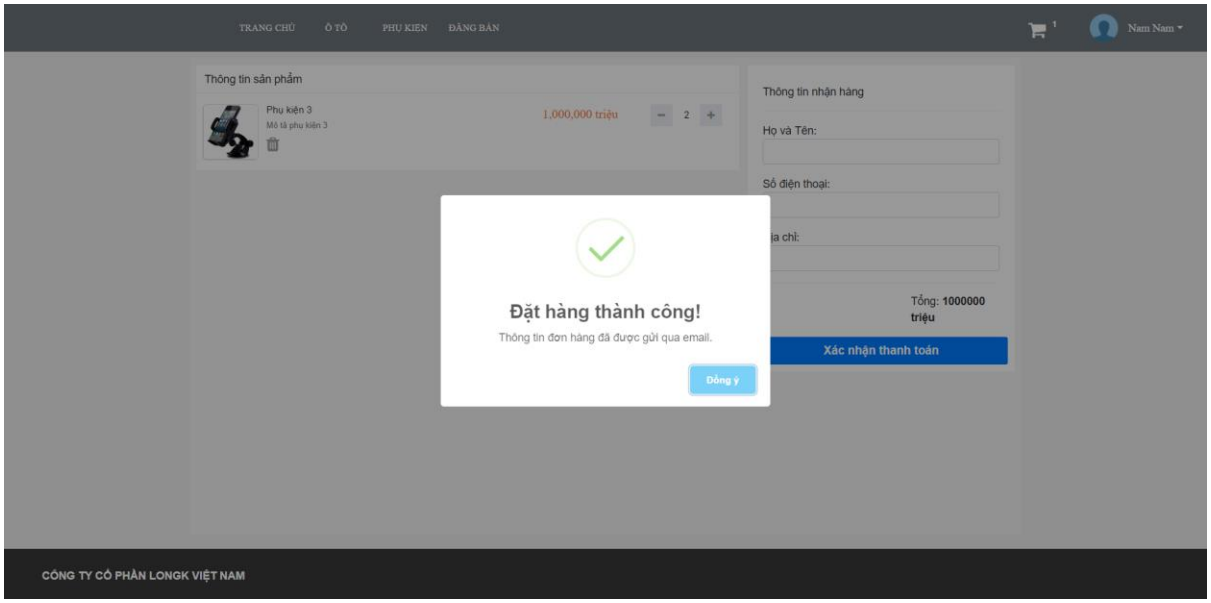



*Figure 4-61:User booking accessories*

Class Diagram

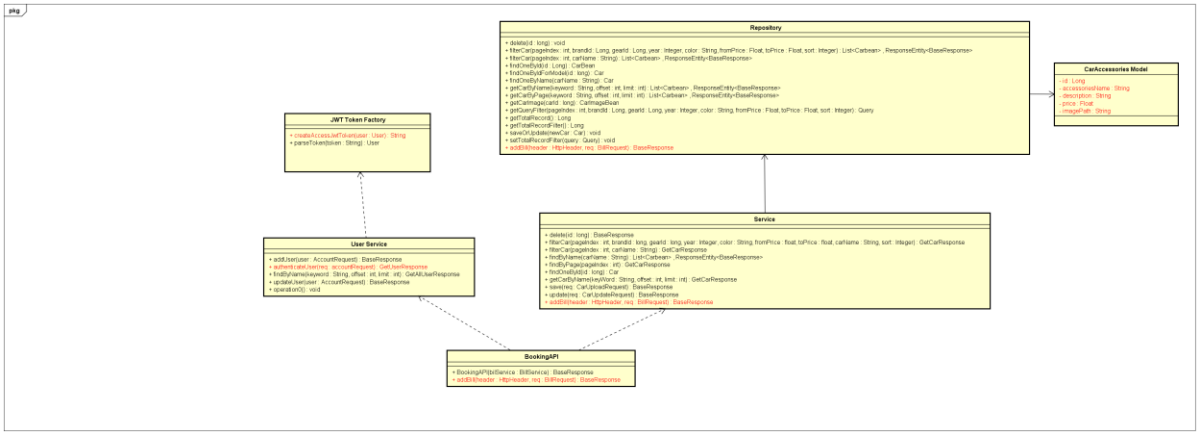



*Figure 4-62:User booking accessories class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | **\src\main\java\fpt\autofun\service** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | id | long | |
| 2 | accessoriesName | String | |
| 3 | Description | String | |
| 4 | Price | Float | |
| 5 | imagePath | String | |
| *Operation* | | | |
| getCarAccessories | | | |
| Return Type | getCarAccessories<br>Response | | |
| Parameters | Name | Type | Description |
| | pageIndex | int | |

Sequence Diagram

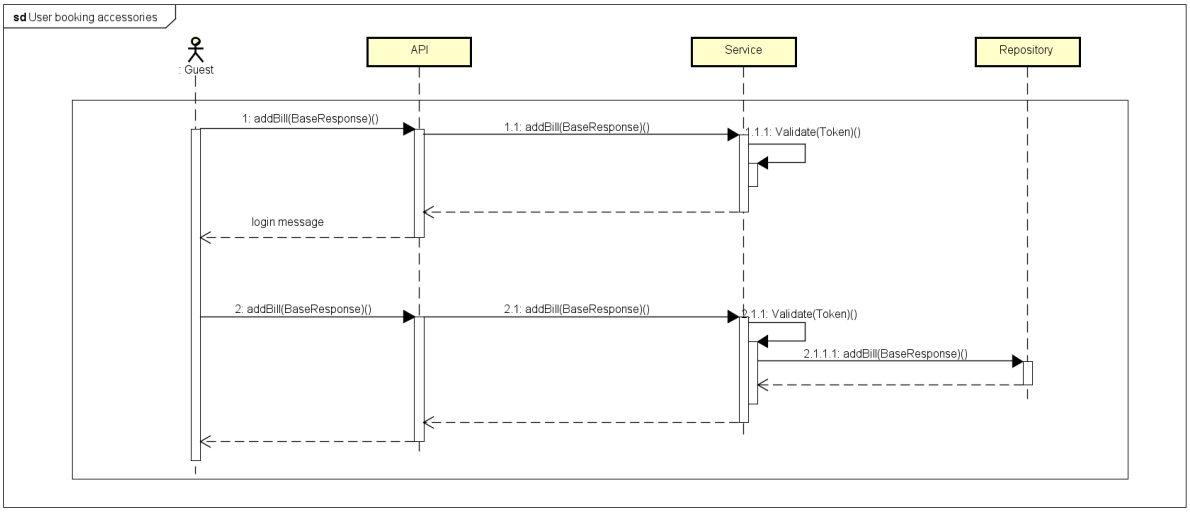



*Figure 4-63:User booking accessories sequence diagram*

## 4.3.22 User post Car

Screen Design

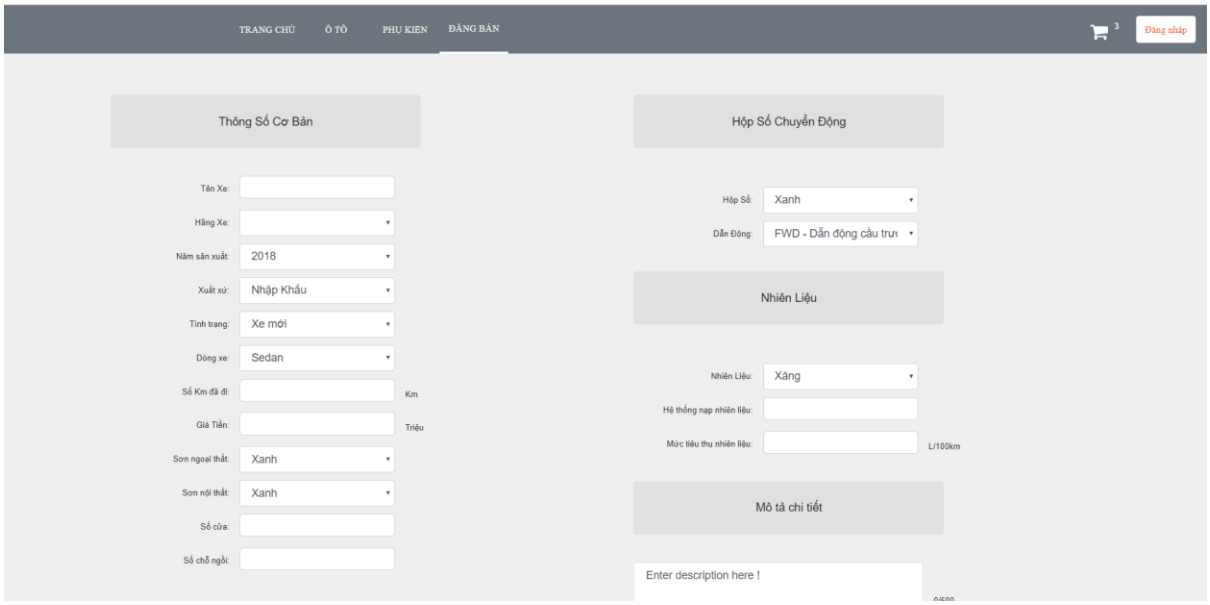



*Figure 4-64:User post car*

Class Diagram

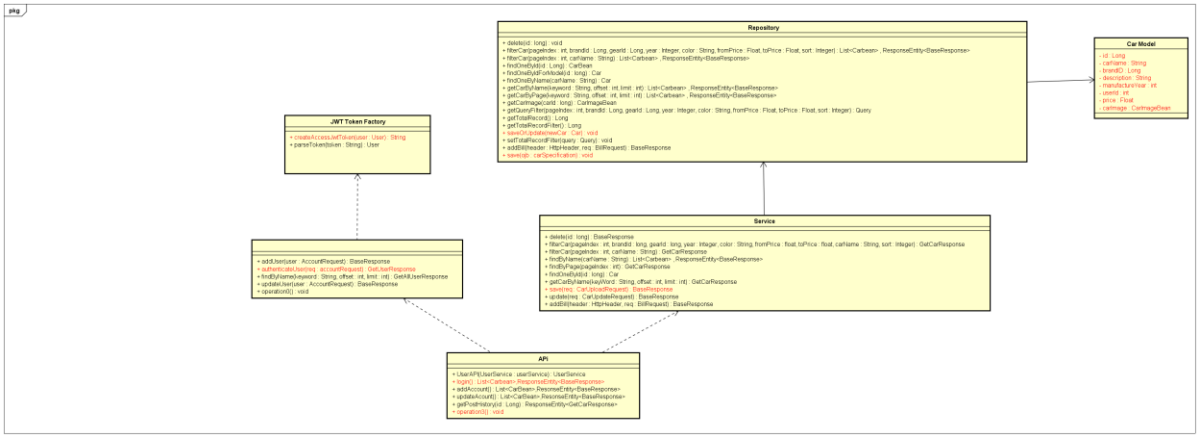



*Figure 4-65:User post car class diagram*

Class Specification

| **Car Service** | | | |
|---|---|---|---|
| Physical address | **/car** | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| 1 | id | Long | |
| 2 | carName | String | |
| 3 | brandId | Long | |
| 4 | description | String | |
| 5 | manufacture | Float | |
| 6 | carImage | String | |
| 7 | userId | Long | |
| 8 | price | Float | |
| *Operation* | | | |
| getCarspecificationByCar | | | |
| Return Type | getCarspecificationReponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadRequest | |

Sequence Diagram

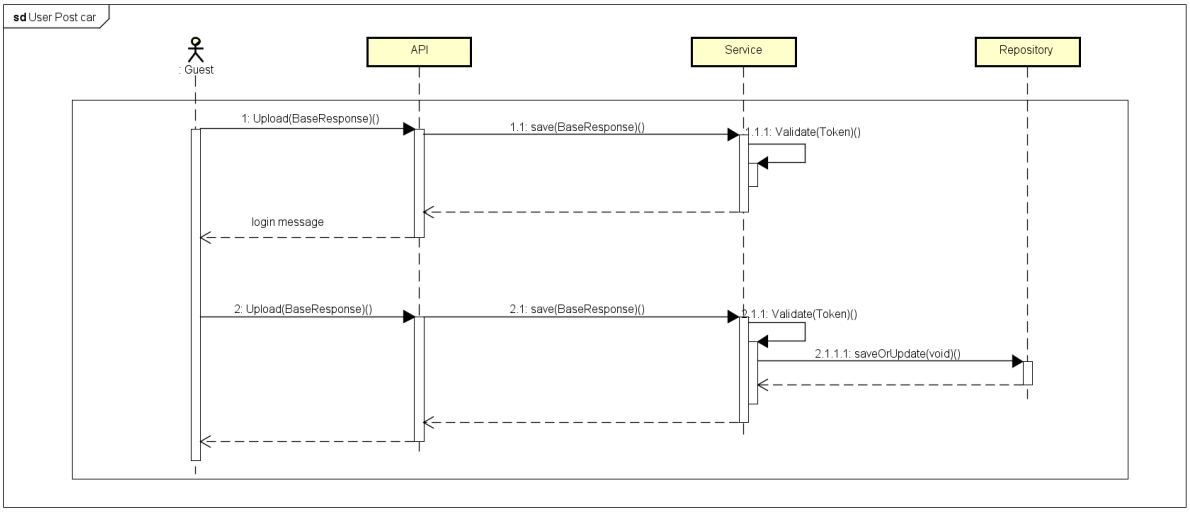



*Figure 4-66:User post car sequence diagram*

## 4.3.23 Admin Add Accessories

Screen Design

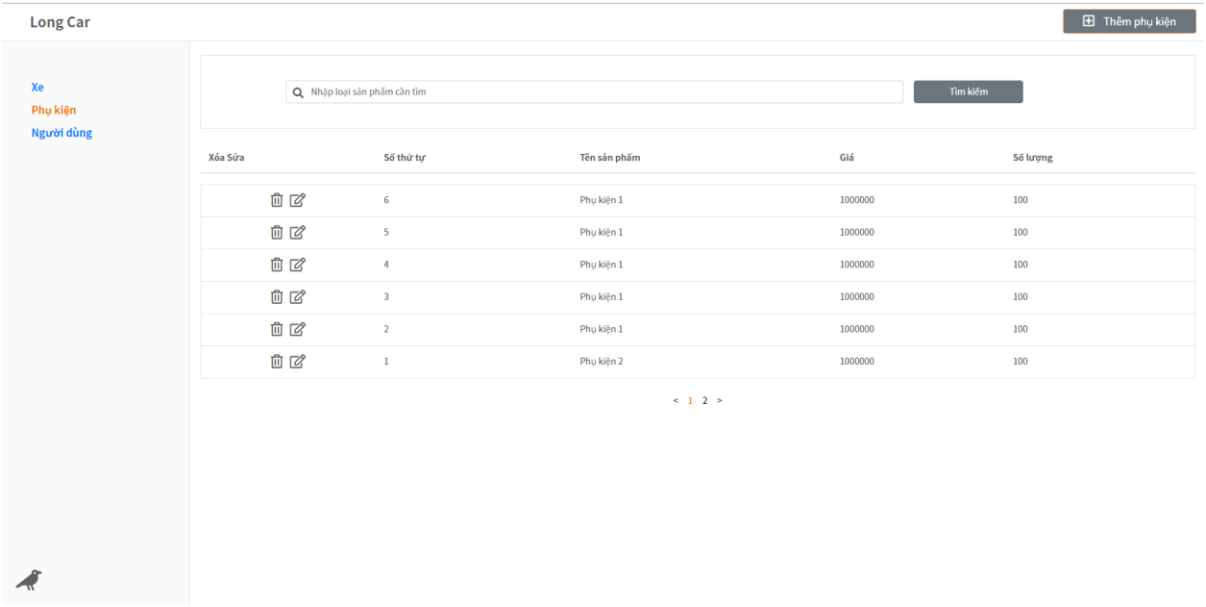



*Figure 4-67:admin Add Accessories*

Class Diagram

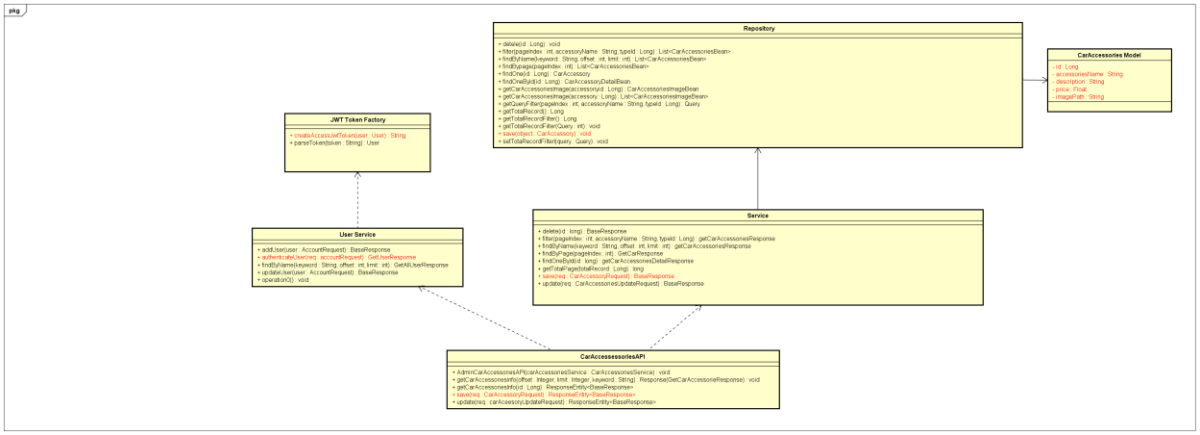



*Figure 4-68:admin Add Accessories class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |

| *Operation* | | | |
|---|---|---|---|
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **Car AccessoriesAPI** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity<BaseResponse> | | |
| Parameters | Name | Type | Description |
| | Req | CarAccessoriesRequest | |

Sequence Diagram

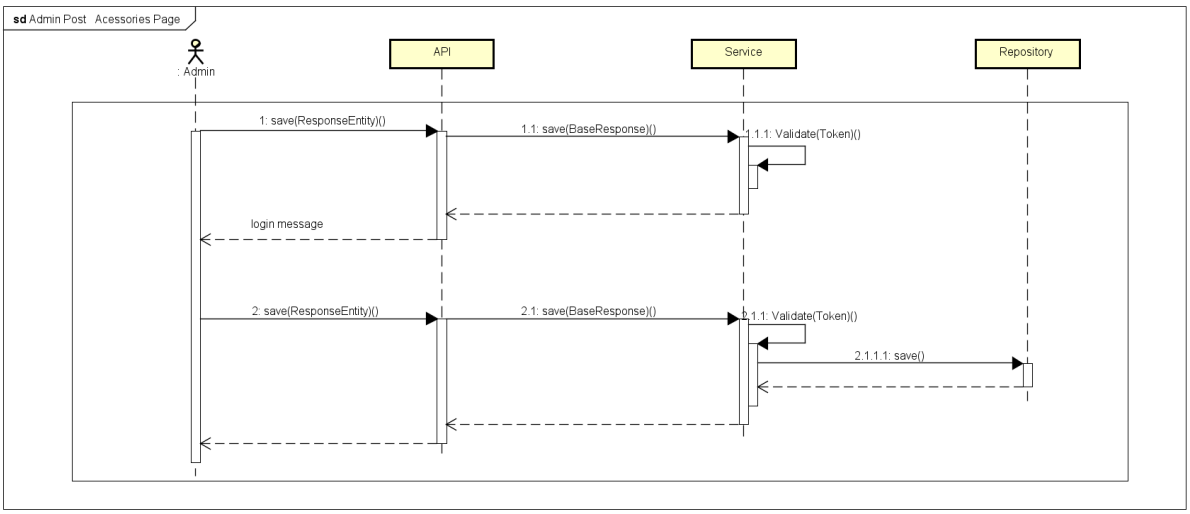


*Figure 4-69: admin Add Accessories sequence diagram*

## 4.3.24 Admin Delete Accessories

Screen Design

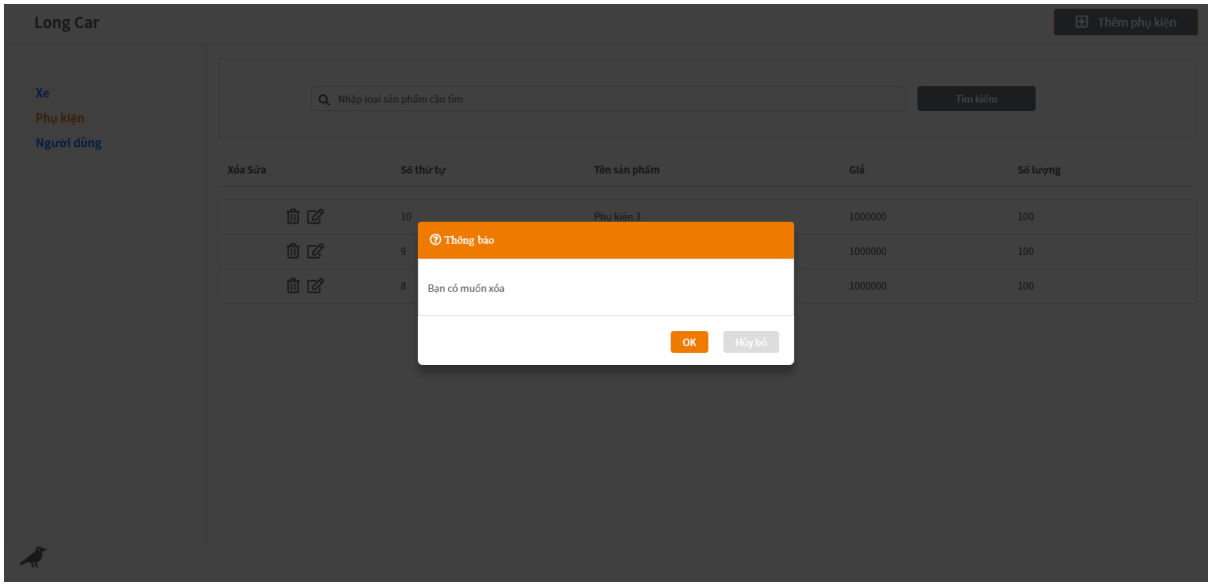



*Figure 4-70:admin Delete Accessories*

Class Diagram

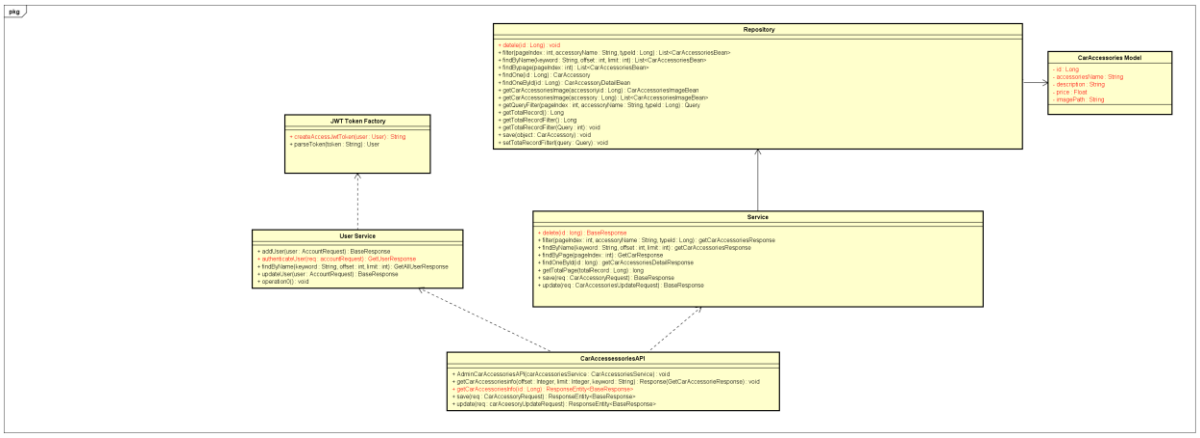



*Figure 4-71:admin Delete Accessories class diagram*

Class Specification

| Car Service | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |

| | | | |
|---|---|---|---|
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | ID | Long | |

Sequence Diagram

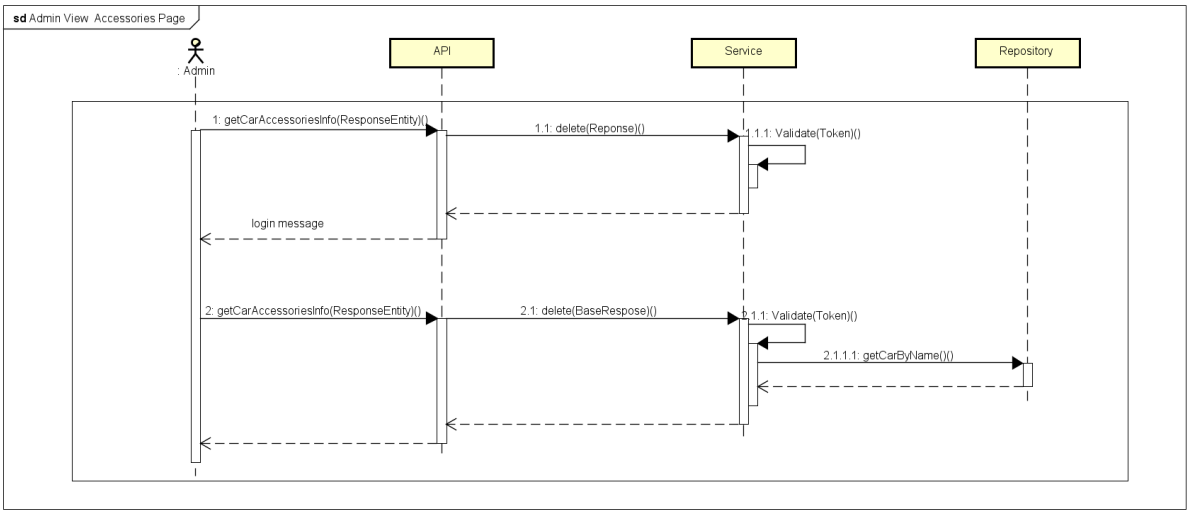



*Figure 4-72:admin Delete Accessories sequence diagram*

## 4.3.25 Admin Search Accessories

Screen Design

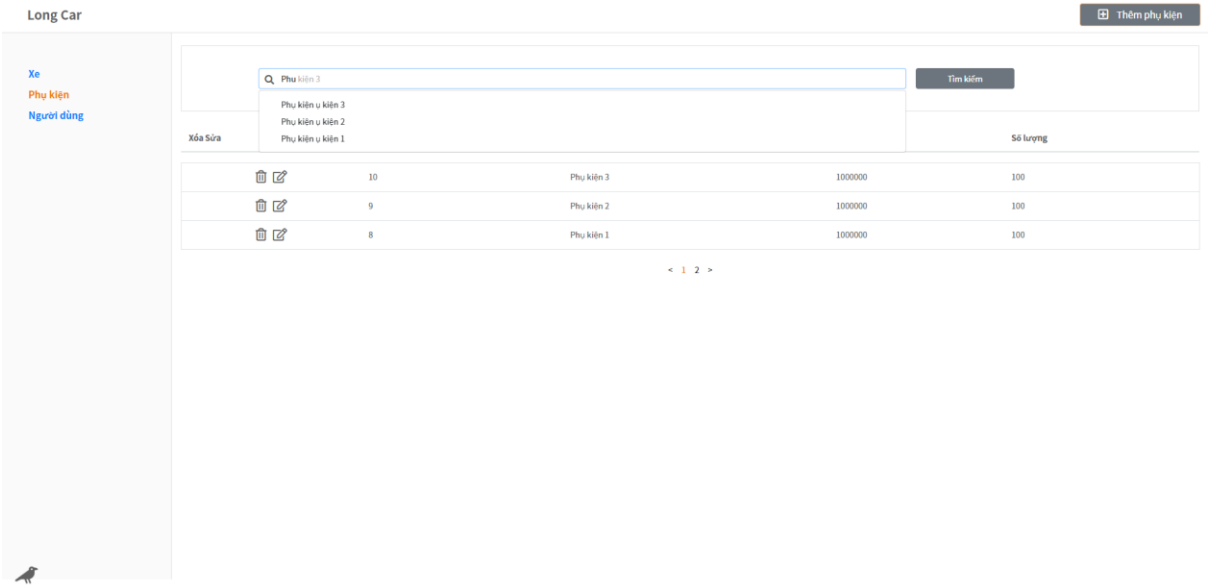


*Figure 4-73:admin Search Accessories*

Class Diagram

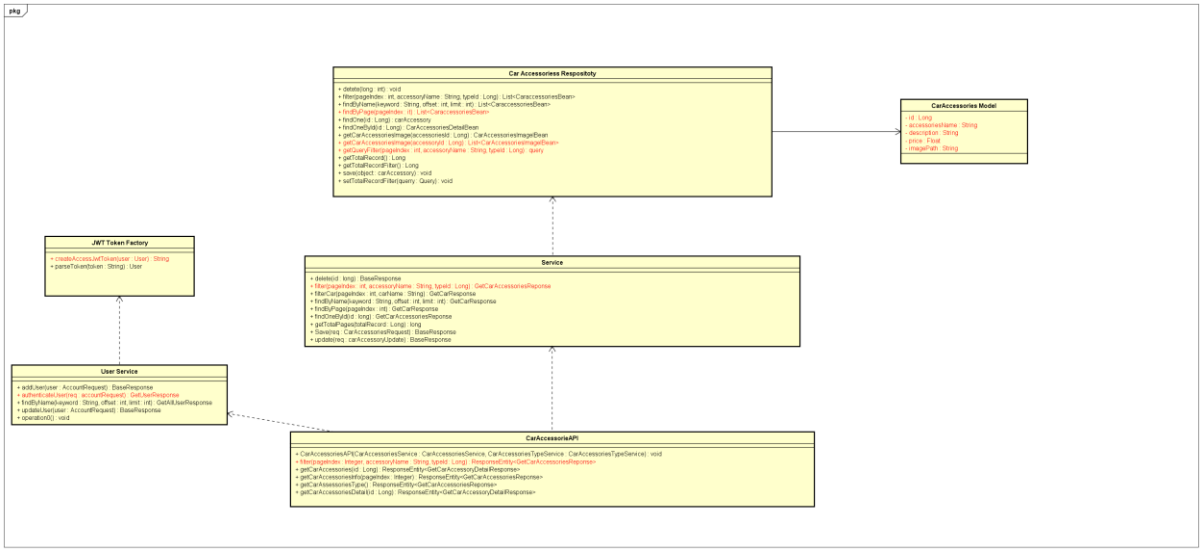



*Figure 4-74:admin Search Accessories class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | GetCarAccessoriesResponse | | |
| Parameters | Name | Type | Description |
| | PageIndex<br>AccessoriesName<br>TypeID | Int<br>String<br>Long | |

Sequence Diagram

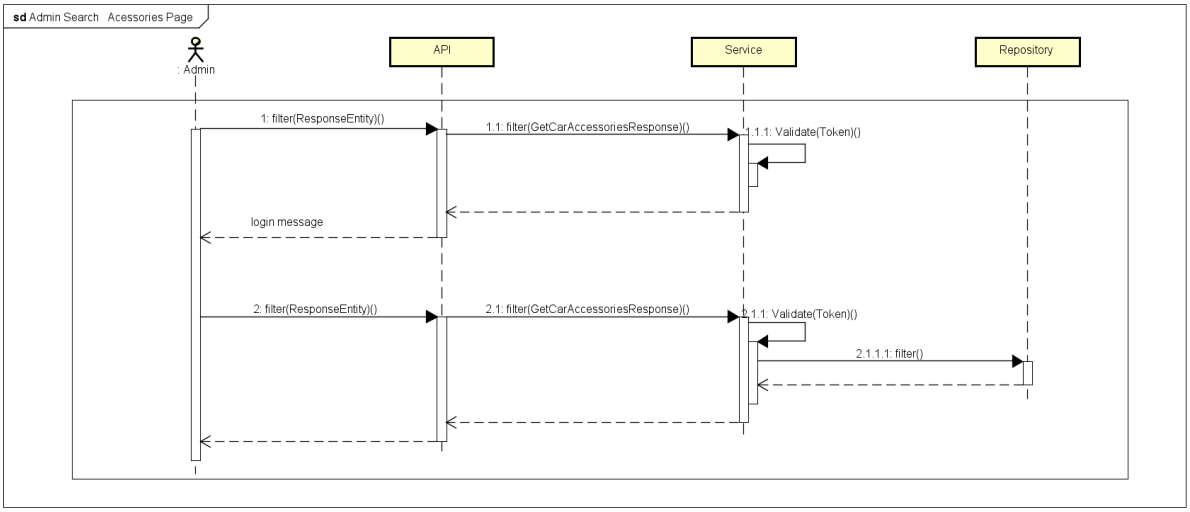



*Figure 4-75:admin Search Accessories sequence diagram*

## 4.3.26 Admin Update Accessories

Screen Design

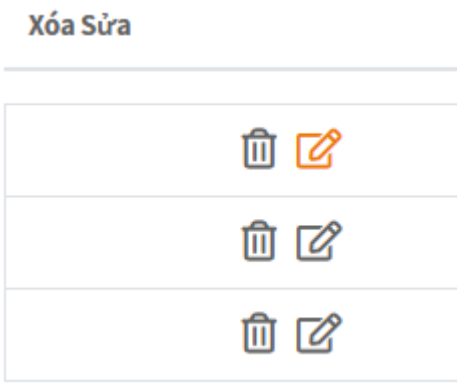



*Figure 4-76:admin Update Accessories sequence diagram*

Class Diagram

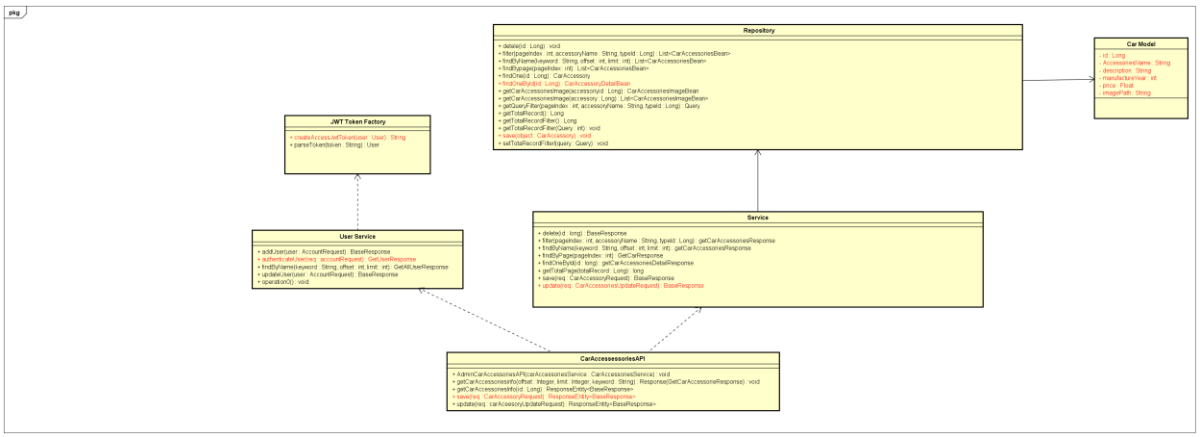



*Figure 4-77:admin Update Accessories class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseRespones | | |
| Parameters | Name | Type | Description |
| | Req | AccountResquest | |

| **Car AccessoriesAPI** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity<BaseResponse> | | |
| Parameters | Name | Type | Description |
| | Req | CarAccessoriesRequest | |

| **Service** | |
|---|---|
| Physical address | |

| Base class | | | |
|---|---|---|---|
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

Sequence Diagram

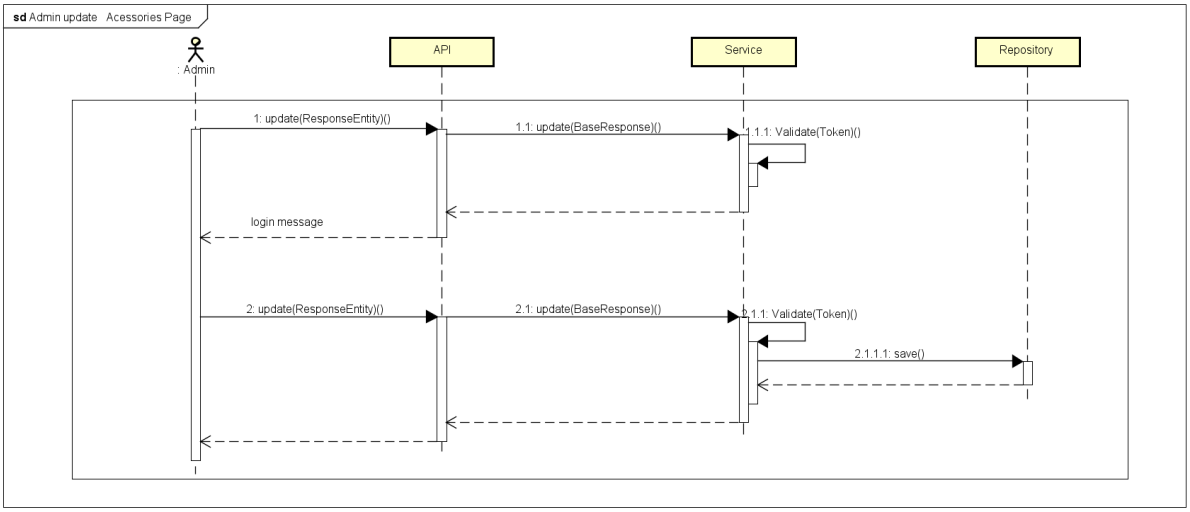



*Figure 4-78:admin Search Accessories sequence diagram*

## 4.3.27 Admin Delete Car

Screen Design

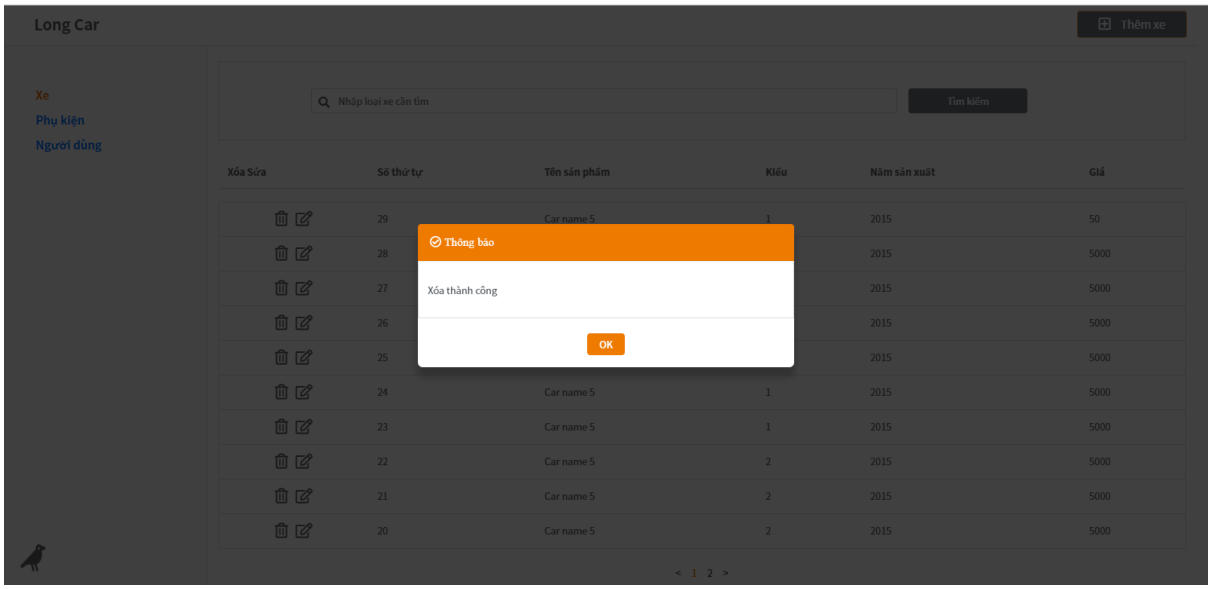



*Figure 4-82:admin Delete car*

## Class Diagram

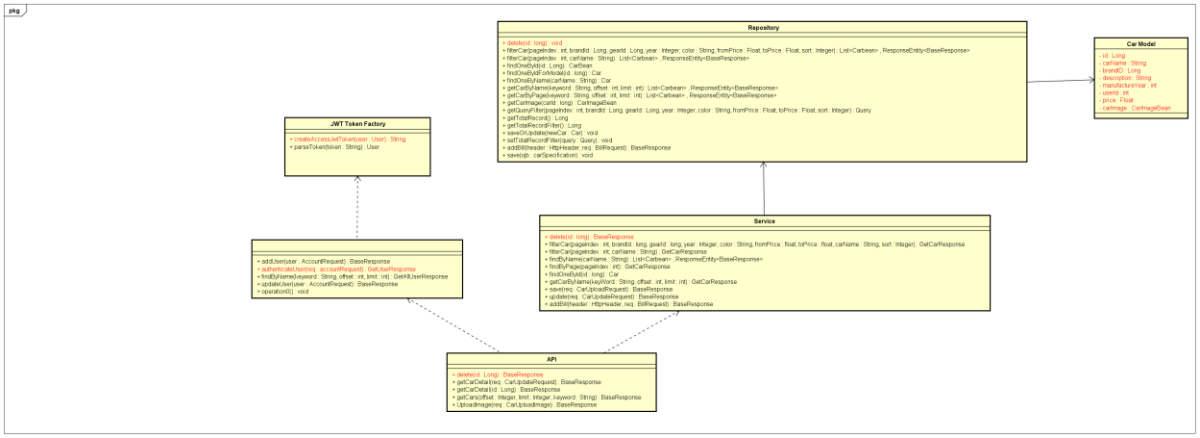



*Figure 4-83:admin Delete car class diagram*

## Class Specification

| Service | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |

| Teen function | | | |
|---|---|---|---|
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **Car AccessoriesAPI** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity<BaseResponse> | | |
| Parameters | Name | Type | Description |

| | | | |
|---|---|---|---|
| | Req | CarAccessoriesRequest | |

Sequence Diagram

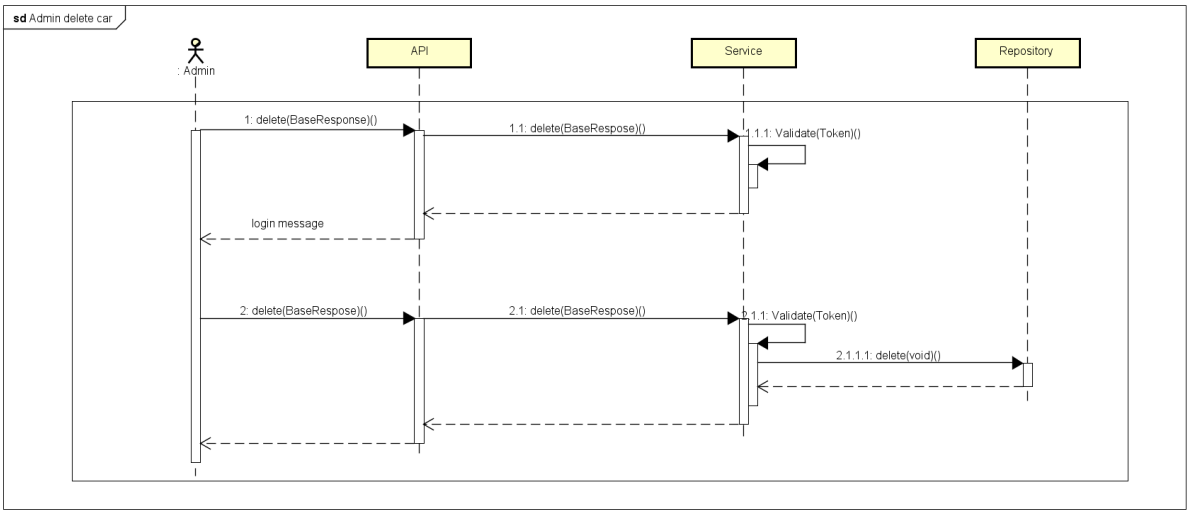



*Figure 4-84:admin Delete car sequence diagram*

## 4.3.28 Admin Post Car

Screen Design

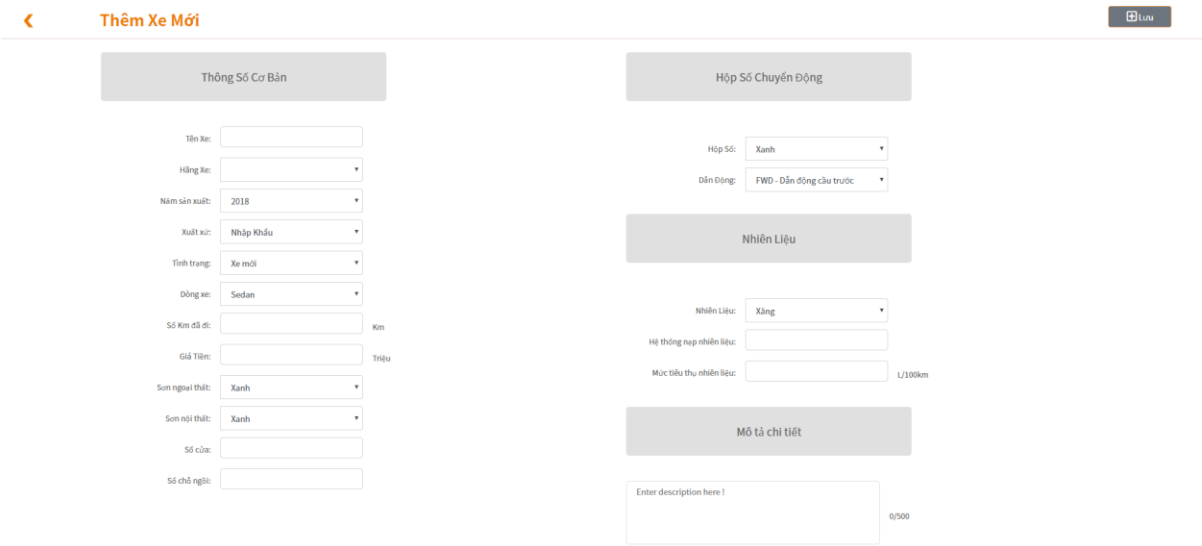



*Figure 4-85:admin Post car*

Class Diagram

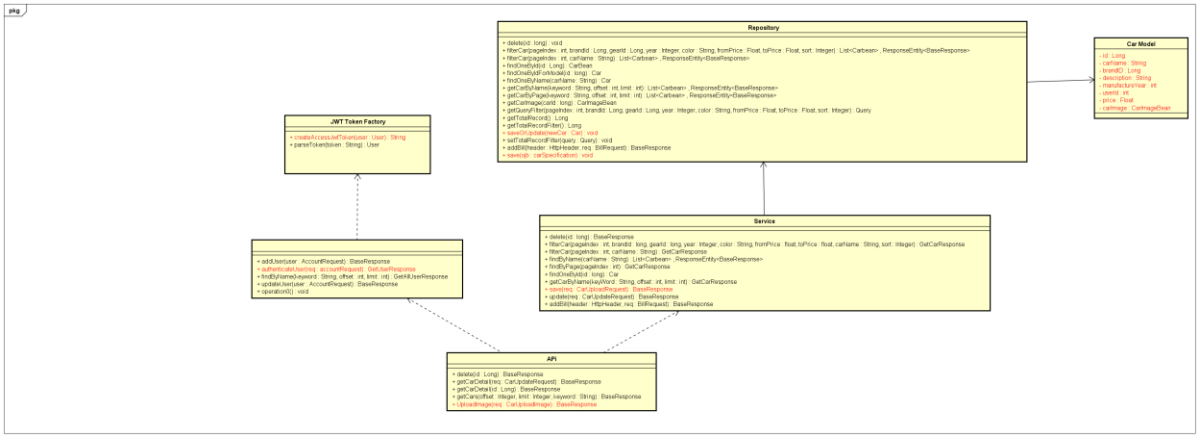



*Figure 4-86:admin Post car class diagram*

Class Specification

| | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountResques | |

| **Admin Car API** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |

| Return Type | BaseResponse | | |
|---|---|---|---|
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

| **Repository** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | void | | |
| Parameters | Name | Type | Description |
| | NewCar | Car | |

| **CarAccessoriesApi** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity | | |
| Parameters | Name | Type | Description |
| | Req | CarAccessoryRequest | |

| Admin Car API | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

Sequence Diagram

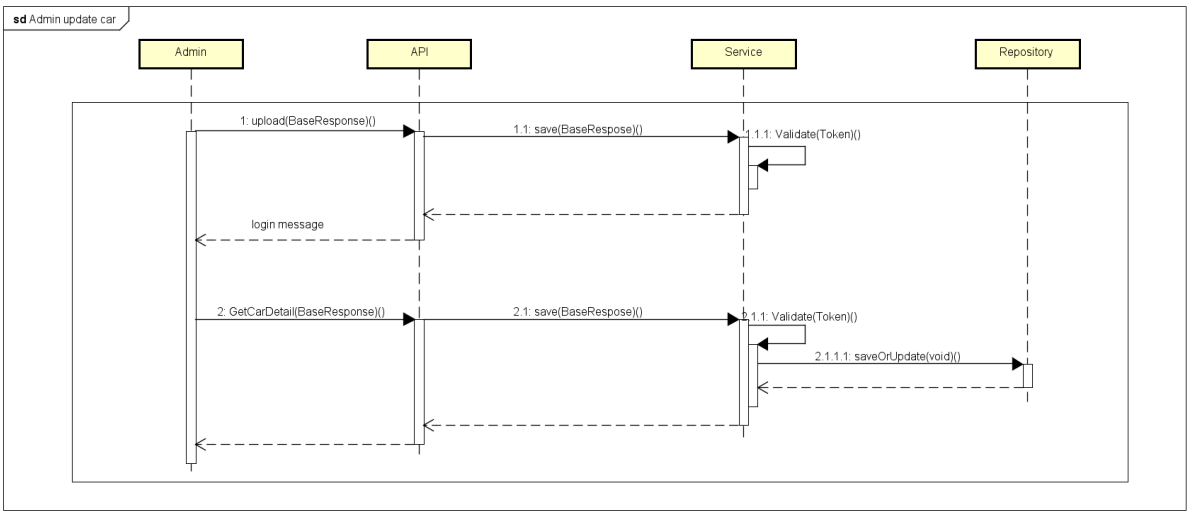



*Figure 4-87:admin Post car sequence diagram*

## 4.3.29 Admin Update Car

Screen Design

| Xóa Sửa | Số thứ tự | Tên sản phẩm |
|---|---|---|
| | 30 | Car name 5 |
| | 28 | Car name 5 |
| | 27 | Car name 5 |

*Figure 4-88:admin Update car*

Class Diagram

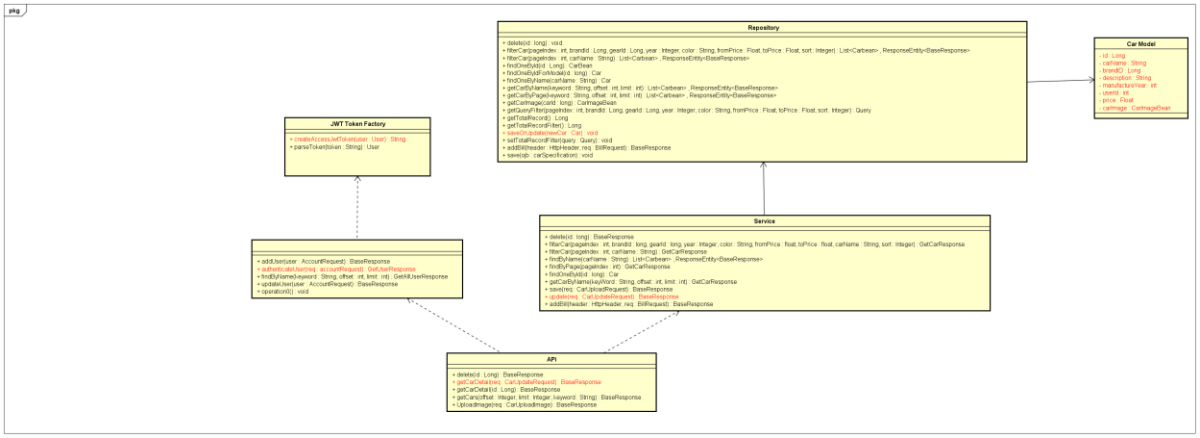



*Figure 4-89:admin Update car class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountResques | |

| Admin Car API | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

| Repository | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | void | | |
| Parameters | Name | Type | Description |
| | NewCar | Car | |

| CarAccessoriesApi | |
|---|---|
| Physical address | |

| Base class | | | |
|---|---|---|---|
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity | | |
| Parameters | Name | Type | Description |
| | Req | CarAccessoryRequest | |

| **Admin Car API** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

Sequence Diagram

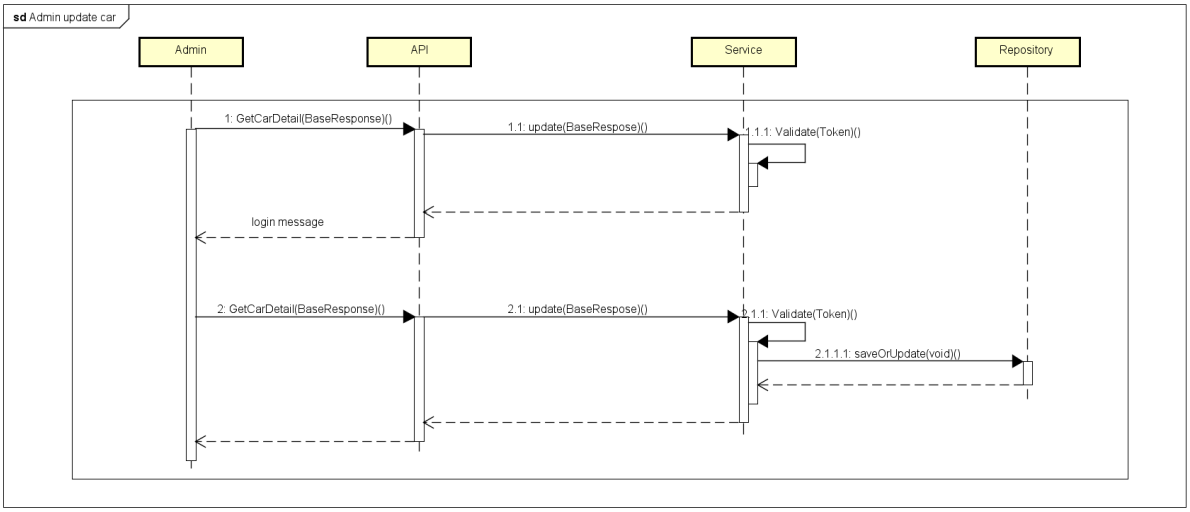



*Figure 4-90:admin Update car sequence diagram*

## 4.3.30 Admin Detail

Screen Design

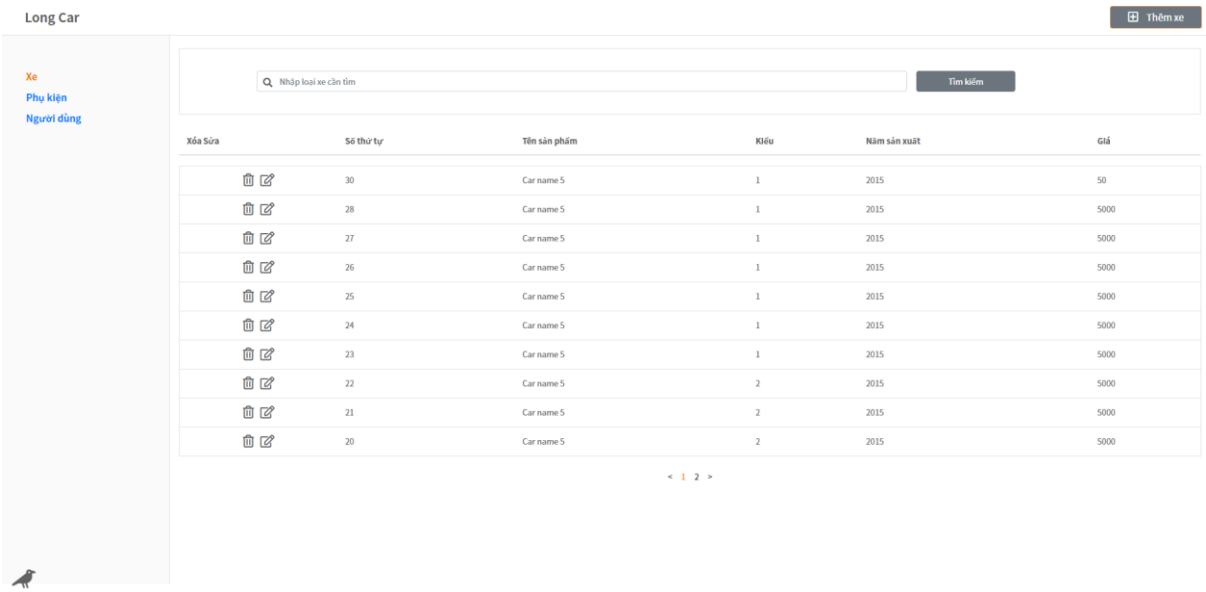



*Figure 4-91:admin Detail*

Class Diagram

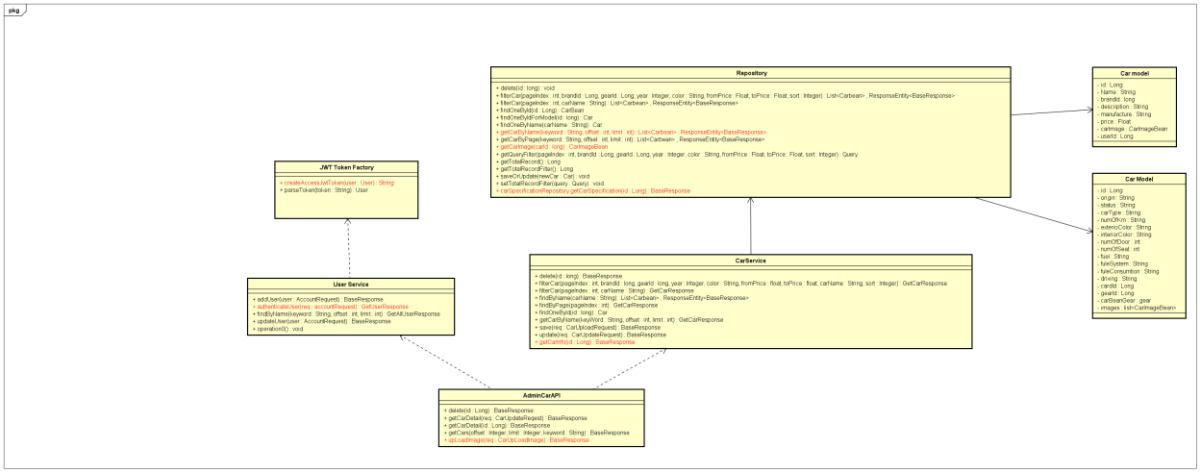



*Figure 4-91:admin Detail class diagram*

Class Specification

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountResques | |

| **Admin Car API** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |

| *Operation* | | | |
|---|---|---|---|
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

| **Repository** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | void | | |
| Parameters | Name | Type | Description |
| | NewCar | Car | |

| **CarAccessoriesApi** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity | | |

| Parameters | Name | Type | Description |
|---|---|---|---|
| | Req | CarAccessoryRequest | |

| **Admin Car API** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUploadImage | |

Sequence Diagram

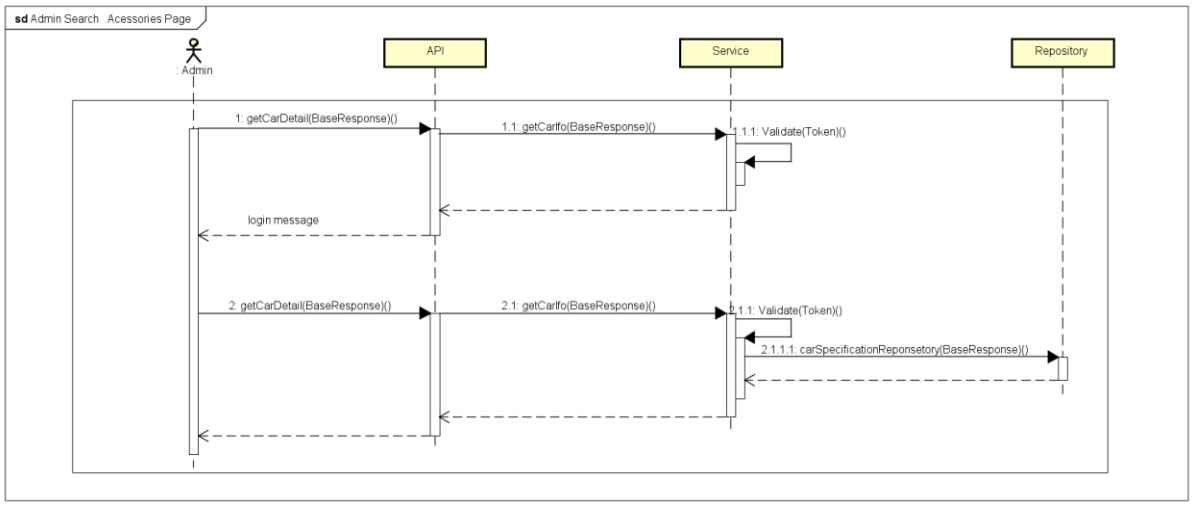



*Figure 4-92:Admin Detail sequence diagram*

### 4.3.31 Admin view Admin Page

Screen Design

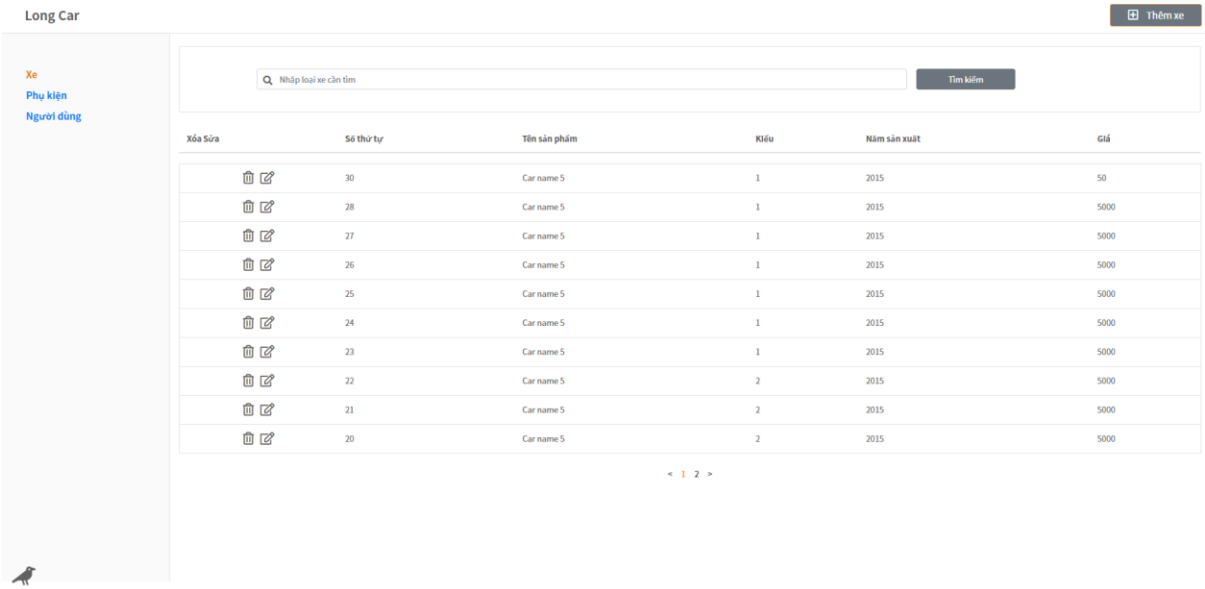



*Figure 4-93:Admin view admin page*

## Class Diagram

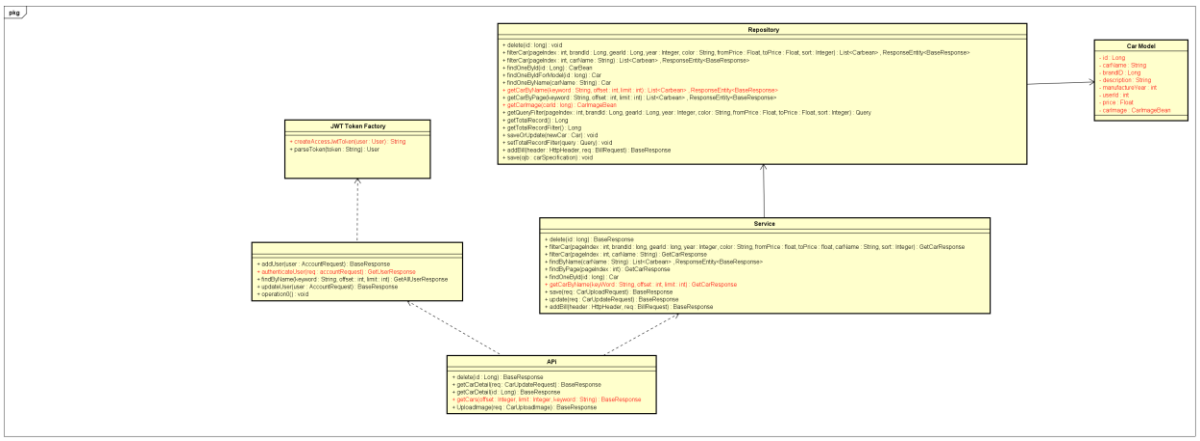



*Figure 4-94:Admin view admin page class diagram*

## Class Specification

| User Service | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| Attributes | | | |
| No | Name | Type | Description |
| 1 | | | |
| Operation | | | |
| | | | |

| Return Type | | | |
|---|---|---|---|
| Parameters | Name | Type | Description |
| | Req | AccountResquest | |

| Car Service | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| Attributes | | | |
| No | Name | Type | Description |
| | | | |
| Operation | | | |
| | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | CarUpdateRequets | |

| User Repository | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| Attributes | | | |
| No | Name | Type | Description |
| | | | |
| Operation | | | |
| | | | |
| Return Type | void | | |

| Parameters | Name | Type | Description |
|---|---|---|---|
| | Car | Newcar | |

Sequence Diagram

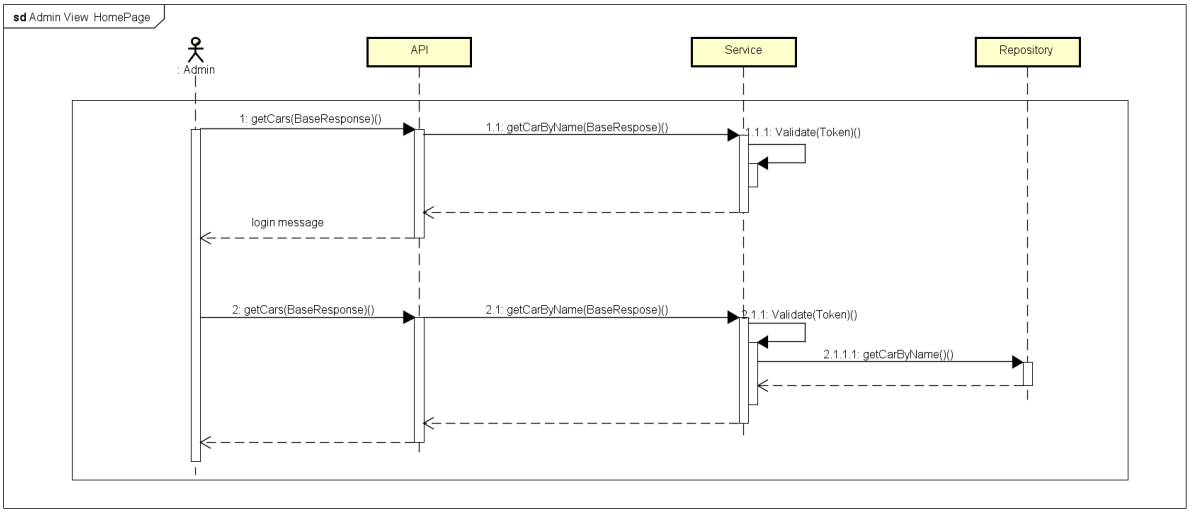



*Figure 4-95:Admin view admin page sequence diagram*

## 4.3.32 View Accessories Admin Page

Screen Design

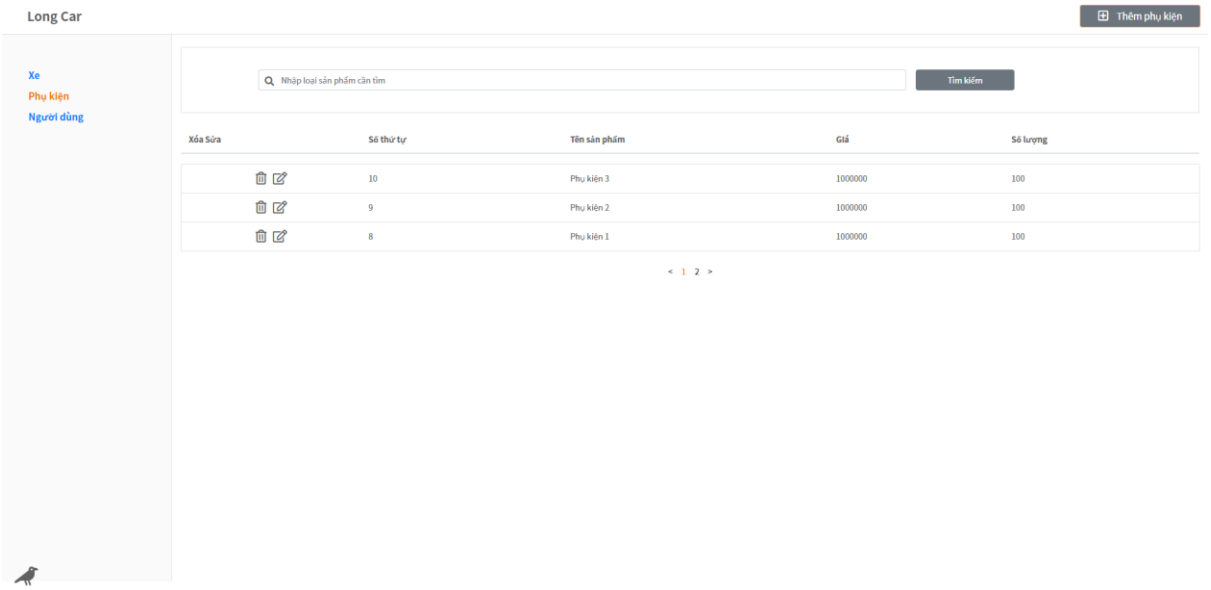



*Figure 4-96: View accessories admin page*

Class Diagram

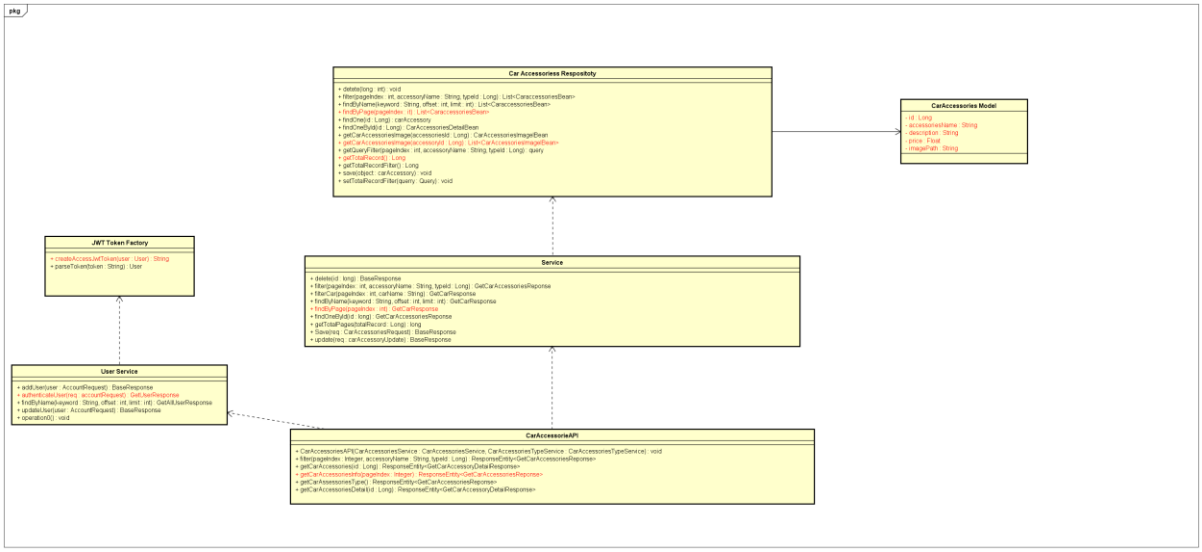



*Figure 4-97: View accessories admin page class diagram*

Class Specification

| **Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **User Service** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |

| | | | |
|---|---|---|---|
| *Operation* | | | |
| Teen function | | | |
| Return Type | BaseResponse | | |
| Parameters | Name | Type | Description |
| | Req | AccountRequest | |

| **Car AccessoriesAPI** | | | |
|---|---|---|---|
| Physical address | | | |
| Base class | | | |
| *Attributes* | | | |
| No | Name | Type | Description |
| | | | |
| *Operation* | | | |
| Teen function | | | |
| Return Type | ResponseEntity<BaseResponse> | | |
| Parameters | Name | Type | Description |
| | Req | CarAccessoriesRequest | |

Sequence Diagram

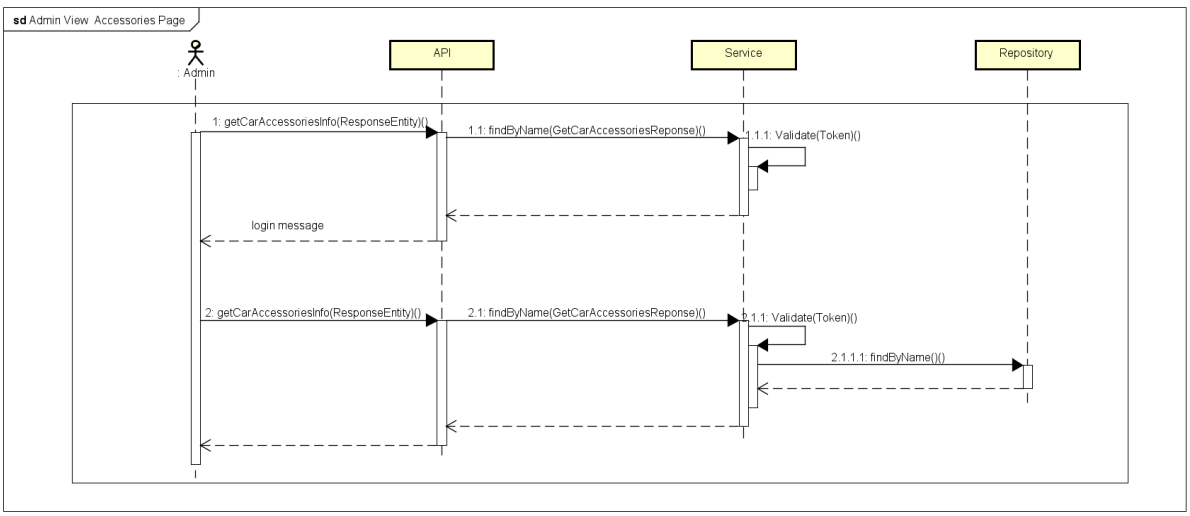


*Figure 4-98: View accessories admin page sequence diagram*

# Chapter 5 : Software Testing Documentation

## 5.1 Introduction

### 5.1.1 Purpose

This chapter provides details related to testing of the system, including the scope of testing, test plan, test case and test report. Through testing, system defects can be discovered and fixed, allowing for better software quality. Using the contents of this chapter, developers and testers can work together to ensure that the system is thoroughly tested and functions as intended.

### 5.1.2 Scope of Testing

5.1.2.1 Test Levels

The following four levels of testing are performed: Unit Test, Integration Test, System Test and Acceptance Test.

| Test Level | Description |
|---|---|
| Unit Test | Unit Tests are performed to ensure that individual units of source code function as intended. Unit tests are primarily used to test complicated algorithms and automatically test important functions.<br>JUnit is used to perform unit tests. |
| Integration Test | Integration Tests are performed to test whether combined units function as intended. |
| System Test | System Tests are performed to test the system as a whole and determine whether or not the system meets the requirements. No specific tools are used for system tests. |
| User Acceptance Test | User Acceptance Test is performed both by team members and other people in order to further ensure that the system works as intended, meets both functional and non-functional requirements, making sure that users feel comfortable when using the system. In addition, Acceptance Test helps to discover bugs and issues not found in the other types of tests. |

Table 5-1: Test Levels

5.1.2.2 Types of Testing

The following types of testing are performed:

Unit testing

Regression testing

Function testing

GUI testing

5.1.2.3 Range of Testing

All features as described in the Software Requirement Specification are tested.

# 5.2 Test plan

## 5.2.1 Testing Tools and Environment

### 5.2.1.1 Testing Tools

The following tools are used for testing:

| Tool | Purpose |
|---|---|
| Google Chrome v62.0 | Use to view the web page, bug logging page, etc.. |
| Mozilla Firefox v57.0 | Use to view the web page, bug logging page, etc.. |
| Excel 2013 | Use to log bugs. |
| JUnit | Use to perform unit tests. |
| Postman | Use to test API |

*Table 5-2: Testing tools*

### 5.2.1.2 Testing Environment

The following table describes the testing environment:

| Type of Testing | Software | Hardware |
|---|---|---|
| Unit Test | Spring tool suite<br>JUnit | Windows 10 Edu<br>Intel Core i5 1.8 Ghz<br>8GB Memory |

| | | |
|---|---|---|
| | | 500GB Disk Space |
| Integration Test<br>System Test | Spring tool suite<br>Google Chrome v68.0<br>Mozilla Firefox v57.0 | Windows 10 Edu<br>Intel Core i5 1.8 Ghz<br>8GB Memory<br>500GB Disk Space<br>Screen resolution: 1920 x 1080 |
| User Acceptance Test | The same as System Test, along with other software and hardware that are different for each user that participated. This includes a wide variety of software and hardware, ranging from PCs running on different types of operating systems to many different mobile devices. A full list of software and hardware used by all testers who participated in User Acceptance Test is not known and not publicly disclosed. | |

*Table 5-3: Testing environment*

### 5.2.2 Resources and Responsibilities

### 5.2.3 Test Strategy

#### 5.2.3.1 Test Model

This project follows the V-Model for testing.

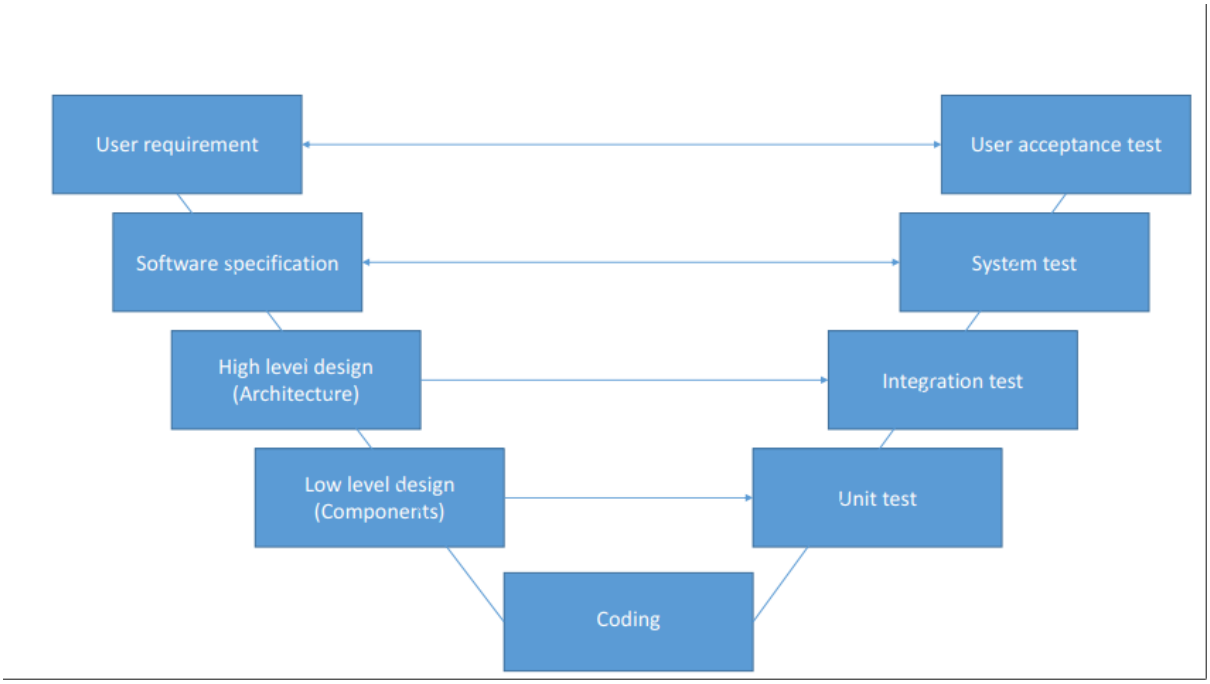



*Figure 5-1: V-Model (Rex Black, Dorothy Graham, Erik van Veenendaal. Foundations of Software Testing: ISTQB Certification. Google Books. 2012)*

5.2.3.2 Test Types

The following types of testing are performed:

Unit testing o Tests individual methods to ensure that they work correctly. o Unit test has to reach 100% code coverage for the units that are tested.

Unit testing is primarily used to test complicated functions and algorithms.

Regression testing o Regression testing ensures that bugs are fixed. o Regression testing ensures that the fixing of a bug does not impact other parts of the system.

Functional testing o Tests whether or not functions of the system work as intended. o As opposed to unit testing, functional testing is a type of black-box testing and as such, internal structure is rarely considered.

GUI testing o Tests whether or not elements of the websites appear and work correctly, and are positioned correctly. o Tests whether or not the website is usable and appears as intended with various browser sizes:

- 1920 x 1080
- 1440 x 900
- 1366 x 768
- 800 x 600

Checks whether fonts appear correctly and are readable. o Checks whether images appear correctly.

### 5.2.3.3 Test Levels

| **Test Type** | **Test Level** | | | |
|---|---|---|---|---|
| | **Unit Test** | **Integration Test** | **System Test** | **User Acceptance Test** |
| Unit testing | ✓ | ✓ | ✓ | |
| Regression testing | ✓ | ✓ | ✓ | |
| Functional testing | | ✓ | ✓ | ✓ |
| GUI testing | | | ✓ | ✓ |

*Table 5-5: Test levels*

| **Test Schedule** | **Start Date** | **End Date** |
|---|---|---|
| *Iteration 1* | | |
| Unit testing | 2018/08/16 | 2018/08/18 |
| Functional testing | 2018/08/17 | 2017/08/20 |
| GUI testing | 2018/08/18 | 2017/08/19 |

| | | |
|---|---|---|
| *Iteration 2* | | |
| Unit testing | 2018/08/23 | 2018/08/25 |
| Functional testing | 2018/08/23 | 2018/08/25 |
| GUI testing | 2018/08/25 | 2018/08/27 |
| Acceptance testing | 2018/08/24 | 2018/08/18 |

*Table 5-6: Test schedule*

#### 5.2.3.4 Deliverables

| **Deliverables** | **Responsibilities** | **Completion date** |
|---|---|---|
| Test plan | Test Leader | 2018/07/10 |
| Test cases | Testers | 2018/08/01 |
| Test case review | Testers + Project Manager | 2018/08/10 |
| Test data | Testers | 2018/08/02 |
| Defect report | All members | 2018/08/11 |
| Final test report | Project Manager | 2018/08/27 |

*Table 5-7: Deliverables*

### 5.2.4 Features to be Tested

All features described in the list of use cases in the Software Requirement Specification are to be tested.

## 5.3 Test Approach

### 5.3.1 Unit testing with Junit

Unit tests are done by the developers in order to test individual units, and approved by the test leader.

This is done to ensure that important functions, such as algorithms for verb conjugation and name conversion, return the correct values.

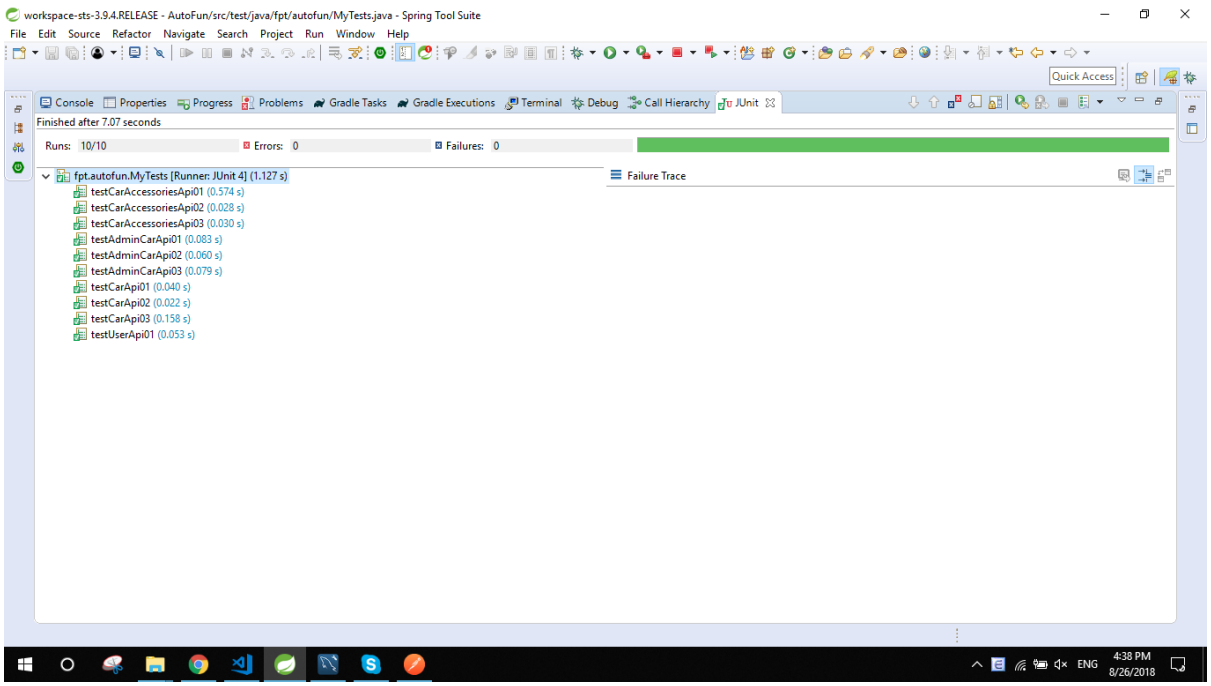



*Figure 5-2: Unit testing with JUnit*

### 5.3.2 Integration and System Test

Integration and system tests are done by testers to ensure that combined units work correctly and that the system as a whole functions as intended. Each test case is tested using Google Chrome v68.0 and Mozilla Firefox v57.0.

GUI testing is also done during this process to ensure that elements load and function correctly, text is readable, and the website interface looks good in various browser sizes.

Test case details are described in the file **LKT_TestReport.xlsx**.

### 5.3.3 User Acceptance Testing

User Acceptance testing is done both by the development team and by other people, primarily the supervisor and friends of the development team. Testers who participate in User Acceptance testing use a wide variety of software and hardware, often without any guidance. This ensures that the system not only works correctly but is also good in terms of usability, as in, the website interface is simple, features are easy to find, and it is easy to use the system without reading instructions or asking other people how to use it. User Acceptance testing also helps to discover bugs that are not found in the other test levels, as testers usually use the system freely without following any testing guidelines, it is easy to come across bugs that the development team is prone to miss.

In addition, two people with color blindness also participate in User Acceptance testing to ensure that the colors used in the website are colorblind-friendly.

The development team uses the following checklist as a guideline for User Acceptance testing:

| ID | Checklist | Yes | No |
|---|---|---|---|
| | **General** | | |
| CL-001 | Text does not have grammatical and spelling errors. | | |
| CL-002 | All buttons are functional. | | |
| CL-003 | All mandatory fields are validated and indicated by asterisk (*) symbol. | | |
| CL-004 | All error messages are displayed using red color. | | |
| CL-005 | All "Delete" functions ask for confirmation. | | |
| CL-006 | All numeric values are formatted properly. | | |
| | **GUI and Usability** | | |
| CL-007 | Screen design follows the standard of the project. | | |
| CL-008 | Features are intuitive and easy to use. | | |
| CL-009 | All important features can be easily accessed. | | |
| CL-010 | All fields on page (e.g. text box, radio options, dropdown lists) are aligned properly. | | |
| CL-011 | Information is presented in the order that the user needs it. | | |
| CL-012 | The static text is clear, concise, and meaningful. | | |
| CL-013 | Use abbreviation only when space is limited. | | |
| CL-014 | All icons and elements are flat. | | |
| | **Database** | | |
| CL-015 | Correct data is getting saved in database upon successful page submit. | | |
| CL-016 | Values columns are not accepting null values. | | |
| CL-017 | Radio button options are saved correctly in database. | | |
| CL-018 | Database fields are designed with correct data type and data length. | | |
| CL-019 | Null values are not allowed for Primary key column. | | |
| | **Security** | | |
| CL-020 | Password is encrypted. | | |

| | | | |
|---|---|---|---|
| CL-021 | Cannot obtain sensitive personal information using the user's Inspector (F12) function. | | |
| CL-022 | Cannot perform unintended actions that negatively affect security using the user's Inspector (F12) function. | | |

*Table 5-8: checklist as a guideline for User Acceptance testing*

### 5.3.4 User Acceptance Testing

**Bug list**

| Defect ID | Module | Description | Type | Severity | Priority | Status | Created Date |
|---|---|---|---|---|---|---|---|
| LKD | Title section | *<Describe defect and expected result>*<br>Ex: Default value of Status field = blank is incorrect<br>Expected result: when create new document, default value of [Status] field must be "Open" | *<Type of defect>*<br>Ex:<br>User Interface | *<Severity of defect>*<br>Ex:<br>Cosmetic | *<Priority of defect>*<br>Ex:<br>High | *<Status of defect>*<br>Ex:<br>Assigned | *<Created date of defect>*<br>Ex:<br>21-Sep-2008 |
| LKD-1 | Search | *link api does not exist*<br>*Expected result: Call api successful* | *Coding Logic* | *Medium* | *High* | | *15-07-2018* |
| LKD-2 | Search | *query sql not true , return error code 500*<br>*Expected result: when input data then click search "button" output data must be display all item of text in search field from database .* | *Coding Logic* | *Medium* | *High* | | *15-07-2018* |
| LKD-3 | Search | *business logic is not true , return fail result*<br>*Expected result: display the correct ressult by name* | *Business Logic* | *Serious* | *High* | | *15-07-2018* |
| LKD-4 | Filter car | *link api does not exist*<br>*Expected result: Call api successful* | *Coding Logic* | *Medium* | | | |
| LKD-5 | Filter by brand | *query sql not true , does not return result by brand*<br>*Expected result: when input data then click button "Filter" output data must be display all item from database by brand .* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LDK-6 | Filter by Year | *query sql not true , does not return result by year*<br>*Expected result: when input data then click button "Filter" output data must be display all item from database by year.* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LKD-7 | Filter by gear | *query sql not true , does not return result by gear*<br>*Expected result: when input data then click button "Filter" output data must be display all* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LKD-8 | Filter by Sort | query sql not true , does not return result by sort<br>Expected result: when click list view sort then display all data of sort in sql, then click button "Filter", then display all item of sort | *Coding Logic* | *Medium* | *High* | | 15/7/2018 |

*Figure 5-3: Bug list*

**Bug list**

| Defect ID | Module | Description | Type | Severity | Priority | Status | Created Date |
|---|---|---|---|---|---|---|---|
| LKD | Title section | *<Describe defect and expected result>*<br>Ex: Default value of Status field = blank is incorrect<br>Expected result: when create new document, default value of [Status] field must be "Open" | *<Type of defect>*<br>Ex:<br>User Interface | *<Severity of defect>*<br>Ex:<br>Cosmetic | *<Priority of defect>*<br>Ex:<br>High | *<Status of defect>*<br>Ex:<br>Assigned | *<Created date of defect>*<br>Ex:<br>21-Sep-2008 |
| LKD-1 | Search | *link api does not exist*<br>*Expected result: Call api successful* | *Coding Logic* | *Medium* | *High* | | *15-07-2018* |
| LKD-2 | Search | *query sql not true , return error code 500*<br>*Expected result: when input data then click search "button" output data must be display all item of text in search field from database .* | *Coding Logic* | *Medium* | *High* | | *15-07-2018* |
| LKD-3 | Search | *business logic is not true , return fail result*<br>*Expected result: display the correct ressult by name* | *Business Logic* | *Serious* | *High* | | *15-07-2018* |
| LKD-4 | Filter car | *link api does not exist*<br>*Expected result: Call api successful* | *Coding Logic* | *Medium* | | | |
| LKD-5 | Filter by brand | *query sql not true , does not return result by brand*<br>*Expected result: when input data then click button "Filter" output data must be display all item from database by brand .* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LDK-6 | Filter by Year | *query sql not true , does not return result by year*<br>*Expected result: when input data then click button "Filter" output data must be display all item from database by year.* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LKD-7 | Filter by gear | *query sql not true , does not return result by gear*<br>*Expected result: when input data then click button "Filter" output data must be display all* | *Coding Logic* | *Medium* | *High* | | *15/7/2018* |
| LKD-8 | Filter by Sort | query sql not true , does not return result by sort<br>Expected result: when click list view sort then display all data of sort in sql, then click button "Filter", then display all item of sort | *Coding Logic* | *Medium* | *High* | | 15/7/2018 |

*Figure 5-3: Bug list*

# 5.4 Test Report

## 5.4.1 Unit Tests

Unit tests are performed using JUnit for important and complicated functions.

| | Phase 1 | | Phase 2 | |
|---|---|---|---|---|
| | **Before** | **After** | **Before** | **After** |
| **Pass** | **13** | **18** | **25** | **30** |
| **Fail** | **5** | **0** | **5** | **0** |
| **Untested** | **0** | **0** | **0** | **0** |
| **Total** | **18** | **18** | **30** | **30** |

*Table 5-8: Unit test report*

For each of the tested units, all branches are tested and code coverage is 100%

### 5.4.2 Integration and System Tests

The number of test cases is shown below:

| Module | Pass | Fail | Untested | Total |
|---|---|---|---|---|
| **Home** | **69** | **0** | **0** | **69** |
| **Sell car** | **87** | **0** | **0** | **87** |
| **Sell accessories** | **52** | **0** | **0** | **52** |
| **post car sales** | **50** | **0** | **0** | **50** |
| **Cart** | **55** | **0** | **0** | **55** |
| **Admin** | **237** | **0** | **0** | **237** |
| **Total** | **548** | **0** | **0** | **548** |

. *Table 5-9: Integration and System test report*

List of test cases (Home and Dictionary modules):

| ID | Description | Proccedure | Expected Output | Result |
|---|---|---|---|---|
| [Home-1] | Test function Login case do not insert data | 1.Do not insert data in textbox " Username or email"<br>2.Do not insert data in textbox " Password"<br>3.Click button "Login" | 1.Textbox "Username or email " display nothing<br>2.Textbox "Password " display nothing<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-2] | Test Login case insert data in [username] but do not insert data in [Password] | 1.Insert data in textbox " Username or email" VD:"Admin"<br>2.Do not insert data in textbox " Password"<br>3.Click button "Login" | 1.Textbox "Username or email " display "Admin"<br>2.Textbox "Password " display nothing<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |

| | | | | |
|---|---|---|---|---|
| [Home-3] | Test Login case do not insert data in [username] but insert data in [Password] | 1.Do not insert data in textbox " Username or email"<br>2.Insert data in textbox " Password"<br>VD:"admin"<br>3.Click button "Login" | 1.Textbox "Username or email " display nothing<br>2.Textbox "Password " display "*****"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-4] | Test Login case insert data do match with DB in [username] and insert data in do match with DB in [Password] | 1.Insert data in textbox " Username or email"<br>VD:"admin"<br>2.Insert data in textbox " Password"<br>VD:"admin"<br>3.Click button "Login" | 1. move to page " Xe"<br>2. Avatar and name of [user] export out of corner must be screen | Pass |
| [Home-5] | Test Login case insert data do not match with DB in [username] and insert data in do not match with DB in [Password] | 1.Insert data in textbox " Username or email"<br>VD:"zxczxczxc"<br>2.Insert data in textbox " Password"<br>VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "zxczxczxc"<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-6] | Test Login case insert data do not match with DB in [username] and insert data in do match with DB in [Password] | 1.Insert data in textbox " Username or email"<br>VD:"zxczxczxc"<br>2.Insert data in textbox " Password"<br>VD:"admin"<br>3.Click button "Login" | 1.Textbox "Username or email " display "zxczxczxc"<br>2.Textbox "Password " display "*****"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |

| | | | | |
|---|---|---|---|---|
| [Home-7] | Test Login case insert data do match with DB in [username] and insert data in do not match with DB in [Password] | 1.Insert data in textbox " Username or email" VD:"admin"<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "admin"<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-8] | Test of the button activity "Close "when complete in form | 1.Insert data in textbox " Username or email" VD:"admin"<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Close" | 1.close the login form<br>2.return to page home | Pass |
| [Home-9] | Test case [Username] has less than 3 characters | 1.Insert data in textbox " Username or email" VD:"xc"<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "xc"<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-10] | Test case [Username] has 3 characters | 1.Insert data in textbox " Username or email" VD:"zxc"<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "zxc"<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |

| | | | | |
|---|---|---|---|---|
| [Home-11] | Test case [Username] has 50 characters | 1.Insert data in textbox " Username or email" VD:"zxc….." 50 characters<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "zxc….."-50 characters<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-12] | Test case has special characters | 1.Insert data in textbox " Username or email" VD:"&^&^&&^" more 50 characters<br>2.Insert data in textbox " Password" VD:"&^&^&^&^"<br>3.Click button "Login" | 1.Textbox "Username or email " display "&^&^&&^"- more 50 characters<br>2.Textbox "Password " display "********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-13] | Test case [Username] has more 50 characters | 1.Insert data in textbox " Username or email" VD:"zxc….." more 50 characters<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display "zxc….."- more 50 characters<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-14] | Test case [Username] has whitespace 2 head | 1.Insert data in textbox " Username or email" VD:" admin "<br>2.Insert data in textbox " Password" VD:"zxczxczxc" | 1.Textbox "Username or email " display " admin "<br>2.Textbox "Password " | Pass |

| | | | | |
|---|---|---|---|---|
| | | 3.Click button "Login" | display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | |
| [Home-15] | Test case [Username] has hyphens, apostrophes, quotes | 1.Insert data in textbox " Username or email" VD:" admin"-' "<br>2.Insert data in textbox " Password" VD:"zxczxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display " admin"-' "<br>2.Textbox "Password " display "*********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-16] | Test case [Password] has less 6 character | 1.Insert data in textbox " Username or email" VD:" admin"<br>2.Insert data in textbox " Password" VD:"zxczx"<br>3.Click button "Login" | 1.Textbox "Username or email " display " admin"<br>2.Textbox "Password " display "*****"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-17] | Test case [Password] has 6 characters | 1.Insert data in textbox " Username or email" VD:" admin"<br>2.Insert data in textbox " Password" VD:"zxczxc"<br>3.Click button "Login" | 1.Textbox "Username or email " display " admin"<br>2.Textbox "Password " display "******"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-18] | Test case [Password] has 30 characters | 1.Insert data in textbox " Username or email" VD:" admin"<br>2.Insert data in textbox " Password" | 1.Textbox "Username or email " display " admin"<br>2.Textbox | Pass |

| | | | | |
|---|---|---|---|---|
| | | VD:"zxczxc..." -30 characters<br>3.Click button "Login" | "Password " display "******"- 30 characters<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | |
| [Home-19] | Test case [Password] has more 30 characters | 1.Insert data in textbox " Username or email" VD:" admin"<br>2.Insert data in textbox " Password" VD:"zxczxc..." - more 30 characters<br>3.Click button "Login" | 1.Textbox "Username or email " display " admin"<br>2.Textbox "Password " display "******"- more 30 characters<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-20] | Test case [password] has whitespace 2 head | 1.Insert data in textbox " Username or email" VD:" admin"<br>2.Insert data in textbox " Password" VD:" zxczxc "<br>3.Click button "Login" | 1.Textbox "Username or email " display " admin"<br>2.Textbox "Password " display "********"<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Home-21] | Test function login with facebook with account facebook is not allowed | Click button "Login with Facebook" | show pop-up login of facebook | Pass |
| [Home-22] | Test function login with facebook with account facebook is allowed | Click button "Login with Facebook" | 1. Autologin with facebook account<br>2. move to page " Xe"<br>3. Avatar and name of [user] export out of corner must be | Pass |

| | | | | |
|---|---|---|---|---|
| | | | screen | |
| [Admin – 23] | Test login "Admin" form. | 1.Do not insert data in textbox " Username or email"<br>2.Do not insert data in textbox " Password"<br>3.Click button "Login" | 1.Textbox "Username or email " display nothing<br>2.Textbox "Password " display nothing<br>3.Form "login" display message "Sai tài khoản hoặc mật khẩu" | Pass |
| [Admin – 24] | Test function Search "Xe" on admin page | 1. Click "Xe" tab on the admin page.<br>2. Insert 1 record do match with DB.<br>VD "BMW"<br>3. Click " Search " button | 1. Textbox display "BMW"<br>2. Table "Xe" display do match product with db | Pass |
| [Admin – 25] | Test function "Thêm xe" on admin page | 1. Choose "Toyota" in combo box "Hãng xe"<br>2. Write "Innova" in textfield "Tên xe"<br>3. Click button "Lưu" | 1. "Hãng xe" display "Toyota"<br>2. "Tên xe" display "Innova"<br>3. Show messenger "Vui lòng nhập năm sản xuất" | Pass |
| [Admin – 26] | Test function remove product on "Xe" page of admin | 1. "Xe" page have 2 product<br>VD: "Xe 1"<br>"Xe 2"<br>2. Click icon "xóa" of product "Xe 1"<br>3. Click button "OK" of dialog box | 1. Product has been removed<br>2. Table "Xe" on admin page display "Xe 2" | Pass |
| [Admin – 27] | Test function edit product on " Xe " page | 1. Choose "Toyota" in combo box "Hãng xe"<br>2. Write "Innova" in textfield "Tên xe"<br>3. Click button "Lưu" | 1. "Hãng xe" display "Toyota"<br>2. "Tên xe" display "Innova"<br>3. Show | Pass |

| | | | | |
|---|---|---|---|---|
| | | | messenger "Vui lòng nhập năm sản xuất" | |
| [Admin – 28] | Test function Search "Phụ kiện" on admin pag | 1. Click "Phụ kiện" tab on the admin page.<br>2. Insert data do not match with DB .VD "zxc"<br>3. Click " Search " button | 1. Textbox display "zxc"<br>2. Table "Phụ kiện" display text "Not found data" | Pass |
| [Admin – 29] | Test function remove product on "Phụ kiện" page of admin | 1. "Phụ kiện" page have 2 product<br>VD: "Phụ kiện 1" "Phụ kiện 2"<br>2. Click icon "xóa" of product "Phụ kiện 1", "Phụ kiện 2"<br>3. Click button "OK" of dialog box | 1. Product has been removed<br>2. Table "Phụ kiện" on admin page show messge " không có Phụ kiện nào " | Pass |
| [Admin – 30] | Test function Search "Người dùng" on admin page | 1. Click "Người dùng" tab on the admin page.<br>2. Insert some record do match with DB<br>VD " bmw bmw bmw "<br>3. Click " Search " button | 1. Textbox dislay "bmw bmw bmw"<br>2. "Phụ kiện" display text " not found data" | Pass |

*Table 5-9: Integration and System test report*

For the full list of test cases and more details, see the file **LKT_TestReport.xlsx**.

### 5.4.3 User Acceptance Test

All checklist items passed (as described in section 5.3.3).

Users are generally satisfied with overall usability of the website.

# Chapter 6 : User manual

## 6.1 Deployment guidelines

### 6.1.1 Environment for development

6.1.1.1 Setup development environment

## 6.2 User guidelines

### 6.2.1 Homepage

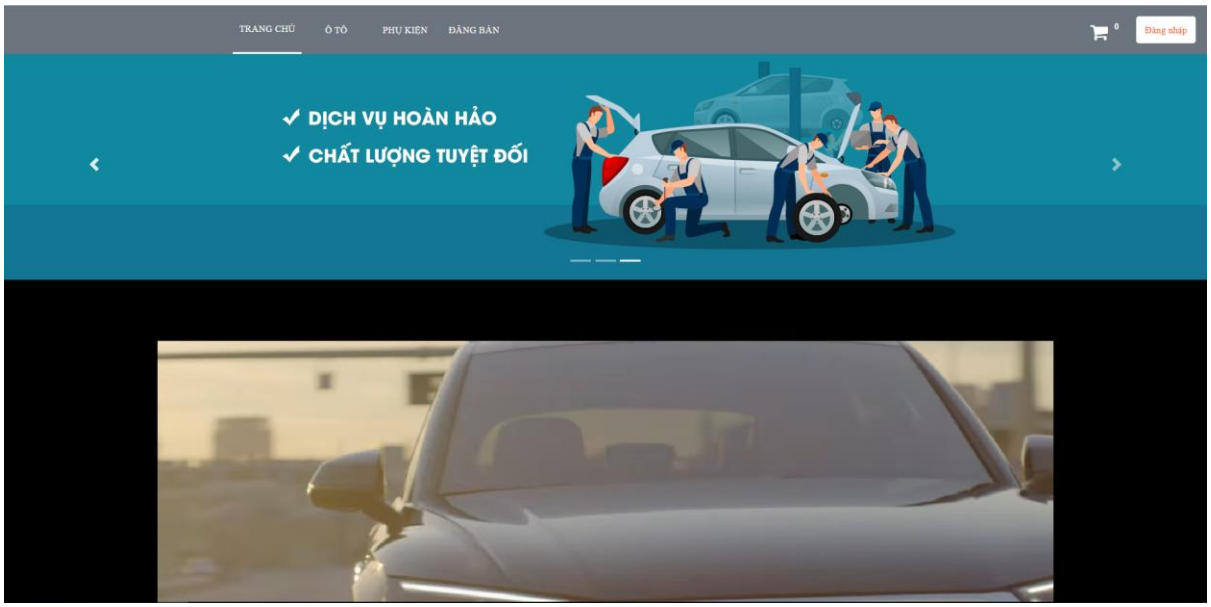



*Figure 6-1: Homepage*

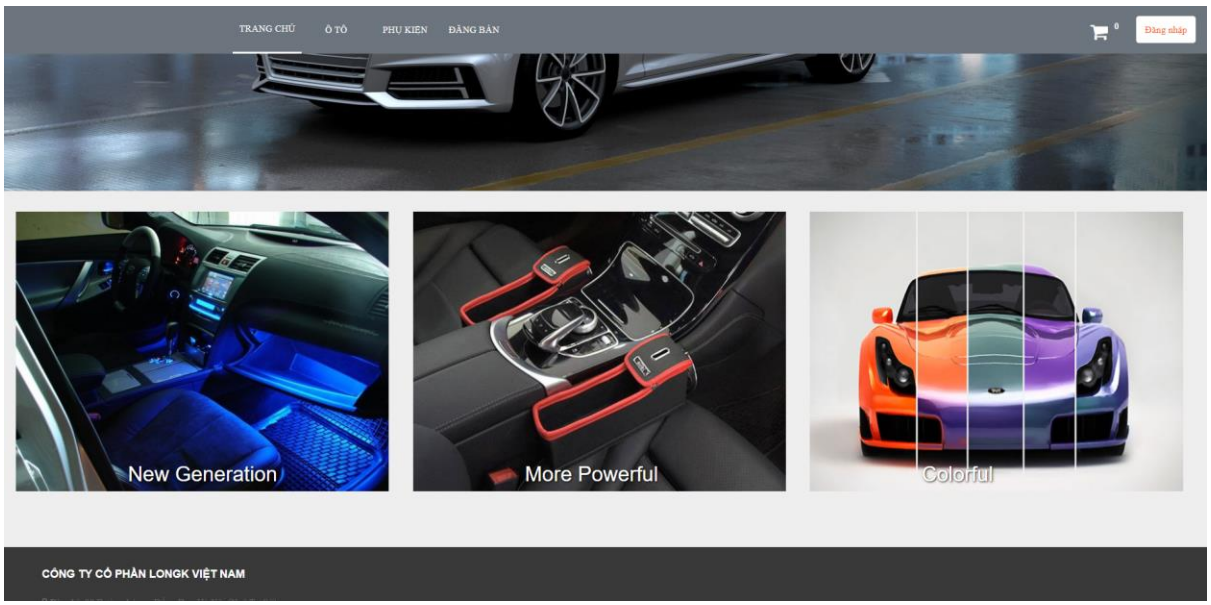



*Figure 6-2: Homepage*

In homepage we have a slide slow of car to introduce some famous Car Brand

### 6.2.2 Car page

Click " Ô TÔ" in the website's header

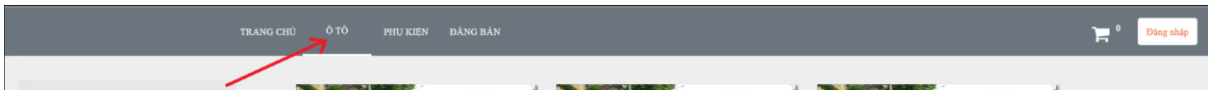



*Figure 6-3: Oto header*

System displays the Car page

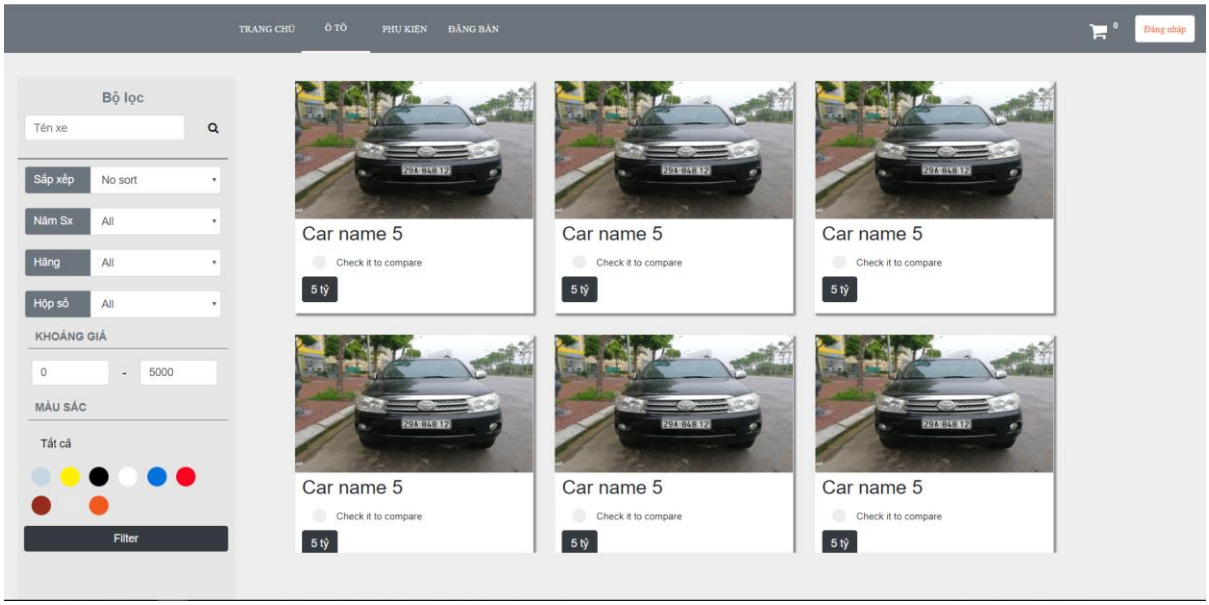



*Figure 6-4: Carpage*

Click to car card to view detail of car

### 6.2.3 Search car

Click to “search” text box write a car name and click search button to find a car

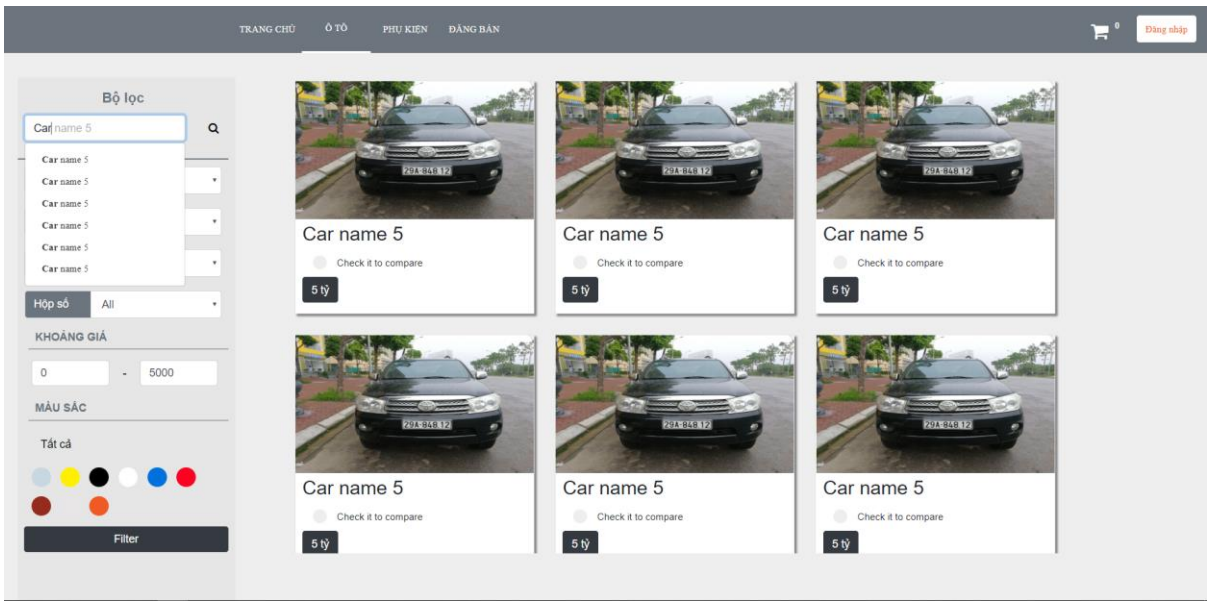



*Figure 6-5: Search car*

If you don’t write any text in search text box then click search button , the website’s will show all car in system.

### 6.2.4 Filter Car

Select option in dropdown list then click Filter button

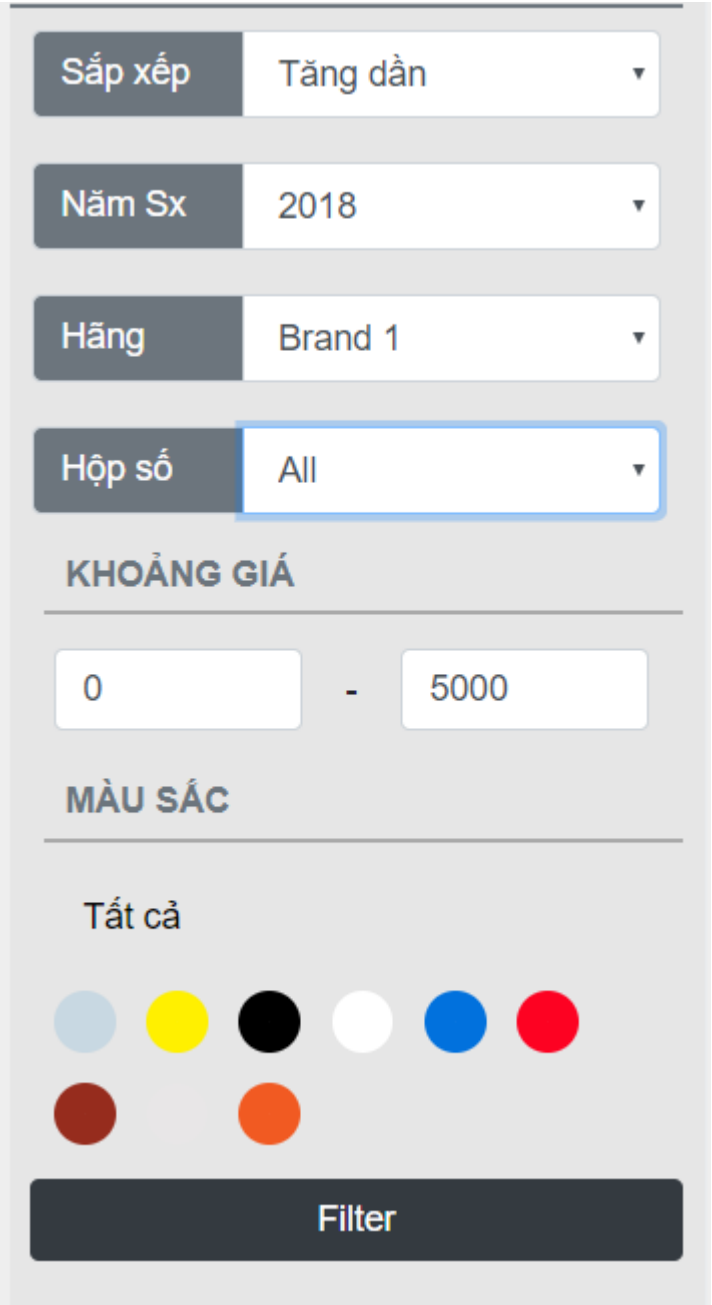



*Figure 6-6: Filter Car*

System show exactly information of car in filter box. If inappropriate requirements , system must show blank in car cart

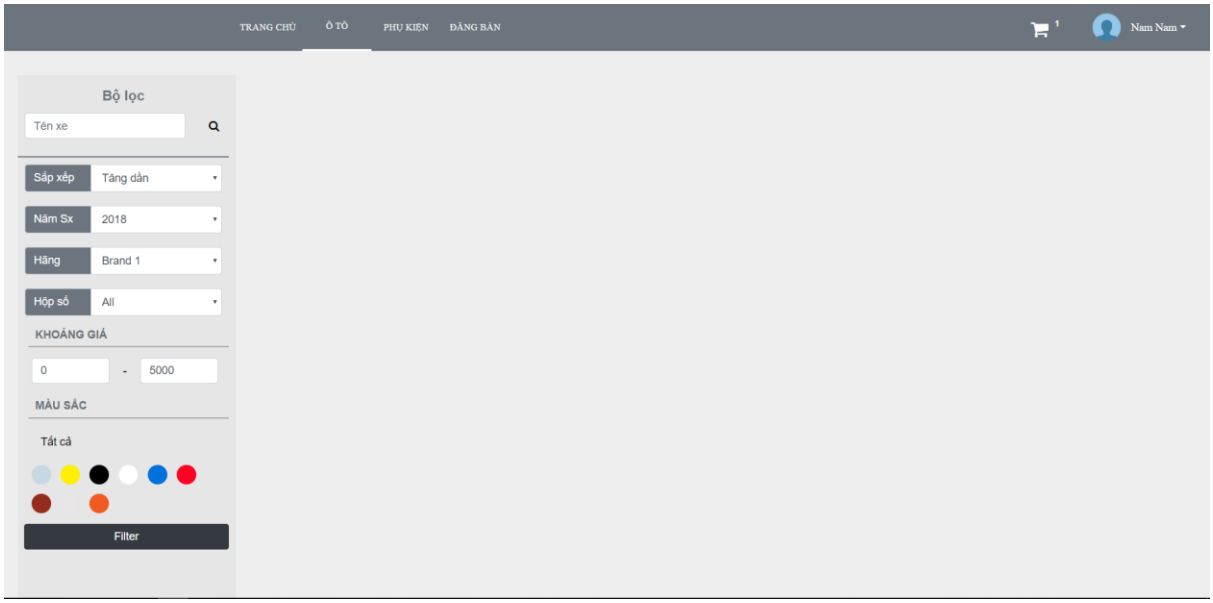



*Figure 6-7: Filter Car*

## 6.2.5 Accessories page

Click " Phụ kiện" in the website's header




*Figure 6-8: Phu kien header*

System displays the Accessories page

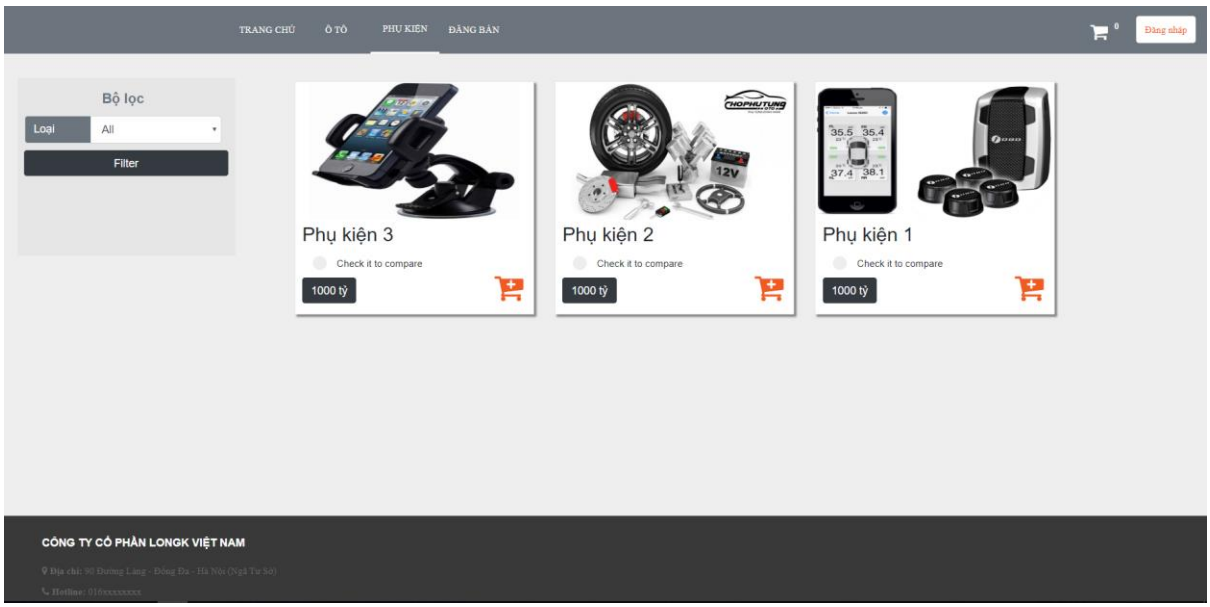



*Figure 6-9: Accessories page*

### 6.2.5.1 Filter Accessories

Select option in dropdown list then click Filter

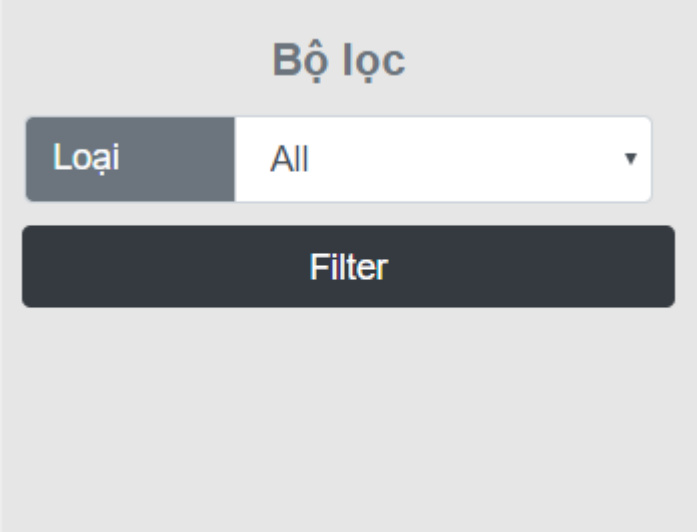



*Figure 6-10: Filter Accessories*

System show exactly information of car in filter box. If inappropriate requirements , system must show blank in accessories cart

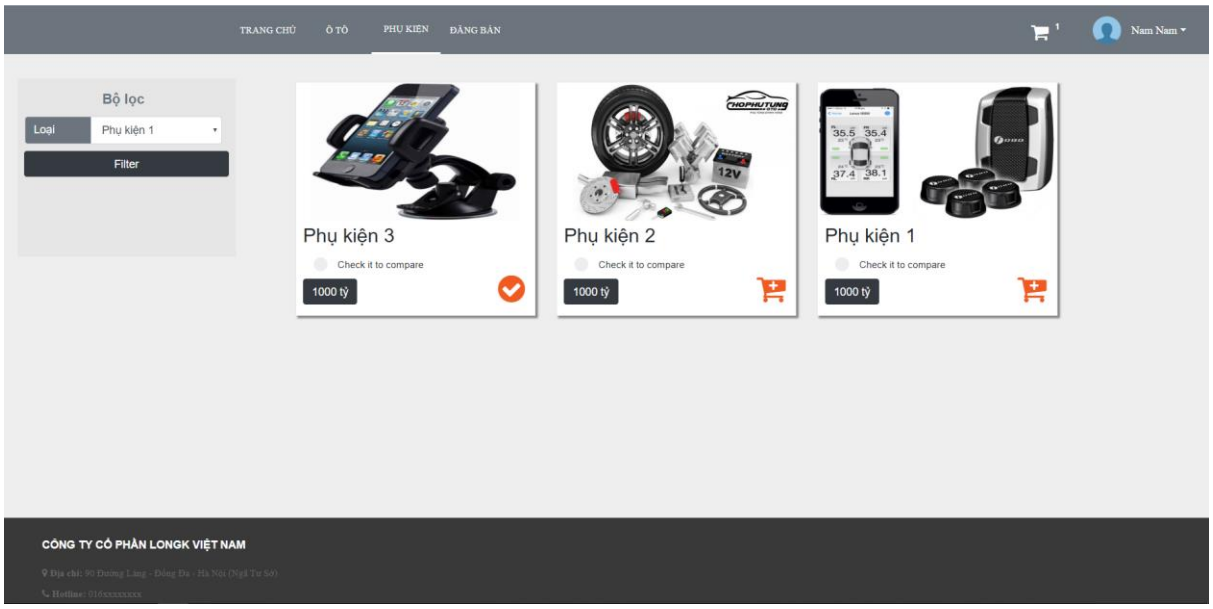



*Figure 6-11: Filter Accessories*

Blank page filter accessories

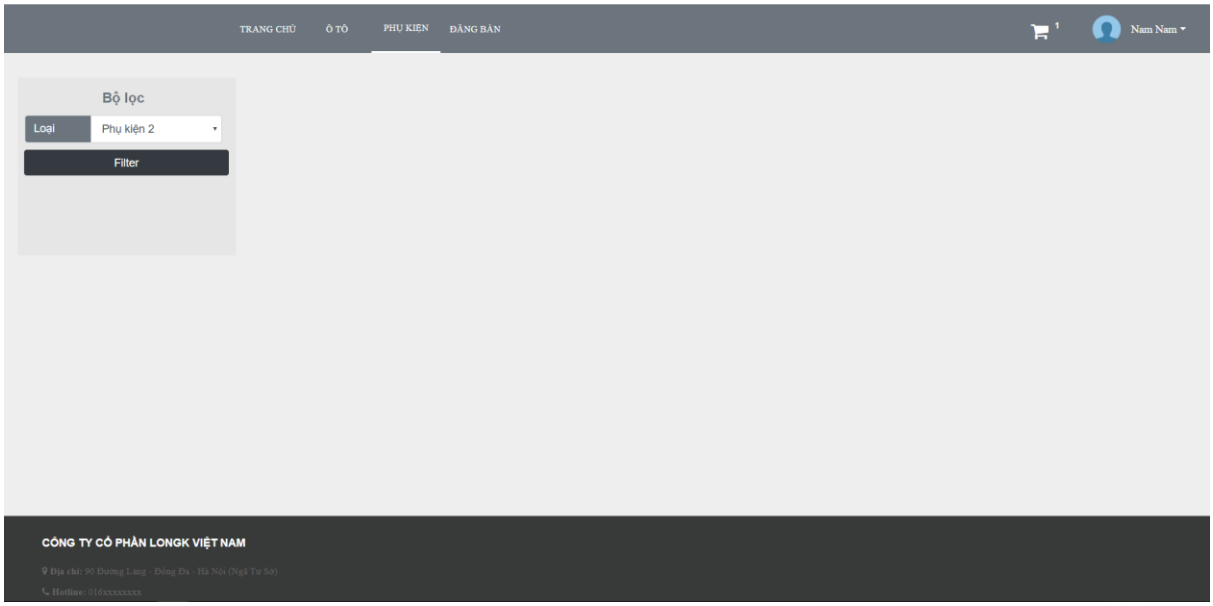



*Figure 6-12: Filter Accessories*

### 6.2.6 Post car page

Click " ĐĂNG BÁN" in the website's header




*Figure 6-13: Menu*

System displays the "Đăng bán" page

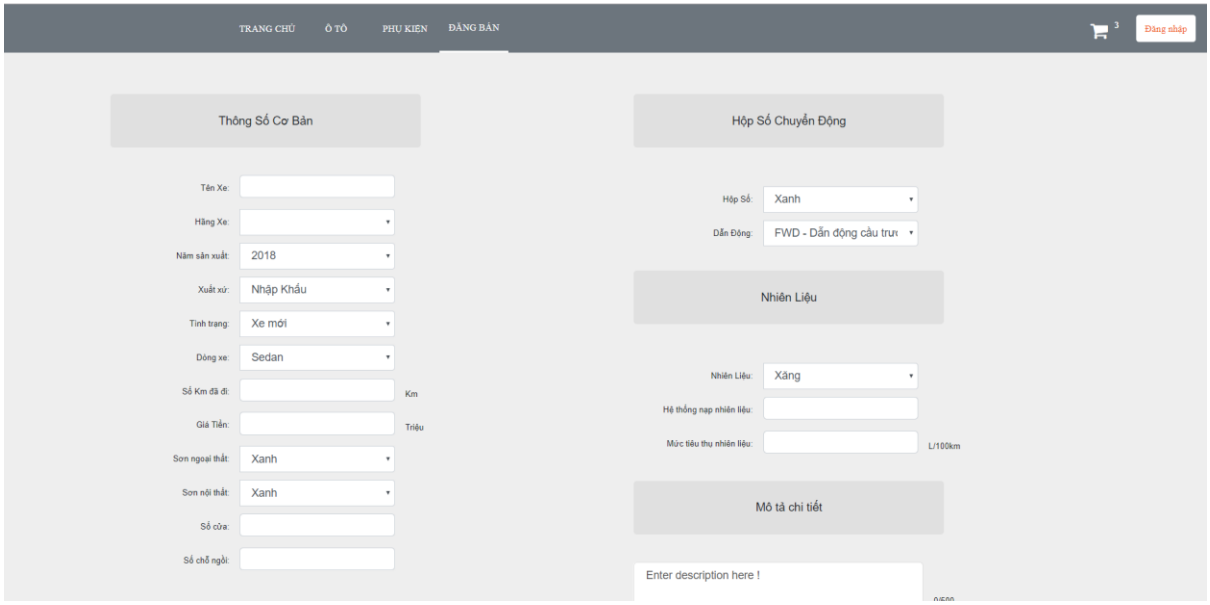



*Figure 6-14: Post car page*

Click “Thêm ảnh” to add image.

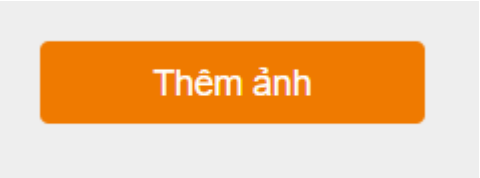



*Figure 6-15: Thêm ảnh*

Select image to add to LongK System and click Open

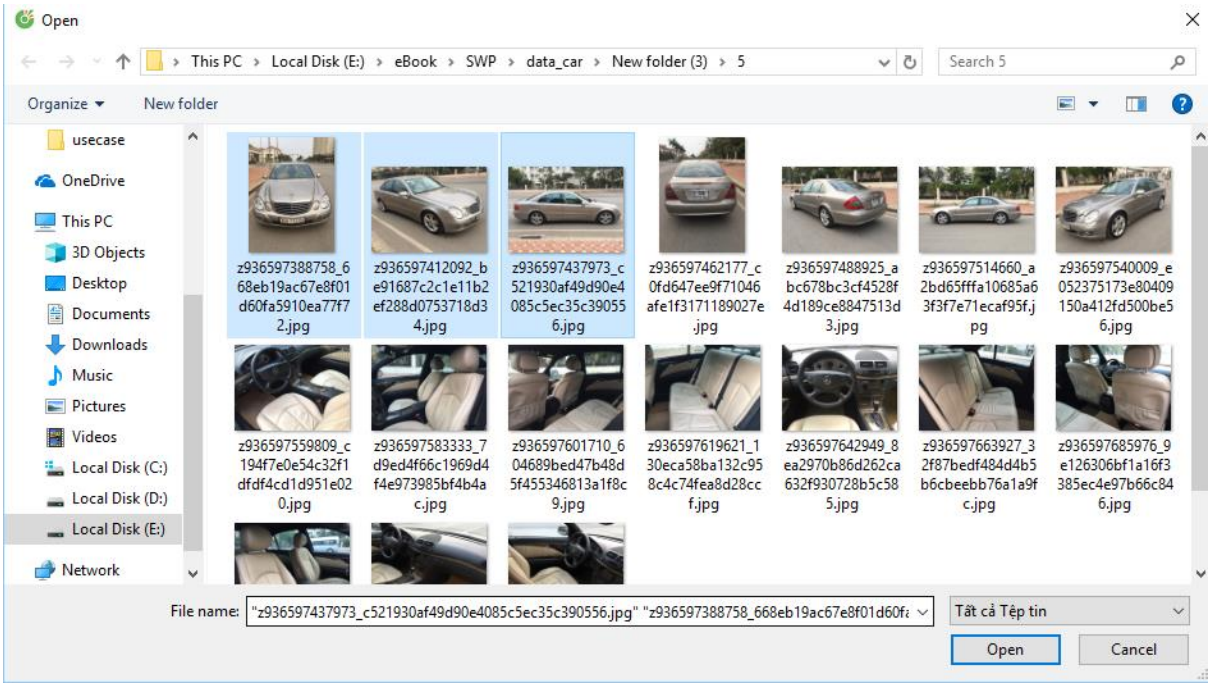



*Figure 6-16: Add to LongK*

System add and display all image you choice and show on car image field.

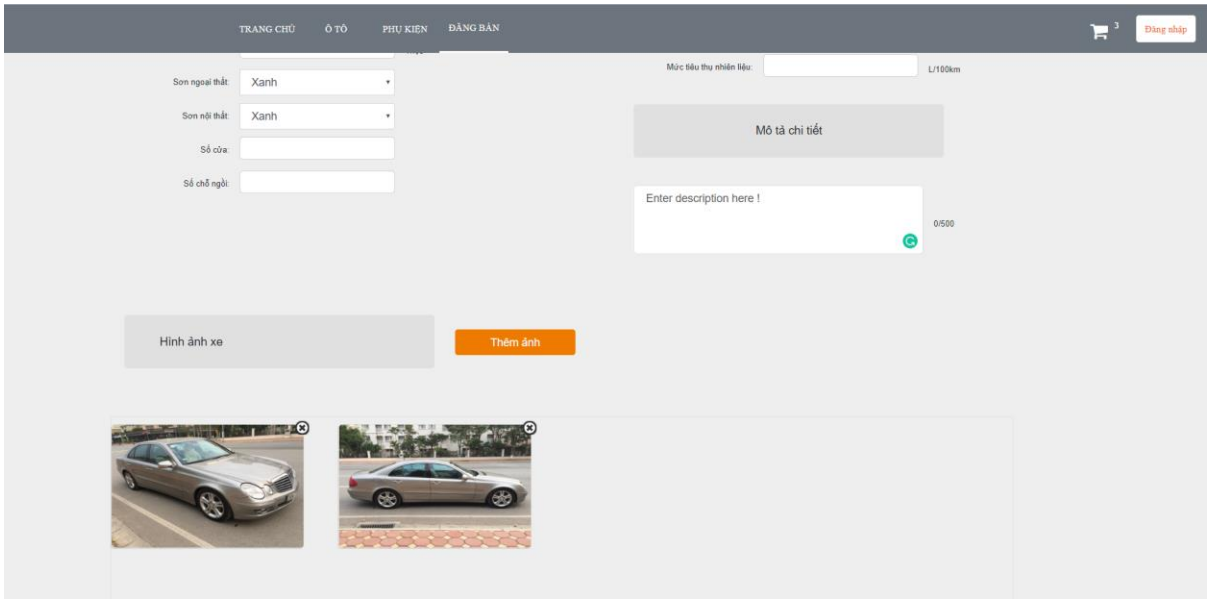



*Figure 6-17: Add to LongK*

### 6.2.7 Add Accessories to cart

Click " Phụ kiện" in the website's header




System displays the Accessories page

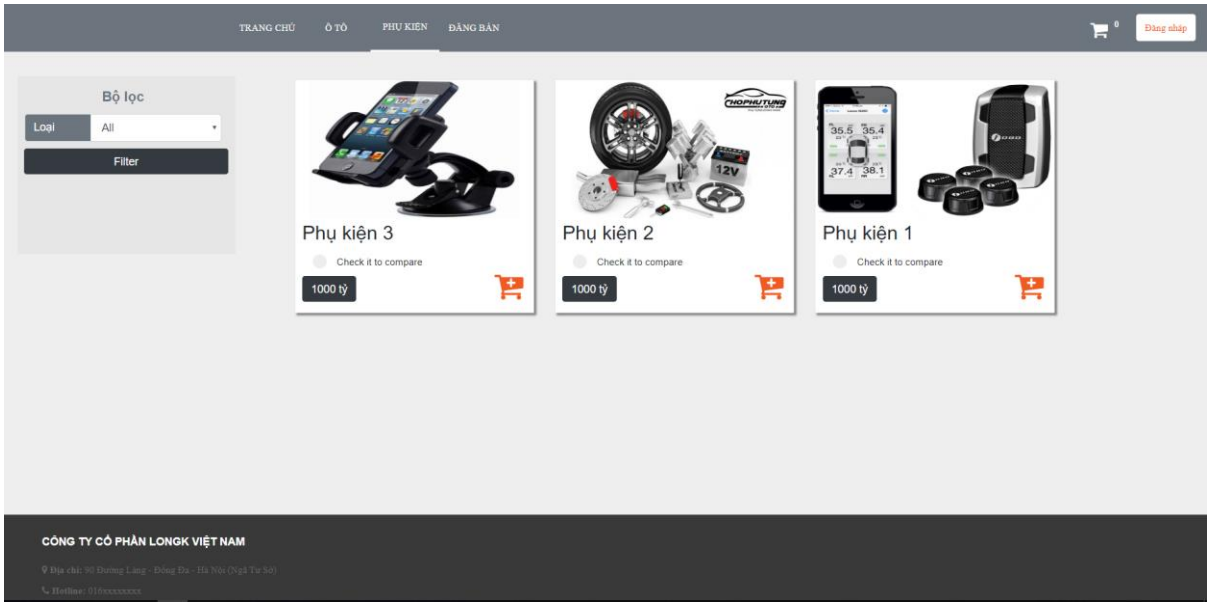



*Figure 6-18: Add Accessories to cart*

Click icon “add cart”



*Figure 6-19: Cart icon*

The system will addition accessories to your cart in the webstie’s header and change status of shopping cart in website’s header.




*Figure 6-20: Cart icon*

Click icon “shopping cart” to check your shopping cart




*Figure 6-21: Cart header*

System display all item in your shopping cart

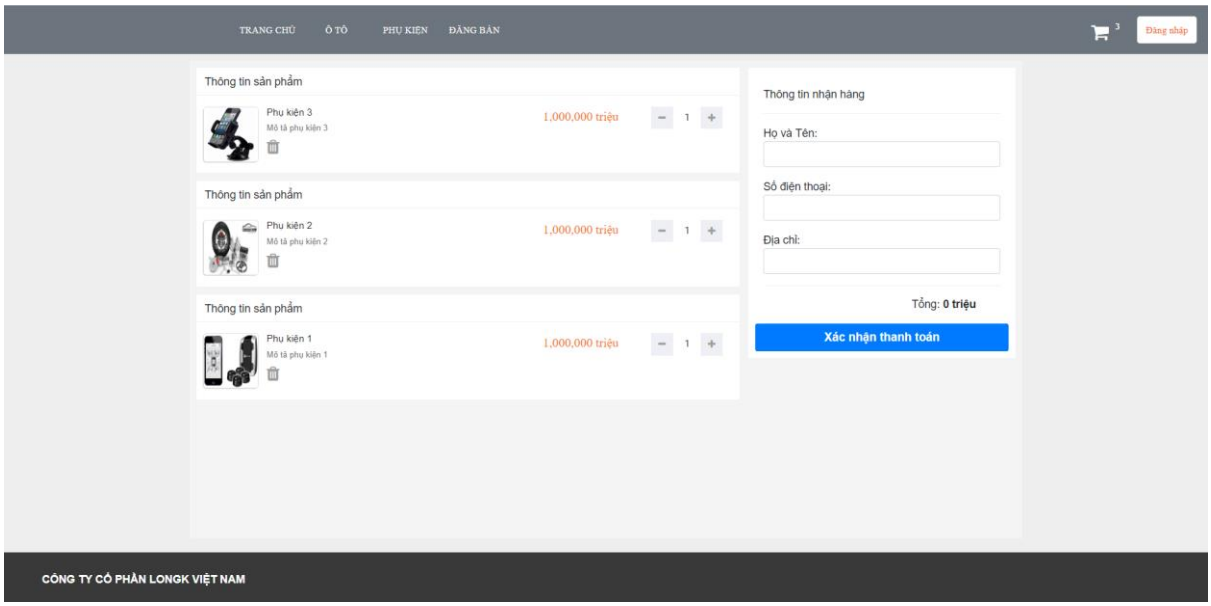



*Figure 6-22: Shopping cart*

### 6.2.8 Check out

Click to “ xác nhận thanh toán “ to checkout ( Pre-condition : You must checkout product when you has already log in on system)

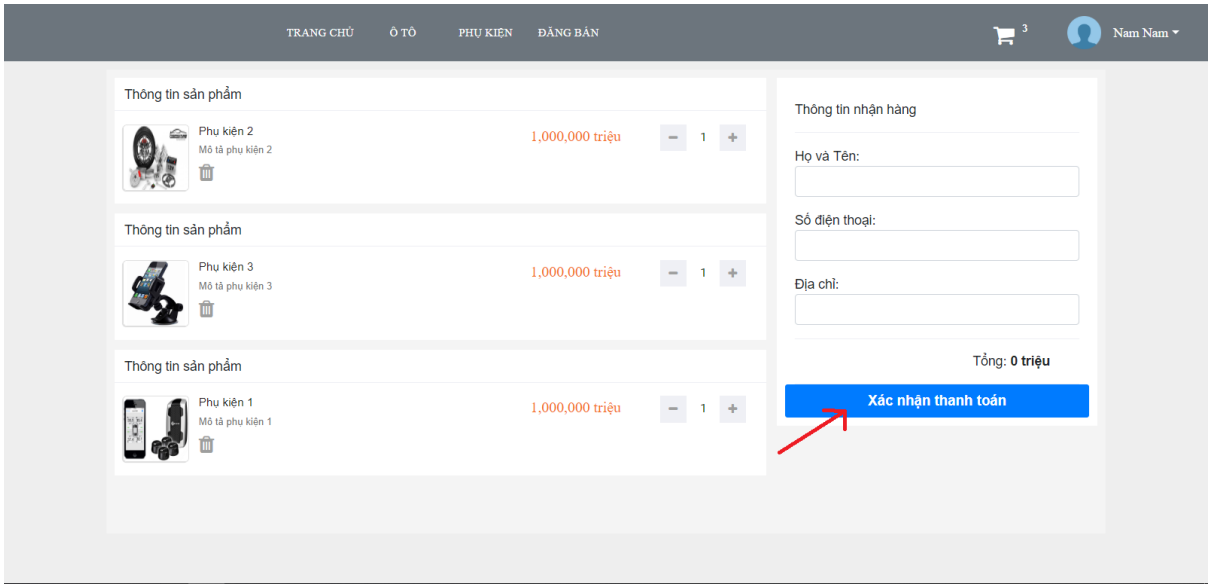



*Figure 6-23: Check out*

System show dialog

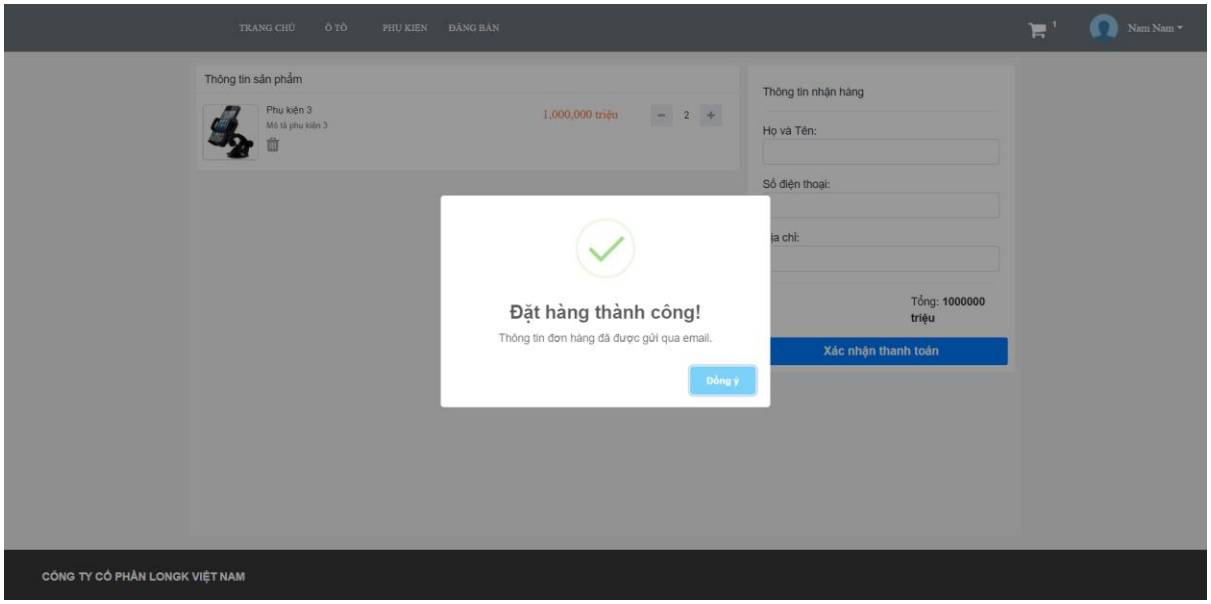



*Figure 6-24: Dialog check out successfully*

System resend Template confirm your oder on your email of user register in LongK Shop car.

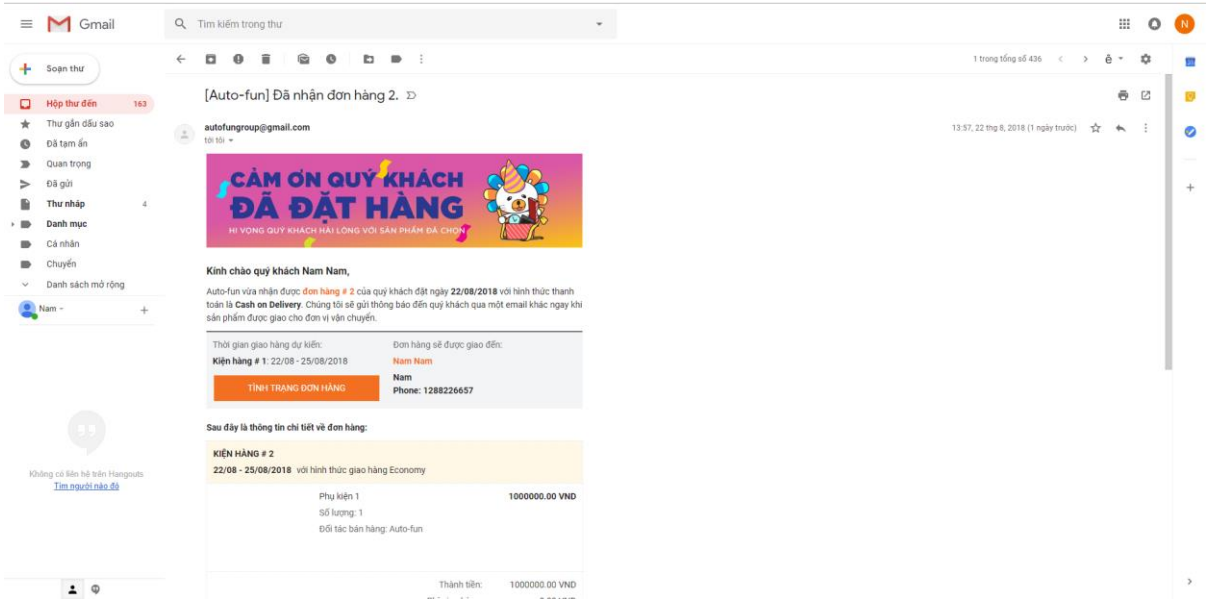



*Figure 6-25: Mail cofirm templete*

## 6.2.9 Register

Click " Đăng nhập" in the website's header

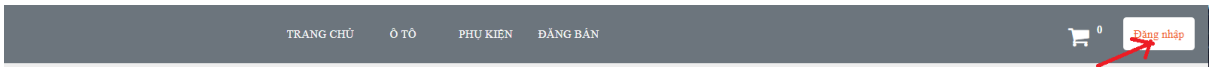



*Figure 6-26: Login button*

System display the Login page

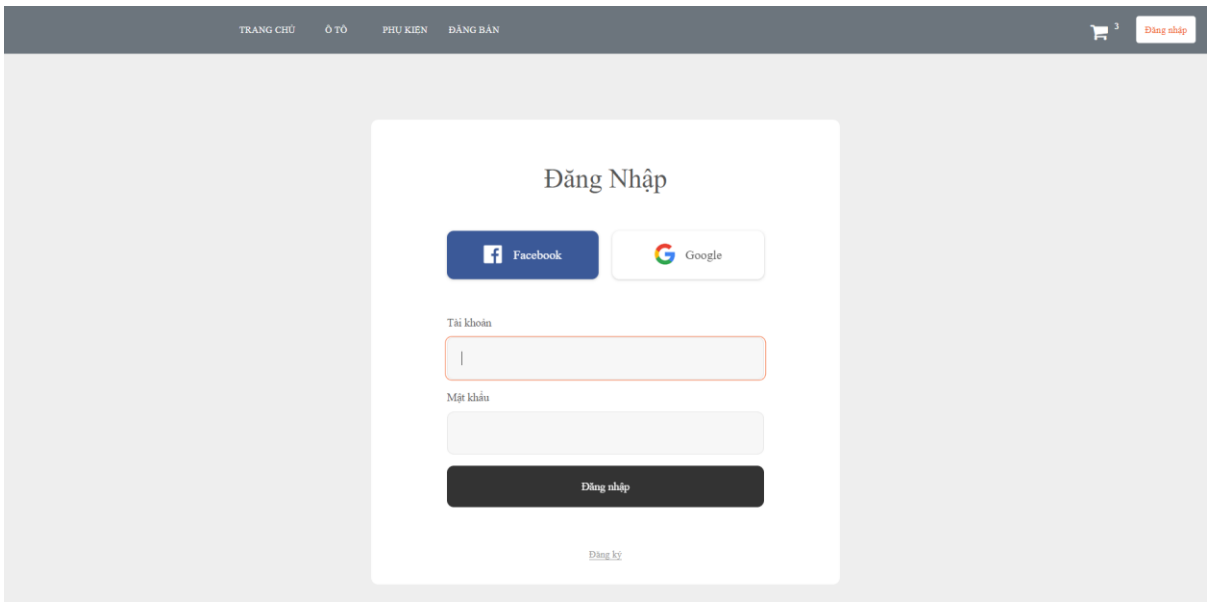



*Figure 6-27: Login page*

Click “Đăng ký” in Login form

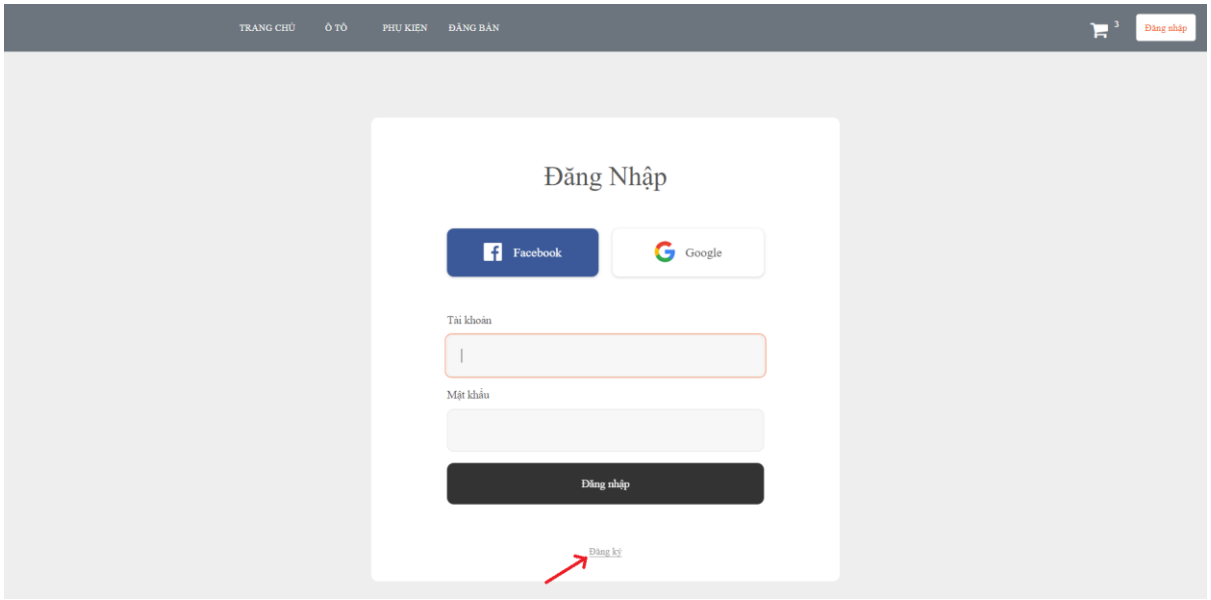



*Figure 6-28: Register*

System display Register page

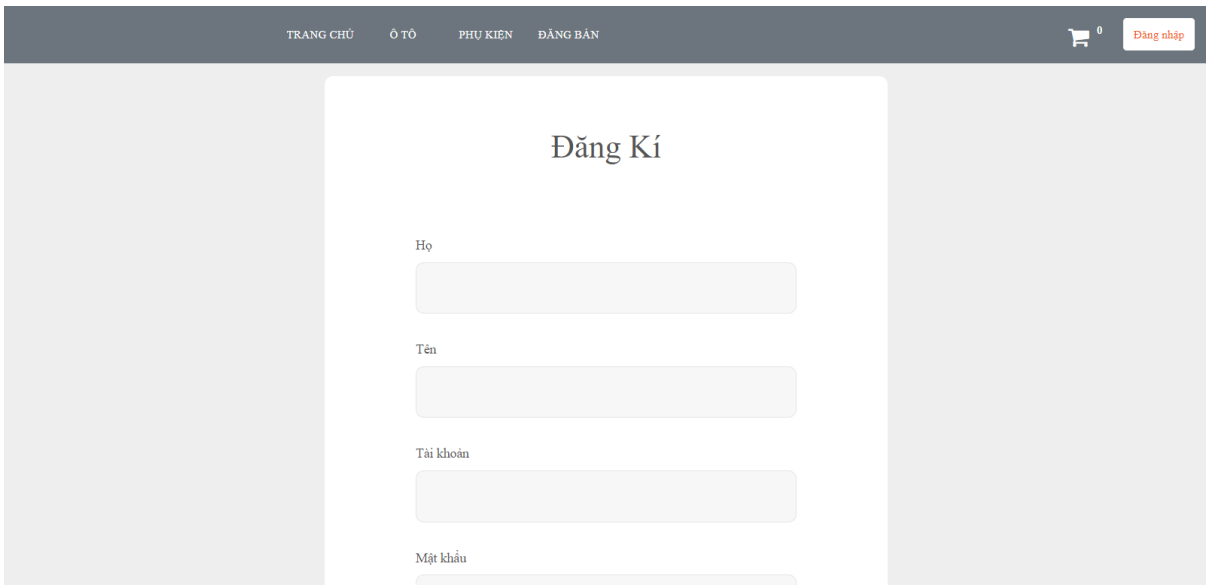



*Figure 6-29: Register page*

Write all information to the text box then click “Đăng ký” button. In email textbox must write email form ( Example: nam@abc.com). “Số điện thoại” just a number. System show dialog when register success.

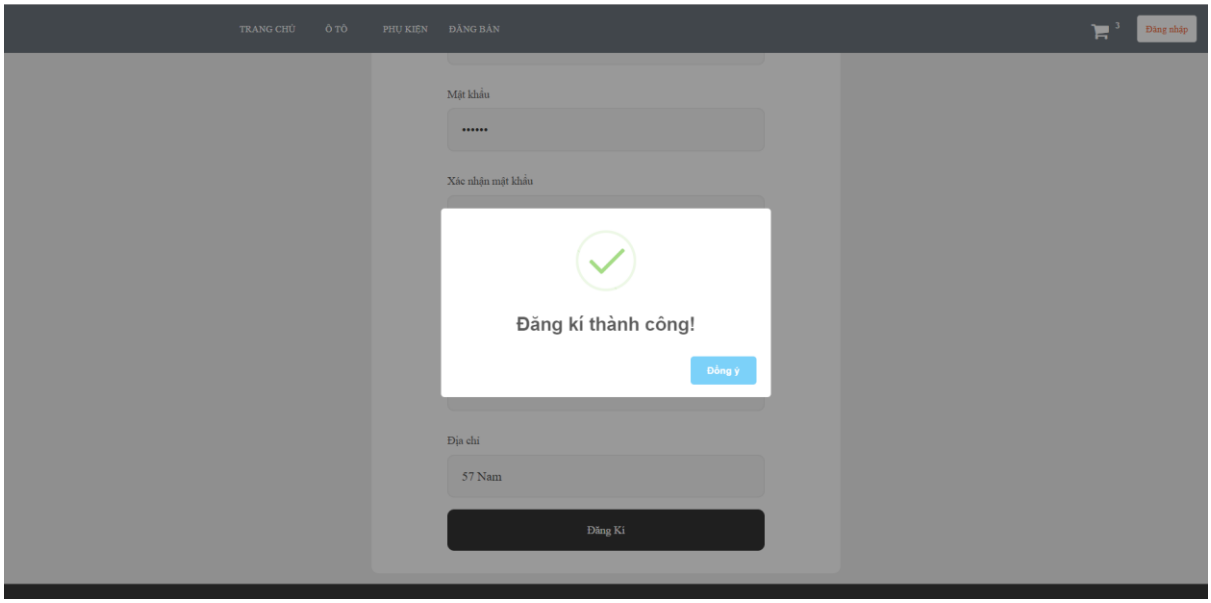



*Figure 6-30: Register dialog succsessfully*

## 6.2.10 Login

Click " Đăng nhập" in the website's header

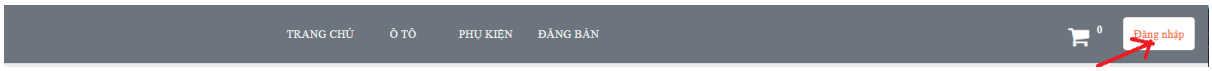



*Figure 6-31: Login header*

System display the Login page

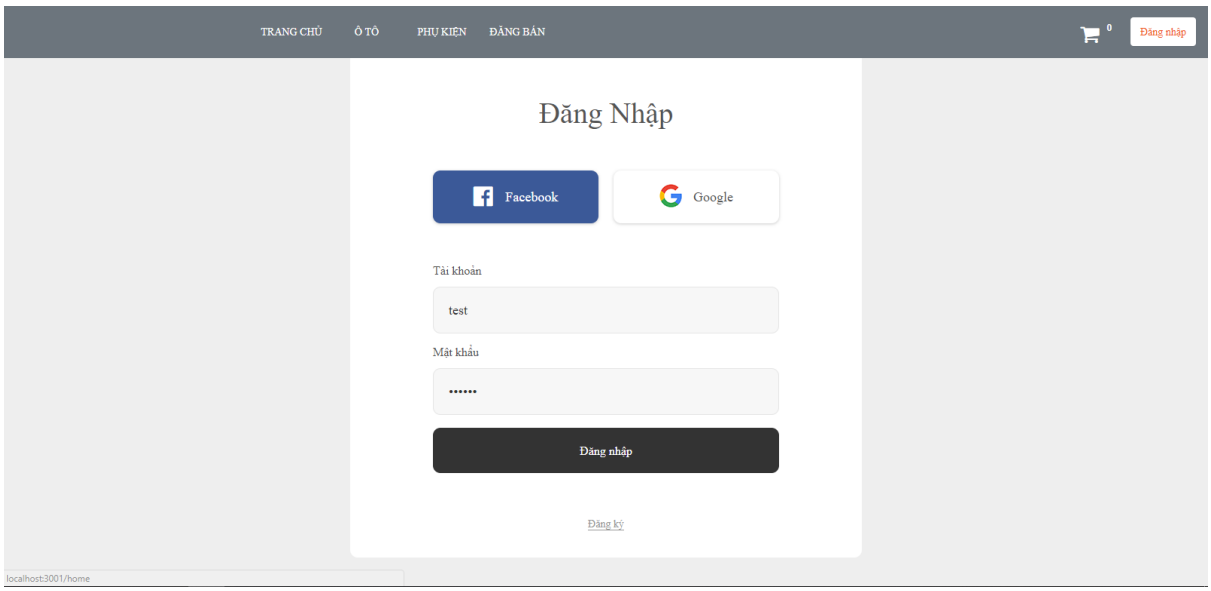



*Figure 6-32: Login page*

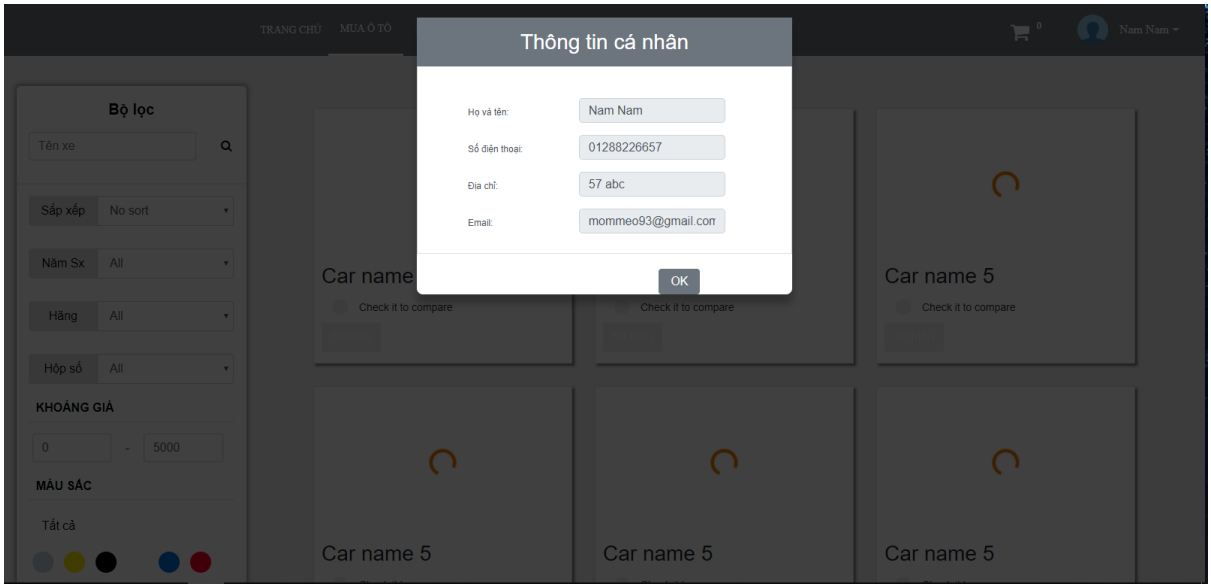
Enter login credentials and click "Đăng nhập" to log in.

### 6.2.11 View Detail User

Click “ Thông tin tài khoản” in the top right corner




System display information of user account




### 6.2.12Logout

Click "Đăng xuất" in the top right corner




*Figure 6-33: Logout*

## 6.2.13 Admin page

Enter a link of admin page to display admin page

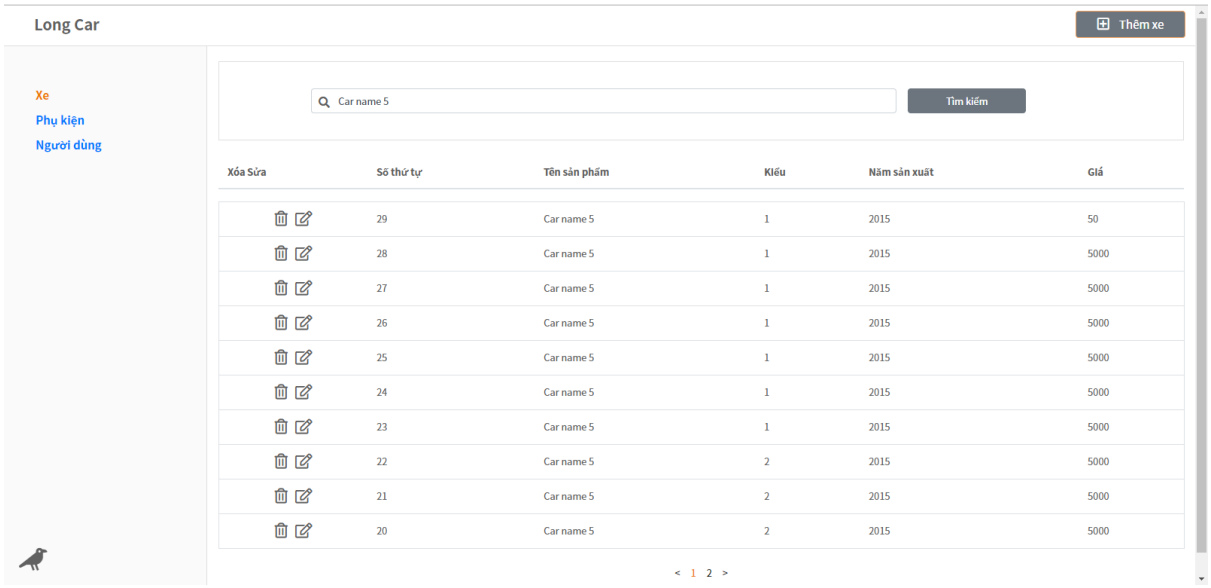



*Figure 6-34: Admin page*

## 6.2.14 Manage Car in admin page

### 6.2.14.1 Add a new car

Click " Thêm xe" on the top right of admin page

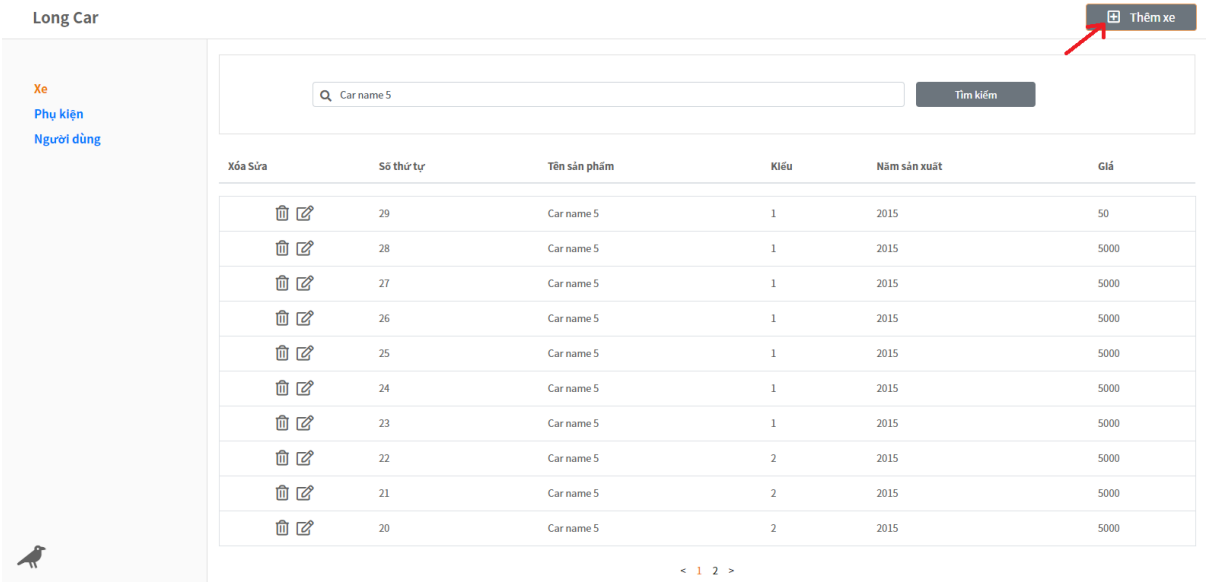



*Figure 6-35: Add a new car*

System display " Thêm xe" page and you must enter all item in "Thêm xe" template to add a new car

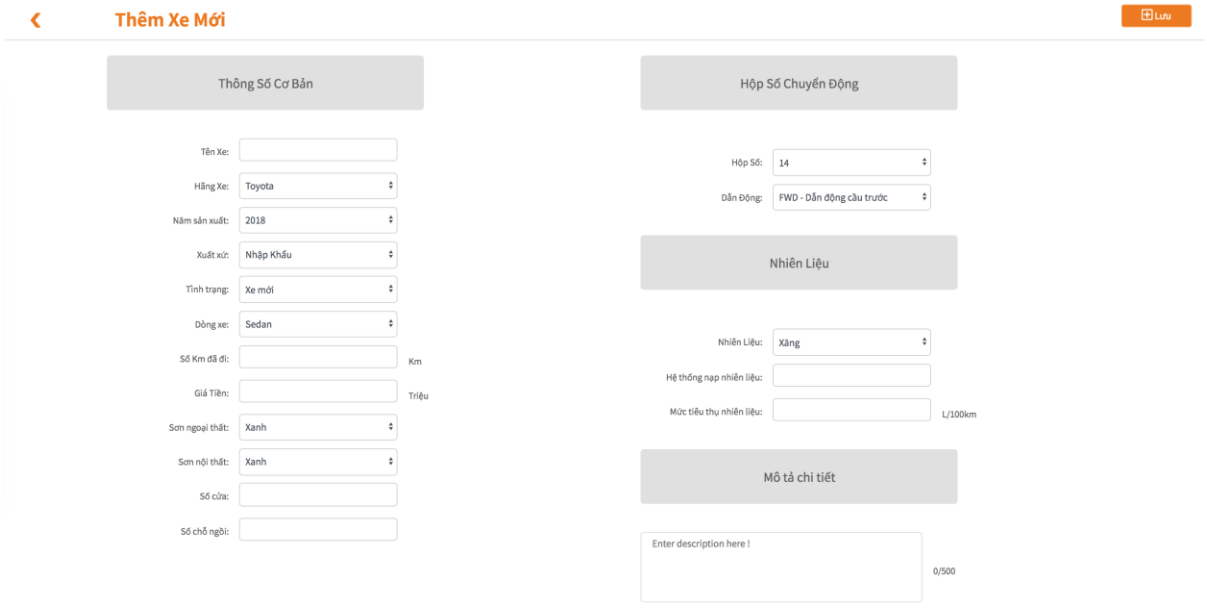



*Figure 6-36: Them xe form*

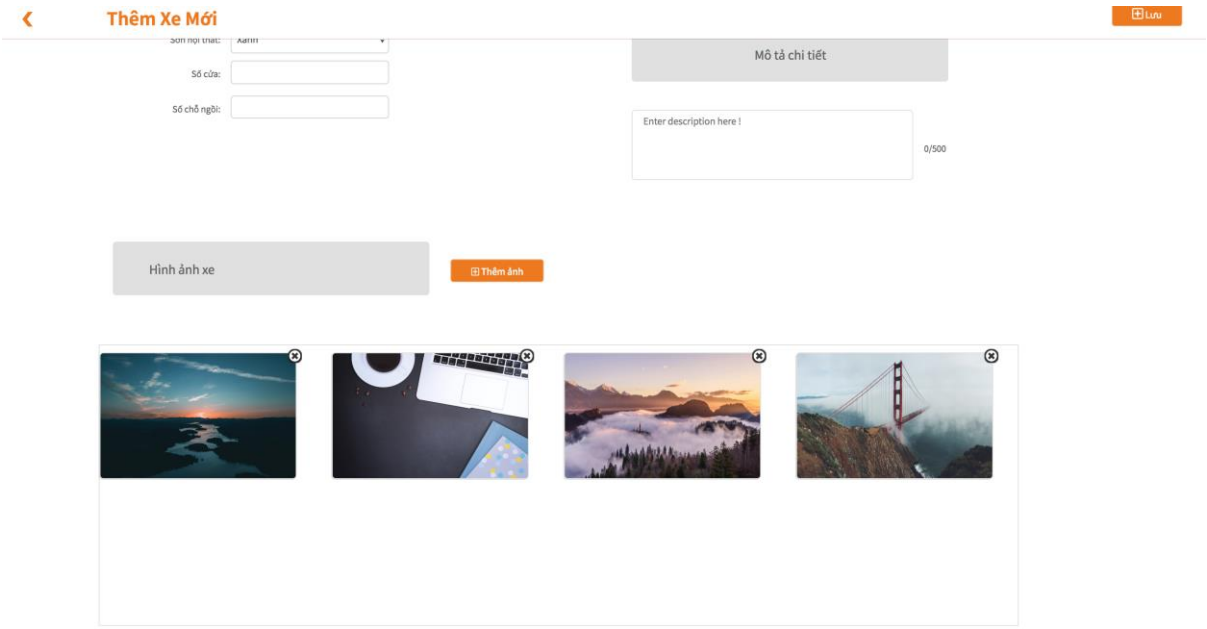



*Figure 6-37: Them xe form*

Click “Luu” button on the top of website’s to finish “thêm xe” task.

### 6.2.14.2 Delete car

In admin page click”Xe”

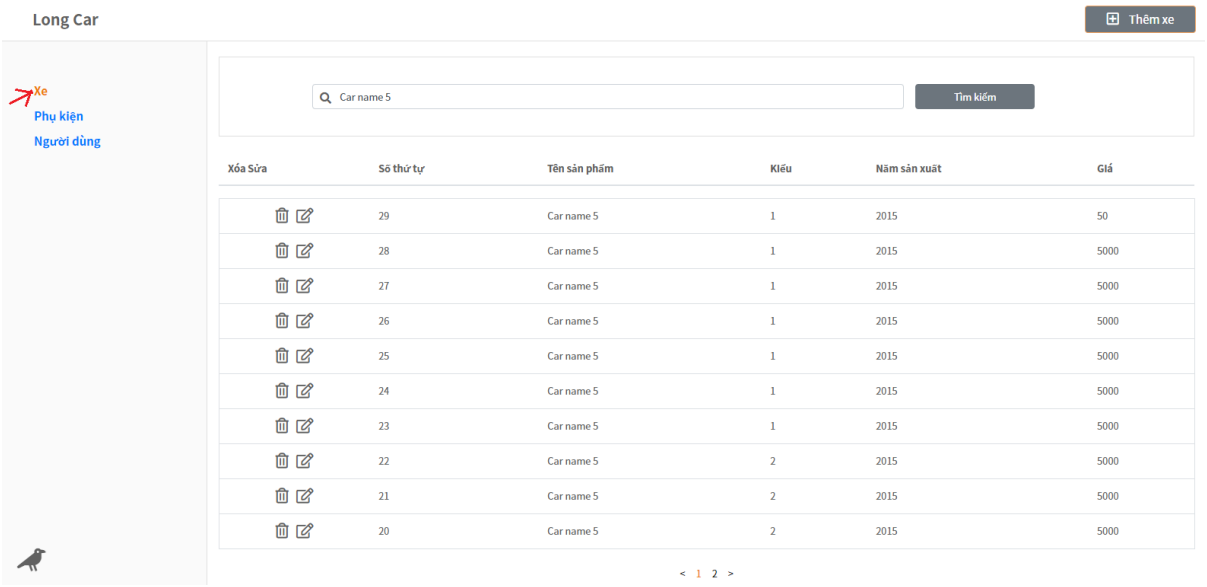



*Figure 6-38: Delete car*

Click icon Recycle bin to delete a car

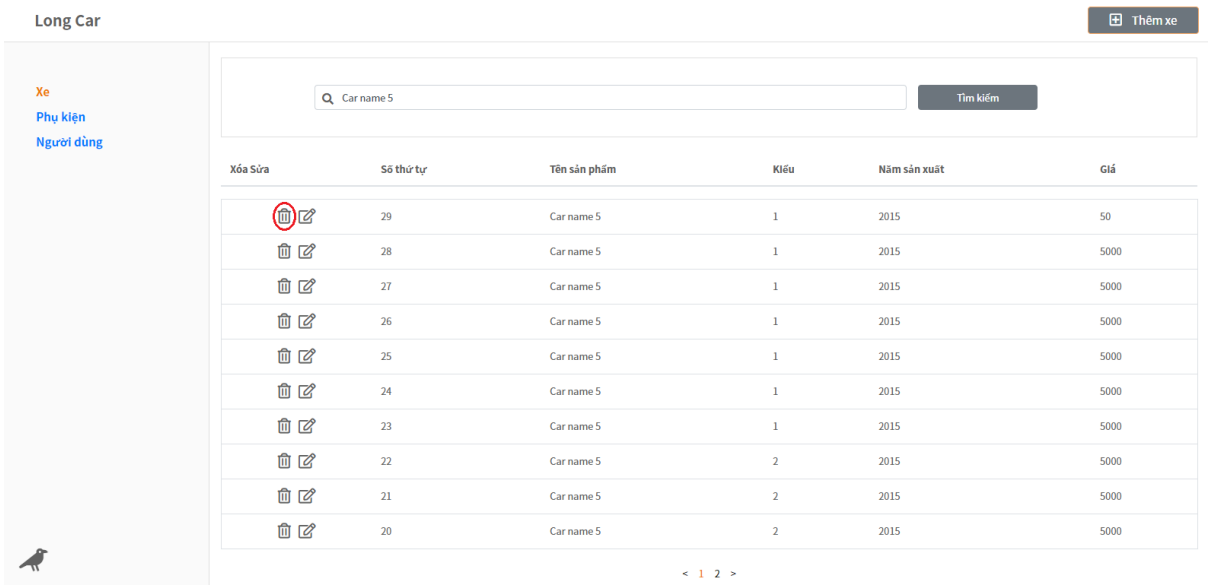



*Figure 6-39: Delete car*

System show a dialog to accept or reject your request

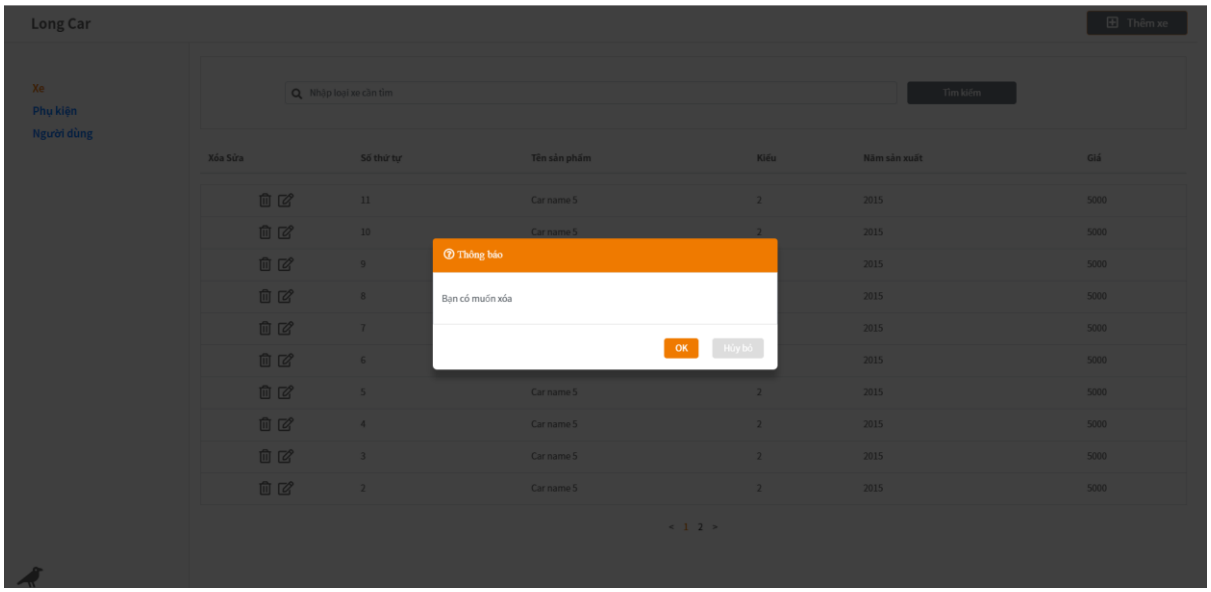



*Figure 6-40: Dialog to accept or reject*

Click “OK” to confirm delete Car

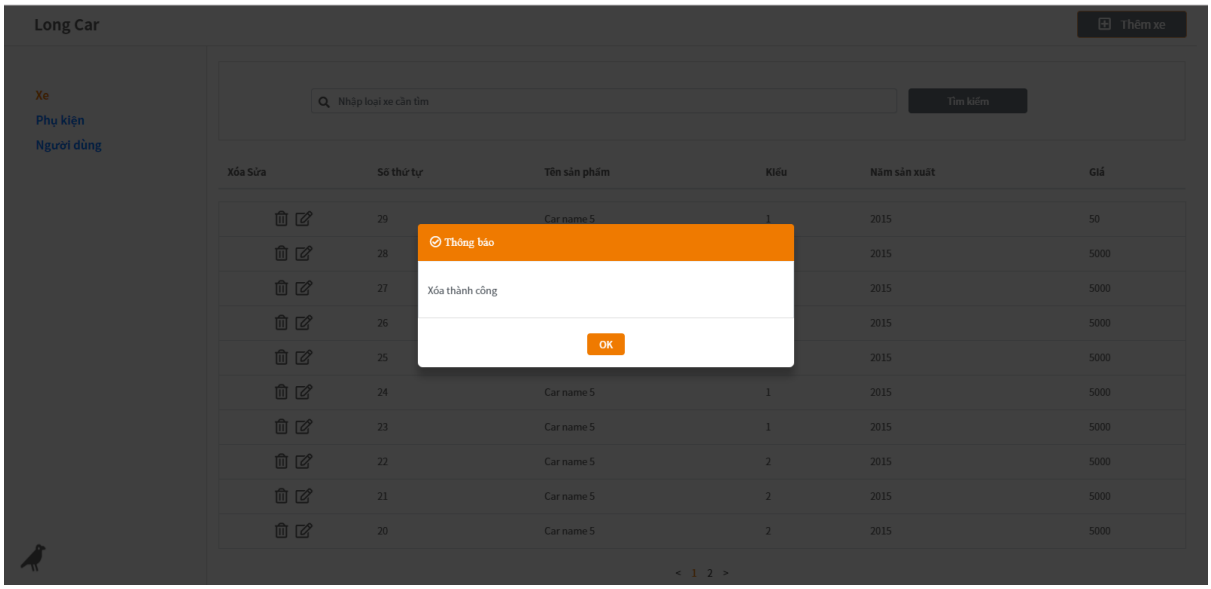



*Figure 6-41: Dialog delete car successfully*

### 6.2.14.3 Update car

### 6.2.14.4 Add Accessorie

Click " Thêm phụ kiện" on the top right of admin page

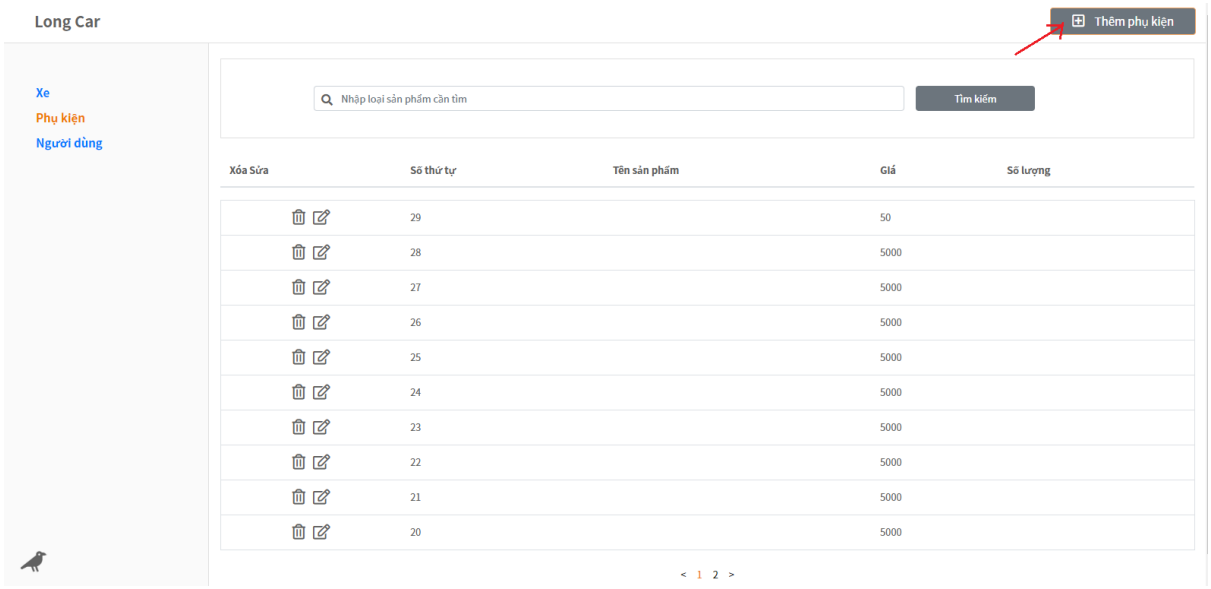



*Figure 6-42: Add accessories*

System display " Thêm phụ kiện" page and you must enter all item in "Thêm phụ kiện" template to add a new car

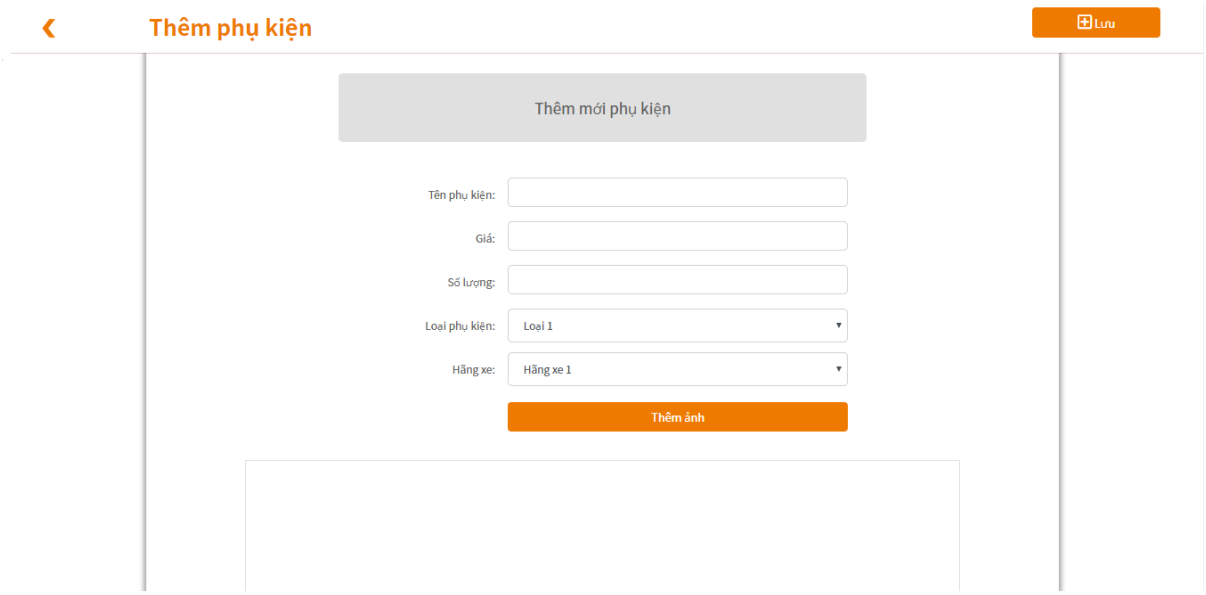



*Figure 6-43: Form them xe*

Click “Lưu” button on the top of website’s to finish “thêm xe” task.

### 6.2.14.5 Delete Accessories

In admin page click”Phụ Kiện”

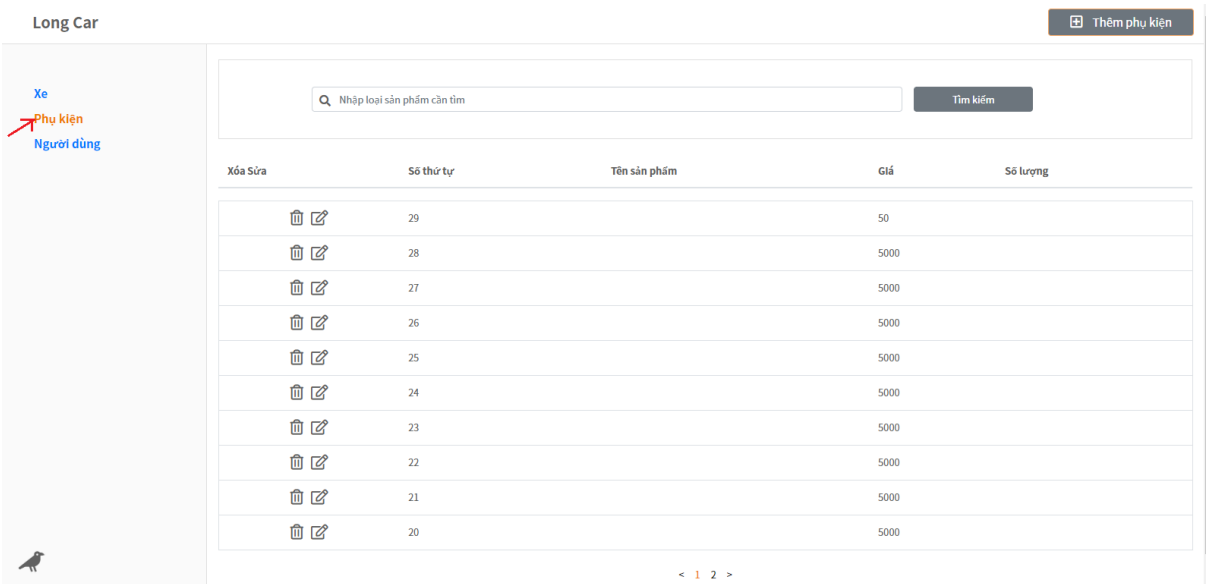



*Figure 6-44: Delete accessorie*

Click icon Recycle bin to delete a car

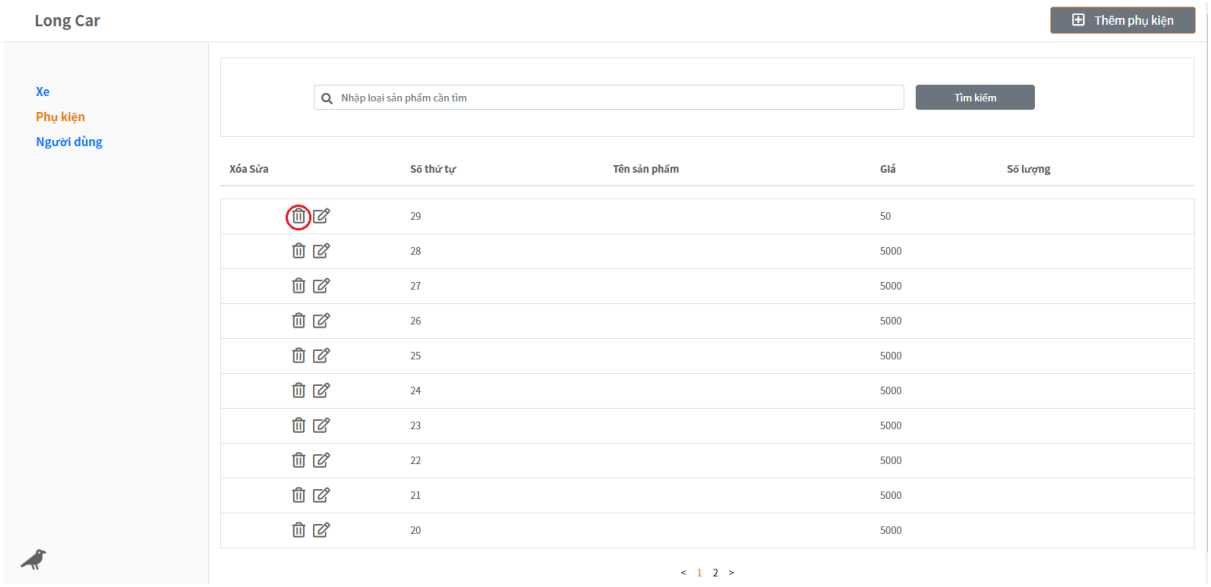



*Figure 6-45: Form them xe*

System show a dialog to accept or reject your request

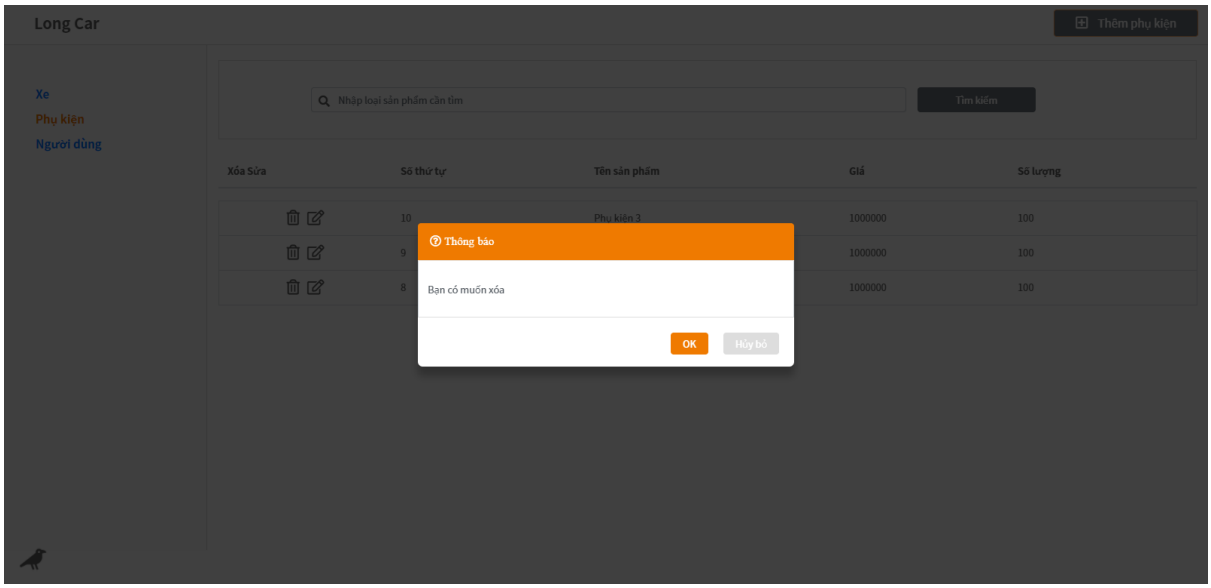



*Figure 6-46: Dialog to accept or reject*

Click “OK” to confirm delete Accessories

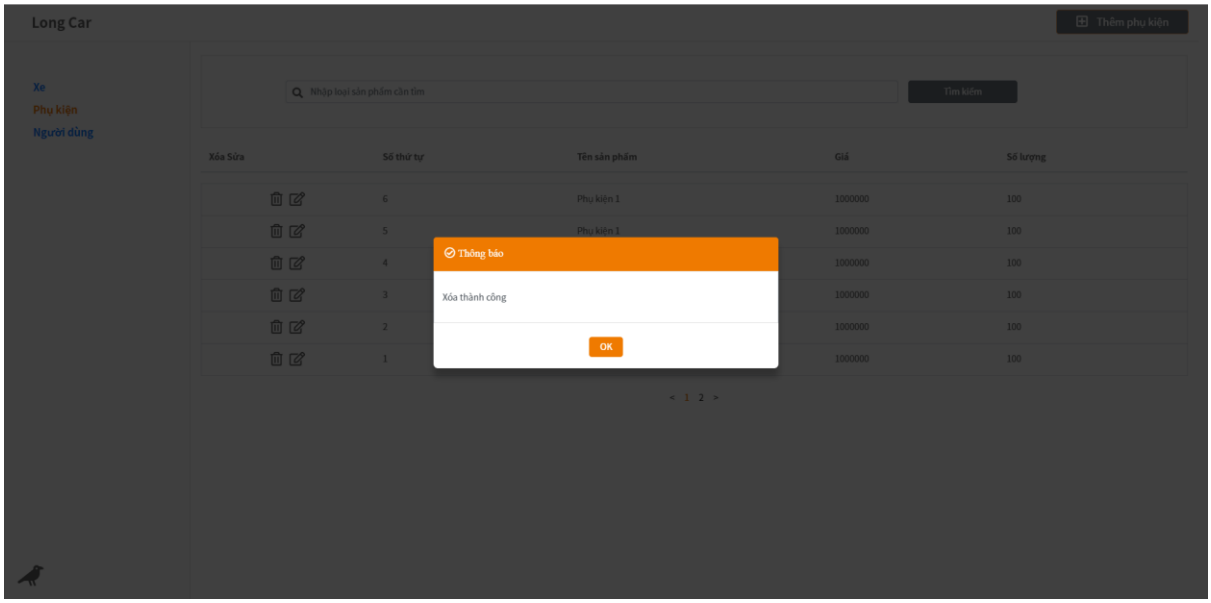



*Figure 6-47: Dialog delete successfully*

### 6.2.14.6 View Account

In admin page click"Người dùng"

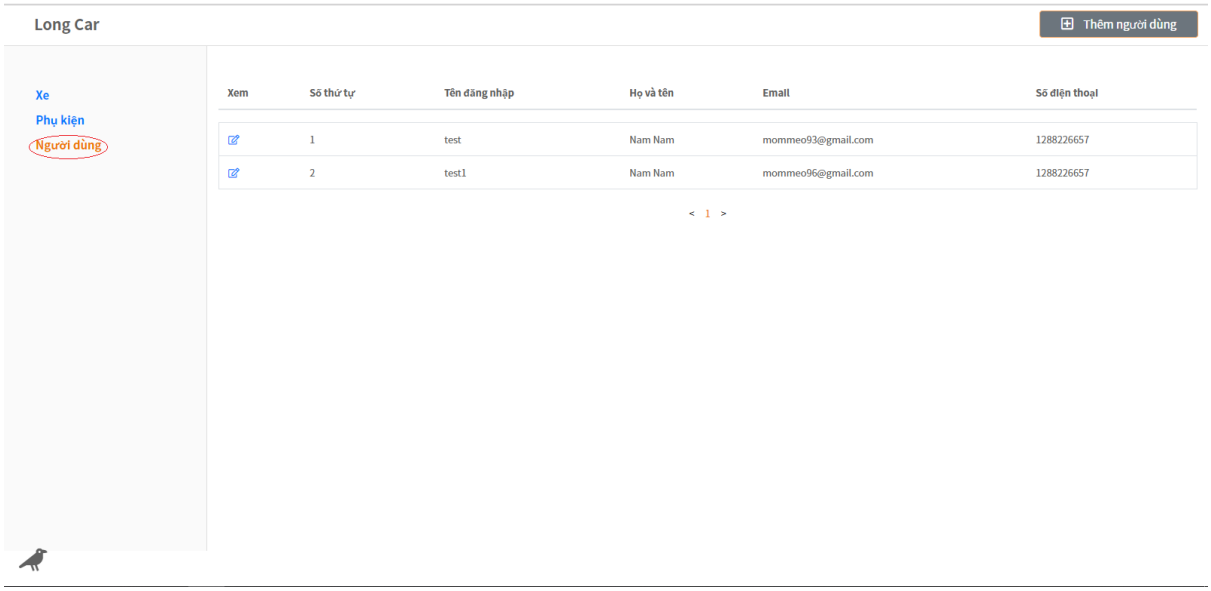



*Figure 6-50: View account*

Click icon View to view information of account

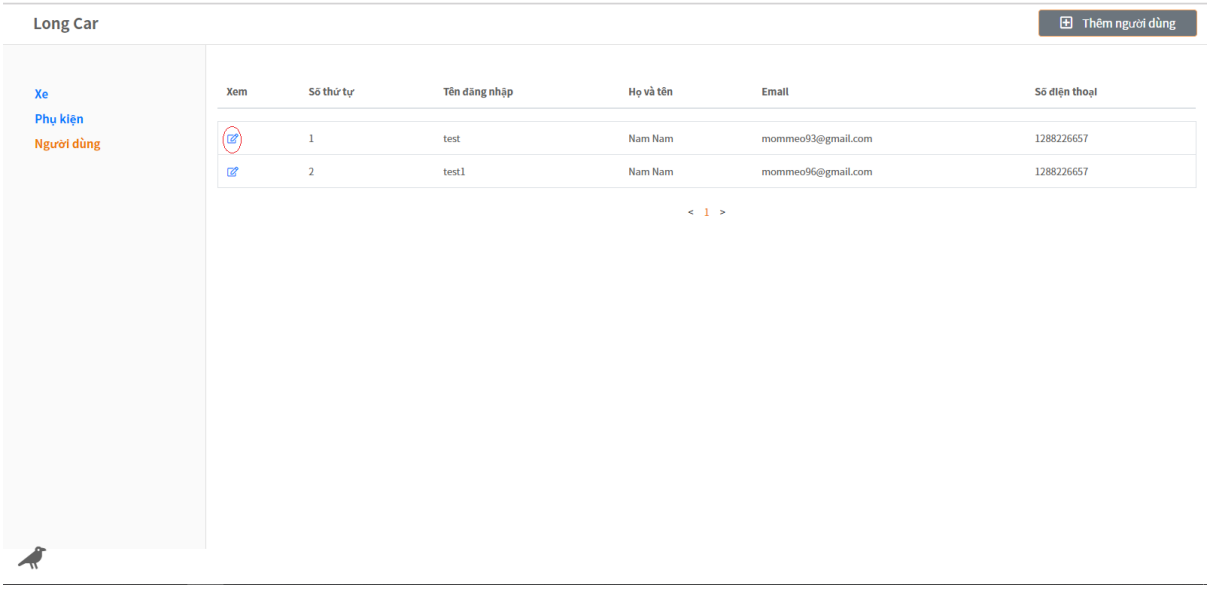



*Figure 6-51: View account*

System show all information of user account